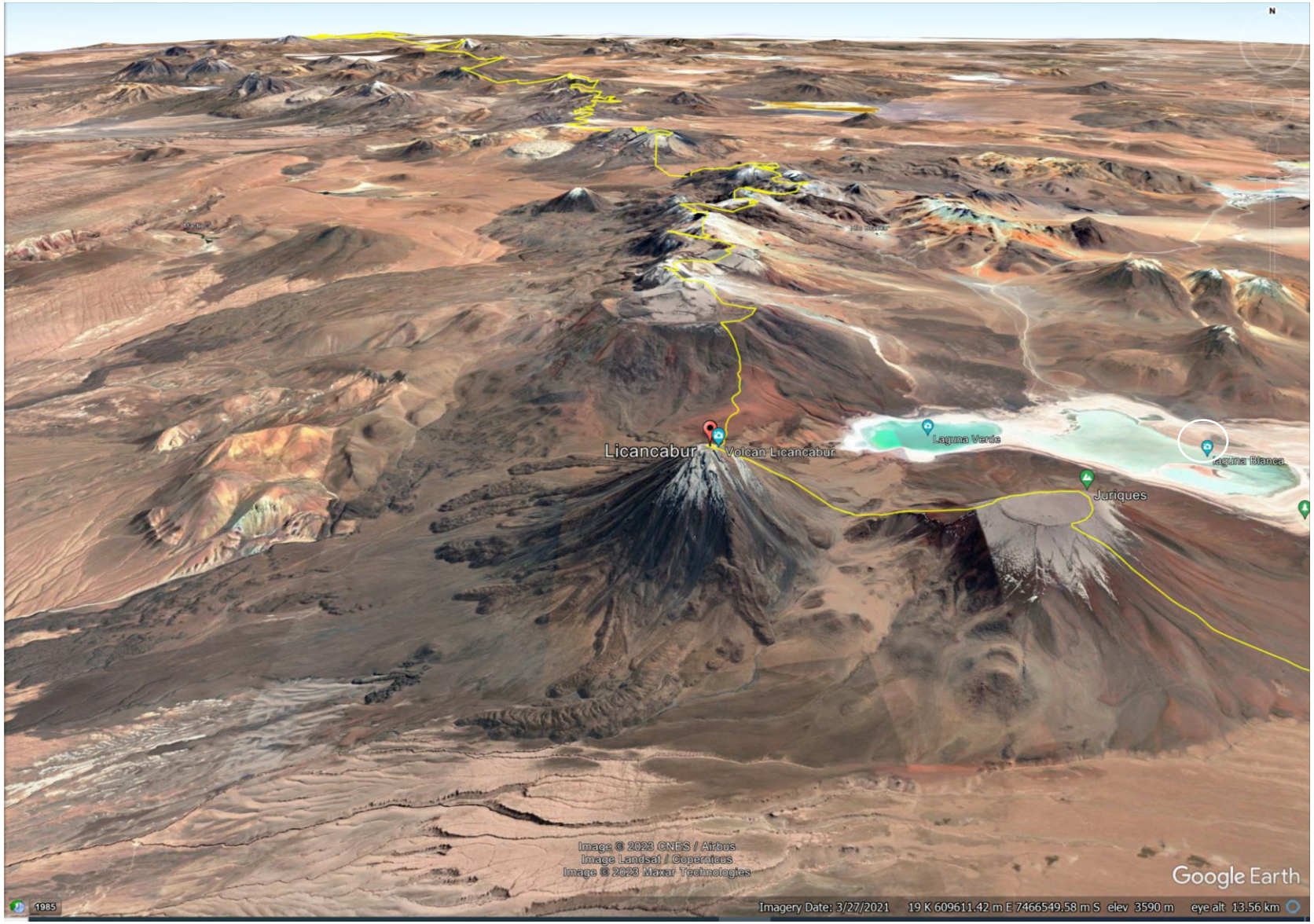
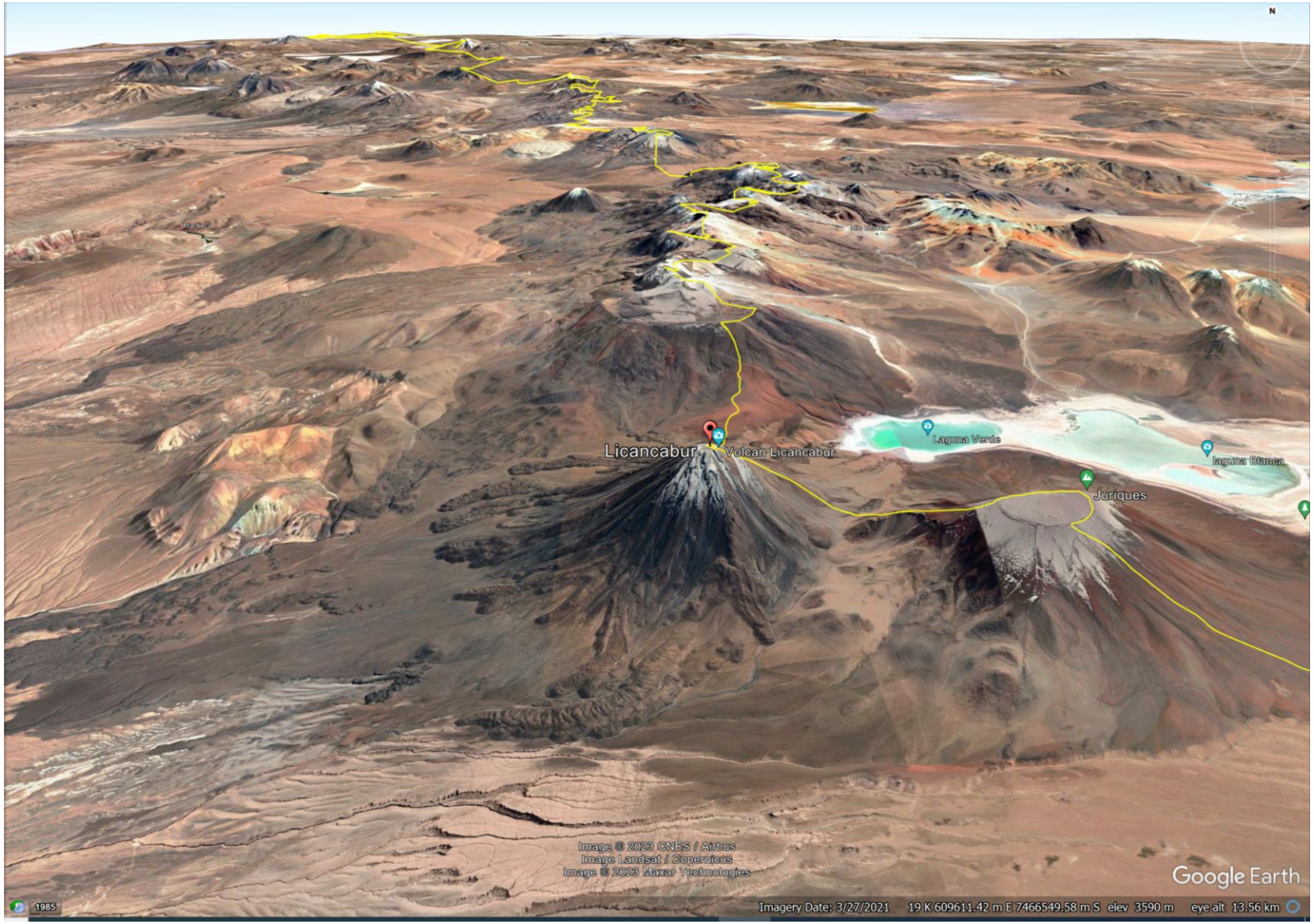


LICANCABUR TITAN2D Exercise





These exercises just use Licancabur as an example!

The model parameters in all the following exercises may or may not be realistic.

Our exercise hazard maps are certainly worse than any made by Chileans!!!!!

- We will use TITAN2D to model a ~10 Mm^3 dome-collapse PDC
- As a reminder, we can use global data to determine appropriate basal friction angles for the volumes we want to model (see next slide for table)

ΔH/L ranges for dome- and column-collapse PDC energy cones from FlowDat regressions

Note:

- Decimal values are used for energy cones
- Degrees are used for TITAN2D basal friction angles
- The regressions at the bottom are for decimal values
- Use the formula Degrees(atan(ΔH/L value)) to convert decimals to degrees in excel

Volume (km^3)	Volume (10^6 x m^3)	ΔH/L best fit Dome collapse	ΔH/L best fit Dome collapse (degrees)	ΔH/L best fit Column collapse	ΔH/L best fit Column collapse (degrees)
0.0001	0.1	0.39	21.1	0.36	19.8
0.001	1	0.29	16.1	0.30	16.5
0.01	10	0.22	12.1	0.24	13.7
0.1	100	0.16	9.1	0.20	11.4

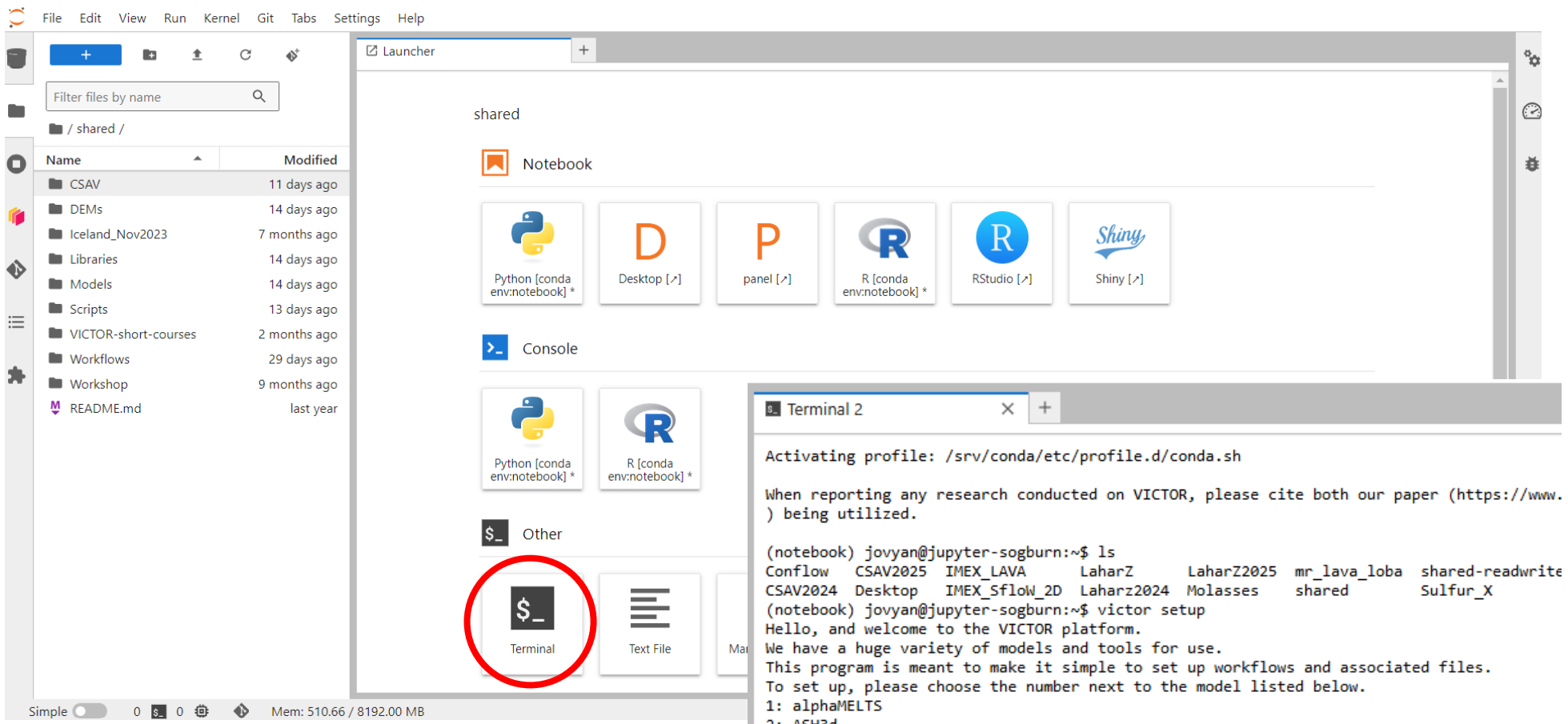
Dome-collapse PDCs (based on 121 flows): $\boldsymbol{\Delta H/L = 0.120\left(V^{-0.127}\right)}$

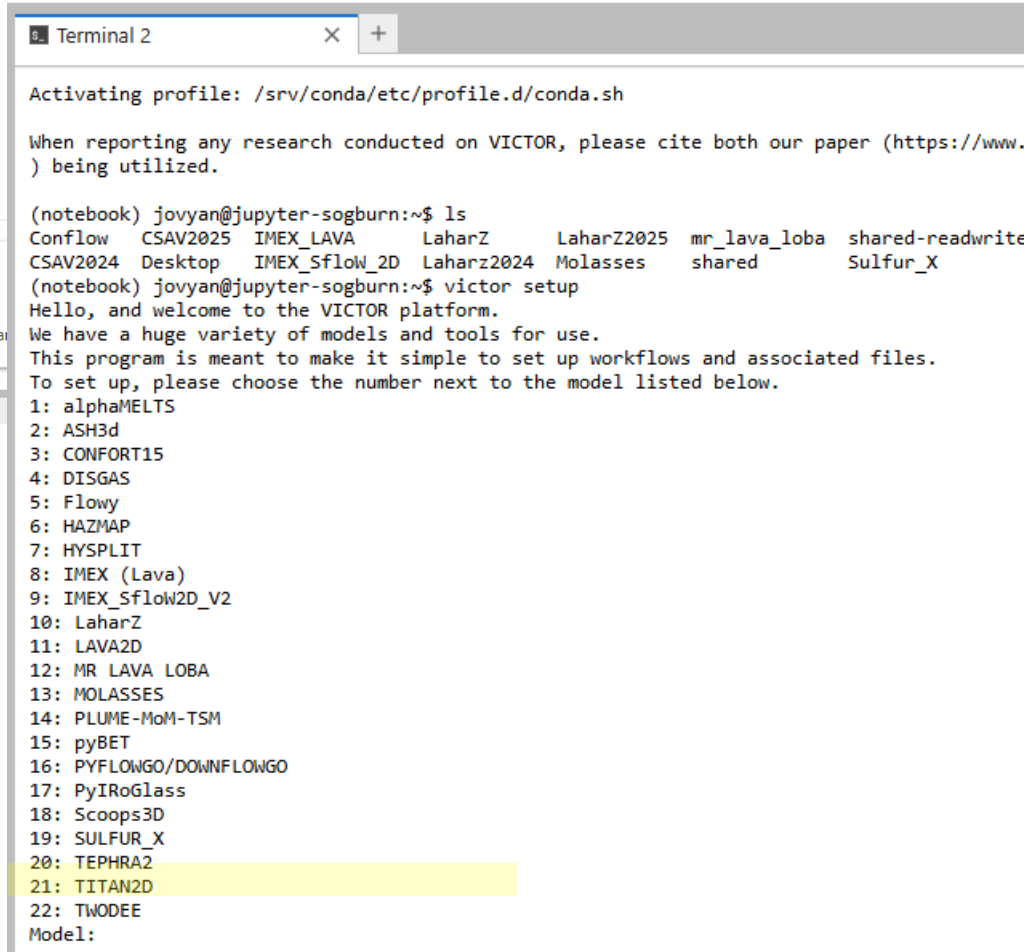
Column-collapse PDCs (based on 40 flows): $\boldsymbol{\Delta H/L = 0.166\left(V^{-0.084}\right)}$

All PDCs (based on 188 flows): $\boldsymbol{\Delta H/L = 0.165\left(V^{-0.085}\right)}$

VDAP

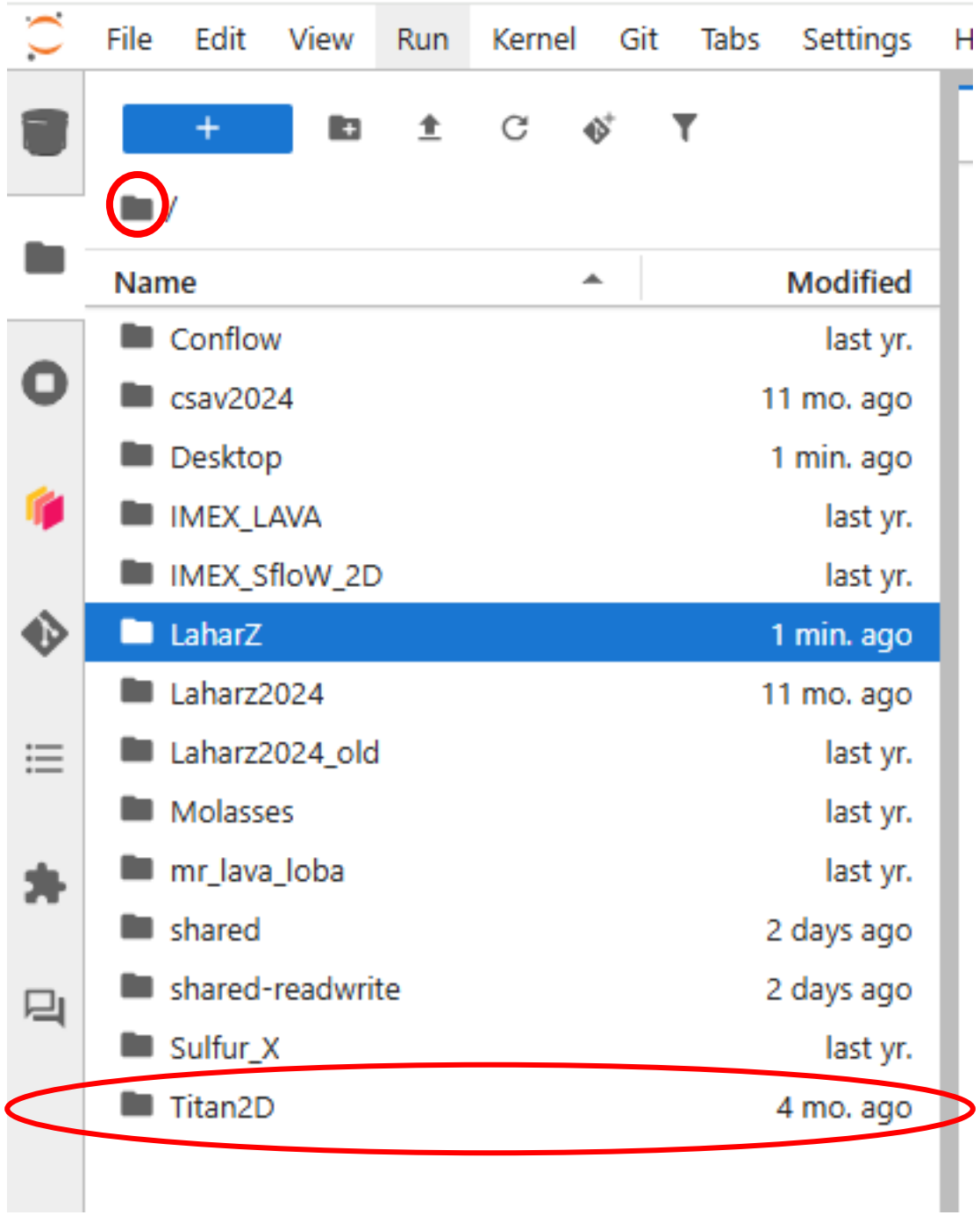
USGS



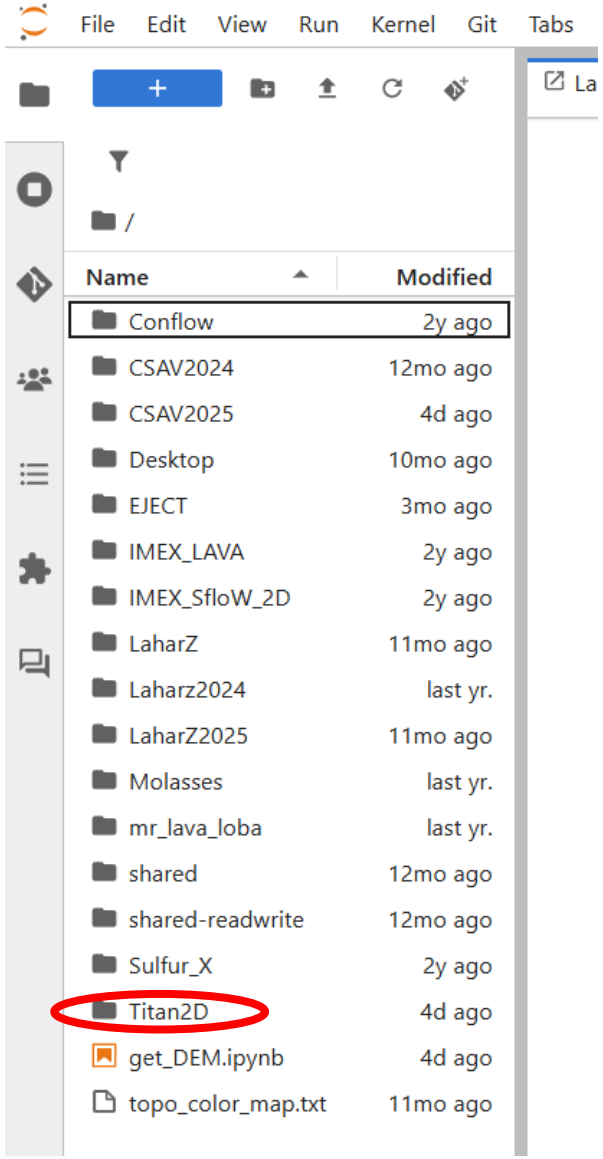


- We want to add the TITAN2D model to your home workspace
- Click the terminal
- Type: victor setup
- Hit enter on your keyboard
- Type the number for TITAN2D
- Hit enter
- Close the terminal

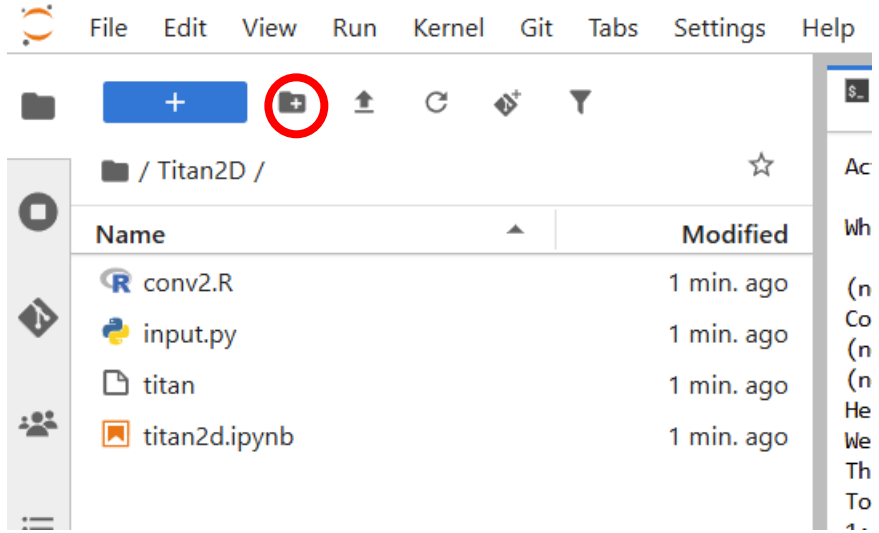
- Check to see if the Titan2D folder was created in your space
- Click the home folder and look for the Titan2D folder (You will have different files than me, don't worry!)



Let's make a folder to work in!

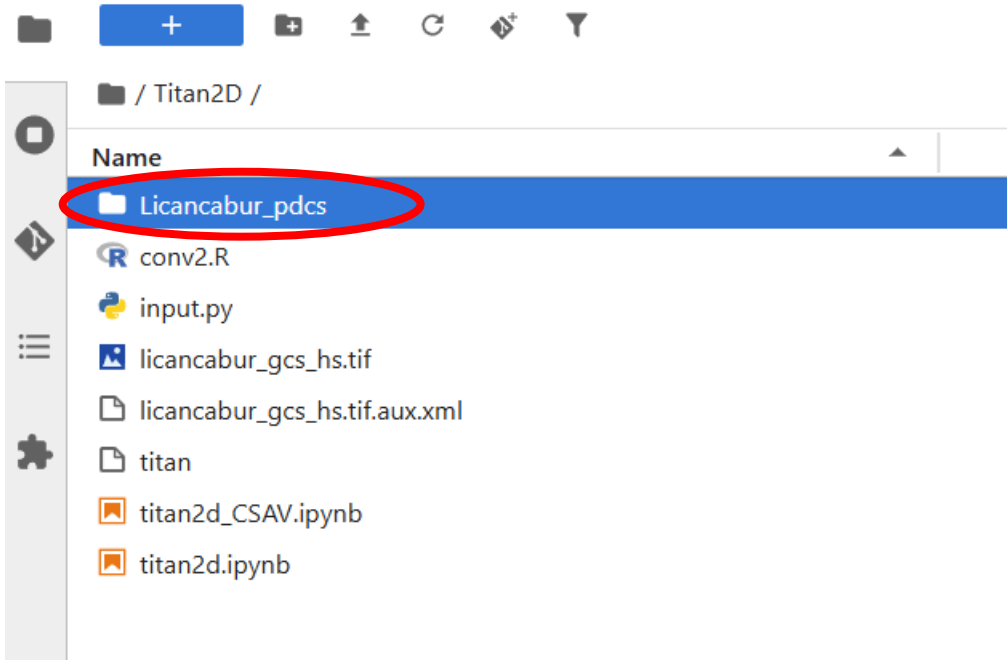


Double click on "Titan2D"

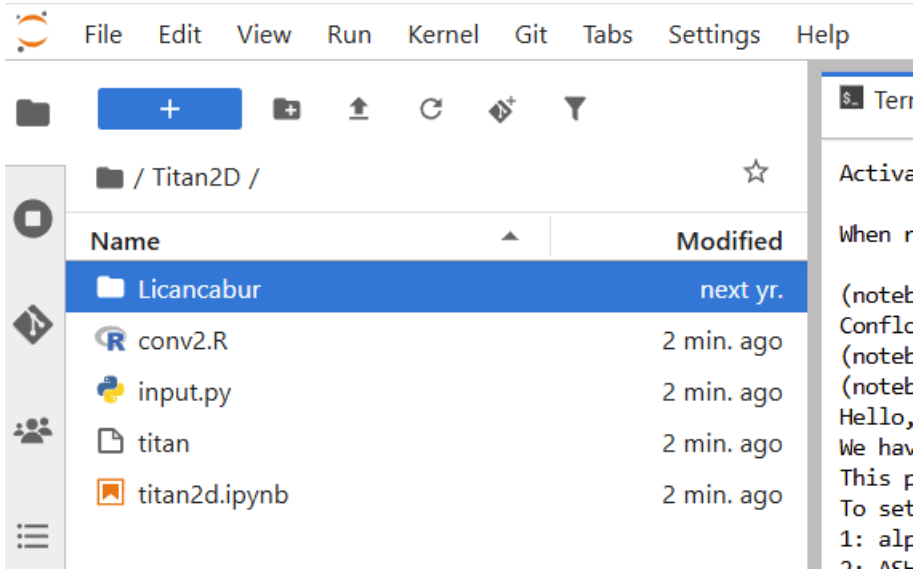


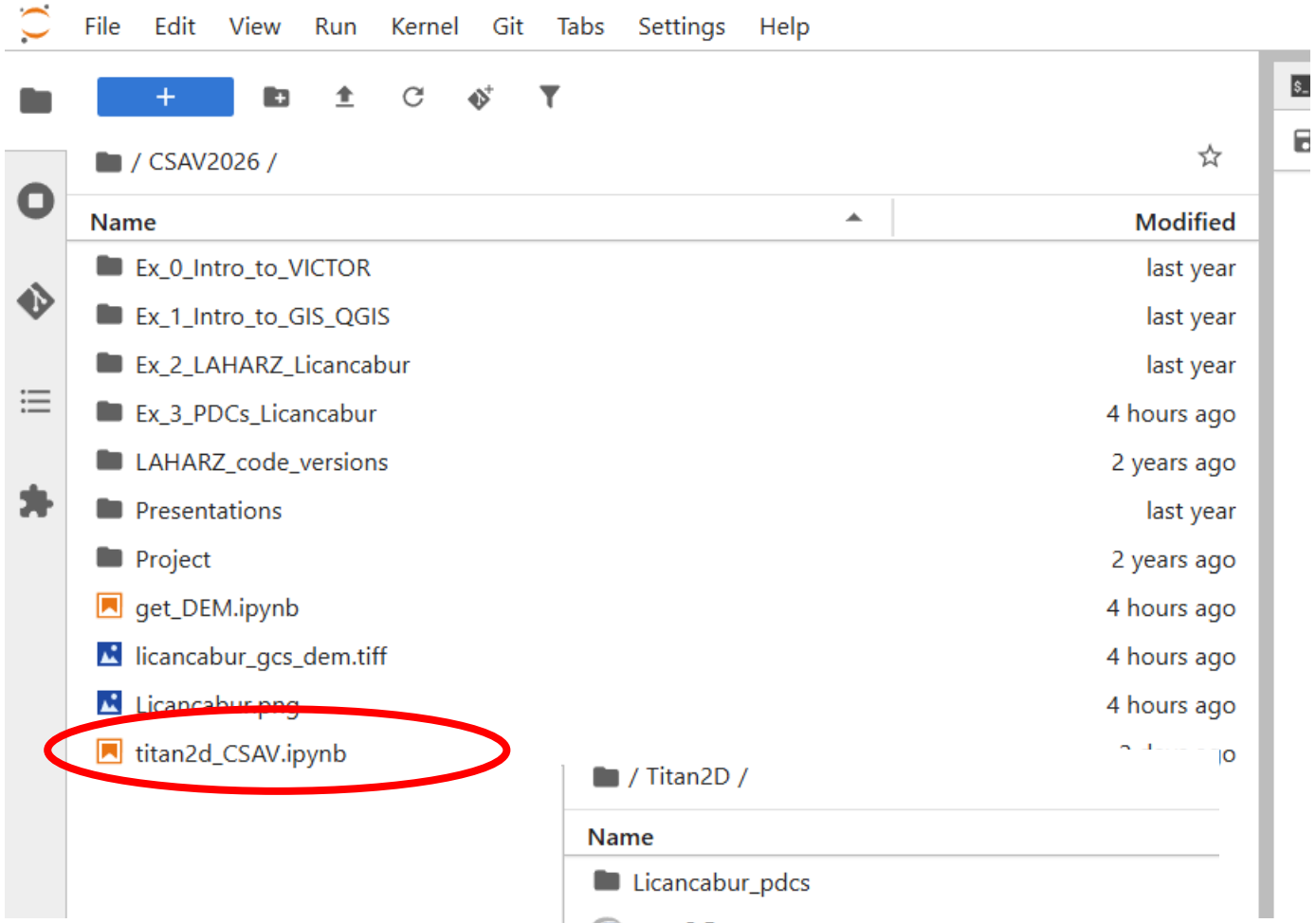
Right click in a blank space and choose "New Folder"

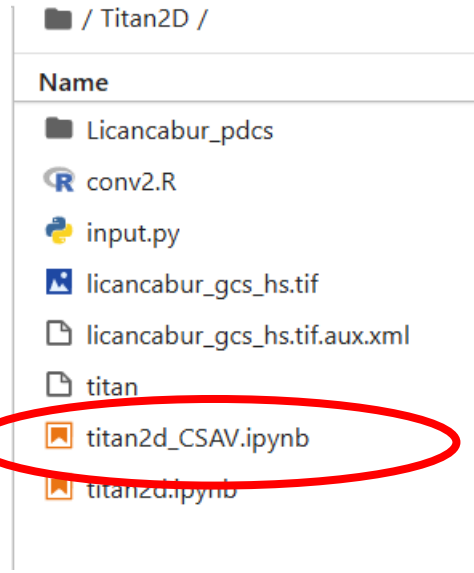
Or click the "New folder" icon at the top



Name the new folder "Licancabur_pdcs"







- You should have a Jupyter notebook in the Titan2D folder called “titan2D.ipynb”
- We want to add one that is specifically for CSAV to this folder
- Go to your CSAV folder and copy the file called ‘titan2d_CSAV.ipynb”
- Paste it into your Titan2D folder

Projecting the DEM to UTM

- The DEM we downloaded for LAHARZ was very, very big (80 km square)
- Using a DEM that big will slow down TITAN
- Let’s get a smaller one using “get_DEM.ipynb” again

- Run get_DEM again
- Change the parameters to 15000

Setting input parameters

In the following cell we use the lat/lon of a target volcano's summit, obtained from the Global Volcano Project database from the volcano'es name, to set the bounding box.

[3]:
```
#Choose a volcano and DEM options

volcano = victor.search_volcano('Licancabur') # Get volcano information from the GVP database

center_lat = volcano[1][0]  # Get volcano latitude
center_lon = volcano[2][0]  # Get volcano longitude

# Set the extent of the region of interest, in meters in each direction from the summit
eastwest_extent = 15000
northsouth_extent = 15000

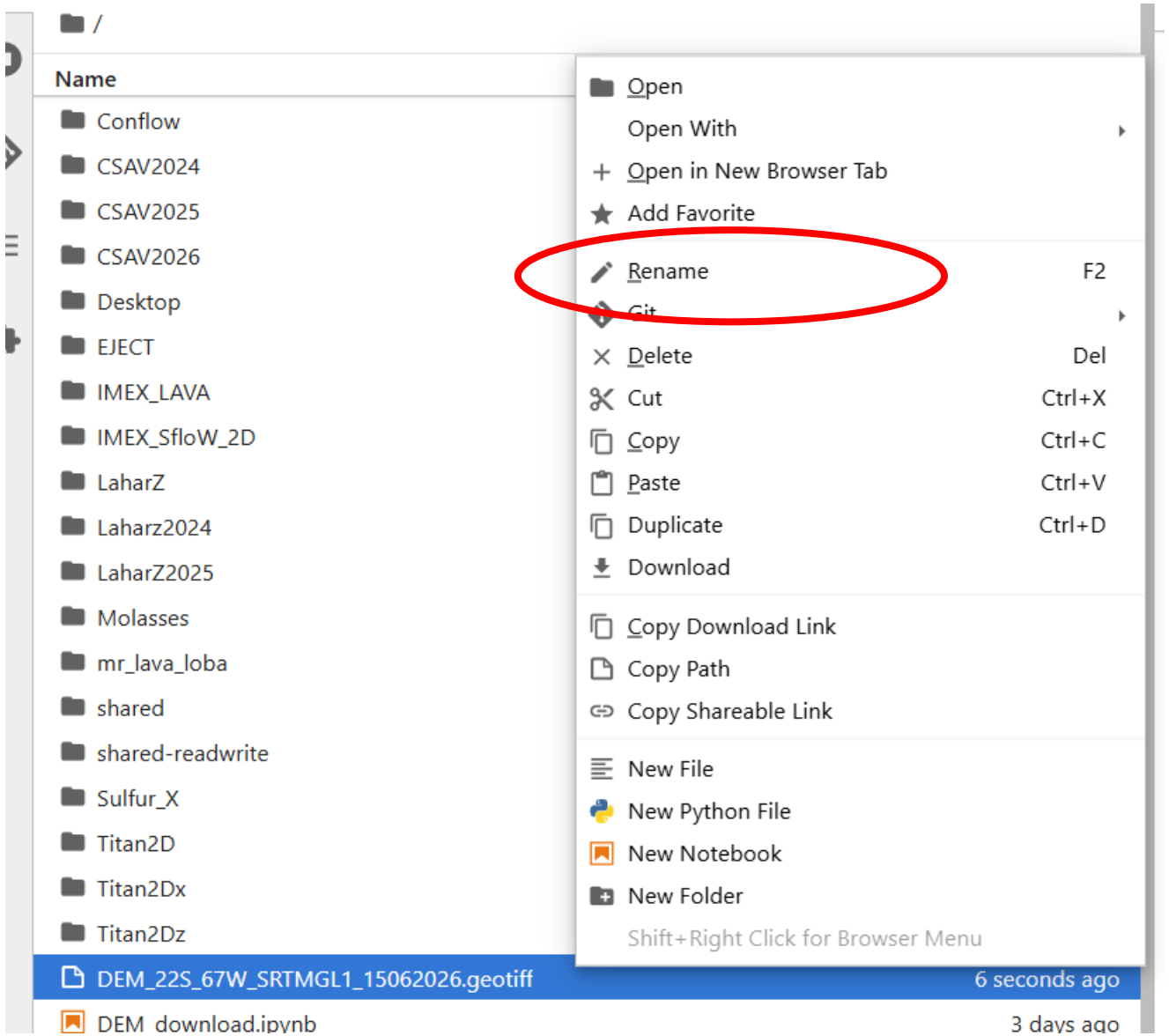
# Set bounding box in lat/lon
utms = utm.from_latlon(center_lat, center_lon)
[north_lat, east_lon] = utm.to_latlon(utms[0]+eastwest_extent, utms[1]+northsouth_extent, utms[2], utms[3])
[south_lat, west_lon] = utm.to_latlon(utms[0]-eastwest_extent, utms[1]-northsouth_extent, utms[2], utms[3])

# Select a DEM format: Input "ascii" for ASCII format (.asc) and "tiff" for geotiff format
outputFormat = "tiff"

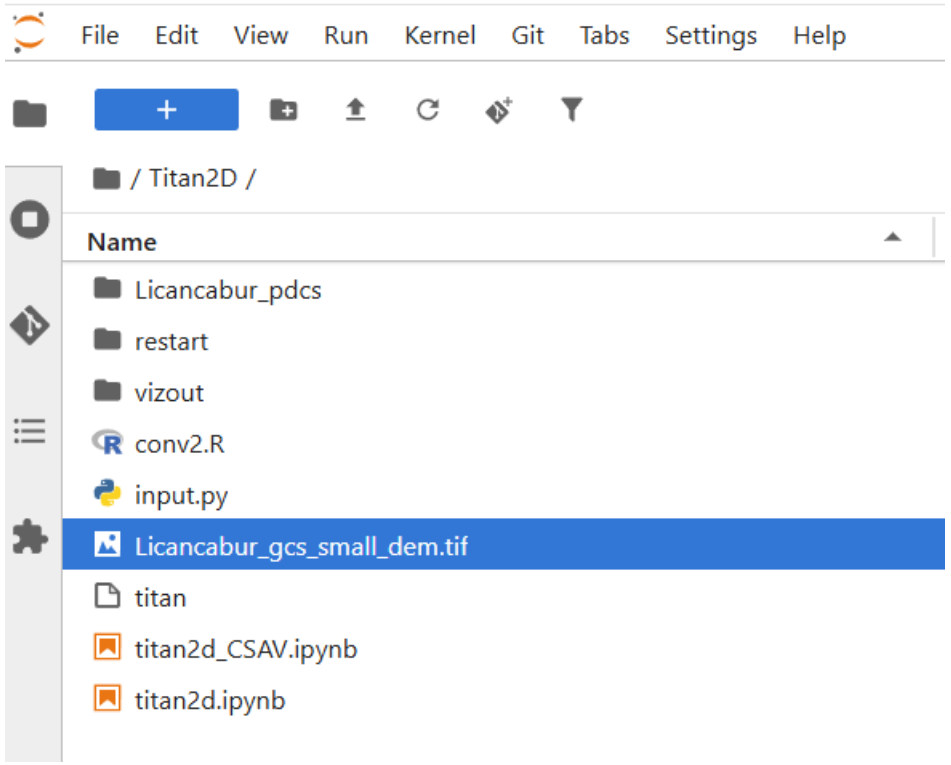
# Enter the name of the output file (without extension)
name = "Licancabur_wgs_dem"

dataset = "SRTMGL1"

print(center_lat, center_lon)
```

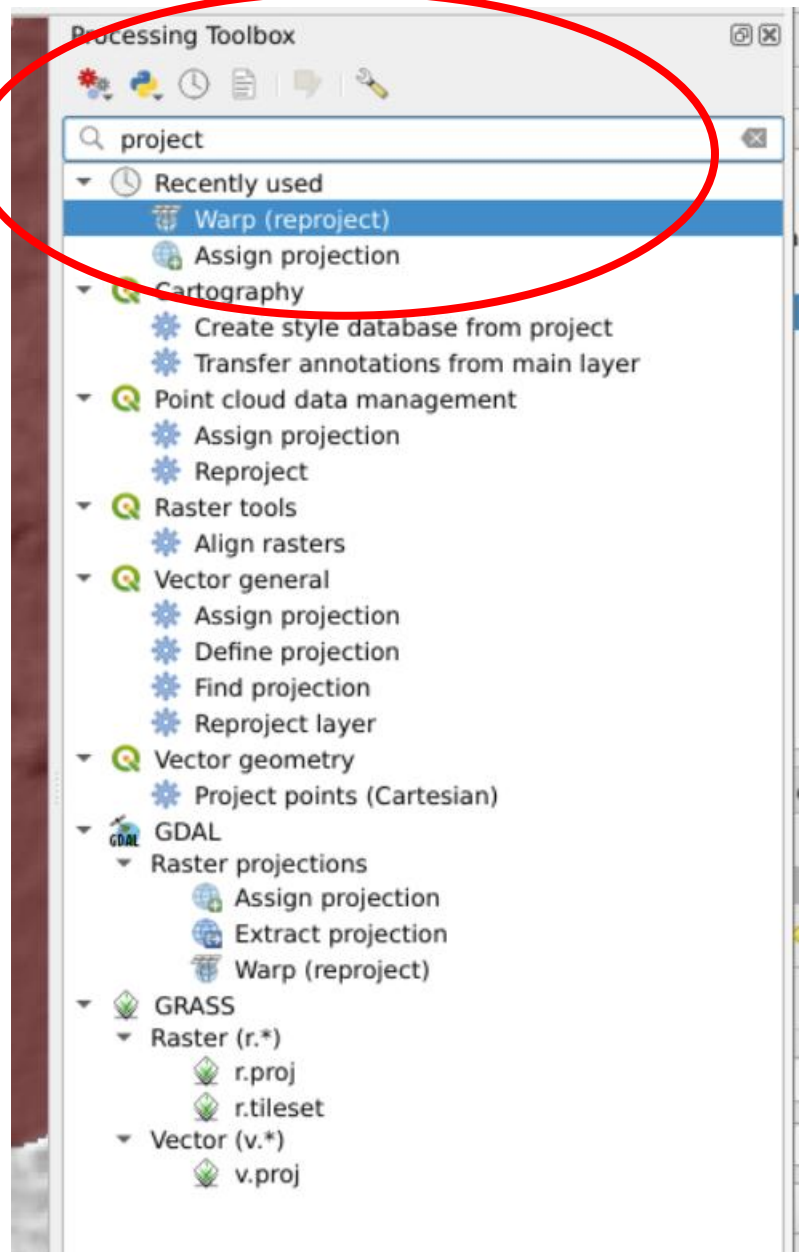


- Rename the file to:
- Licancabur_gcs_small_dem.tif
- Move the file into your Titan2D folder

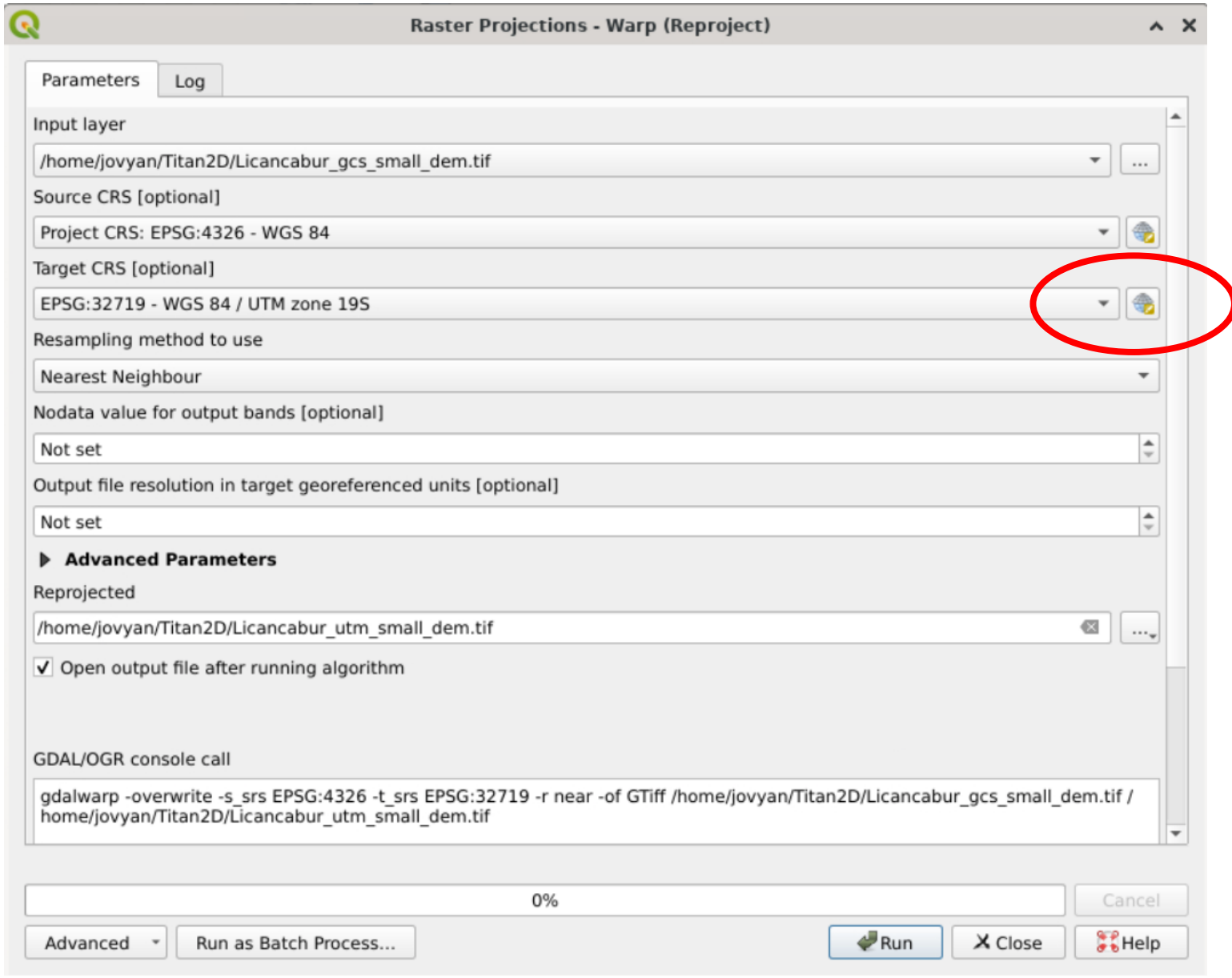


Projecting the DEM to UTM

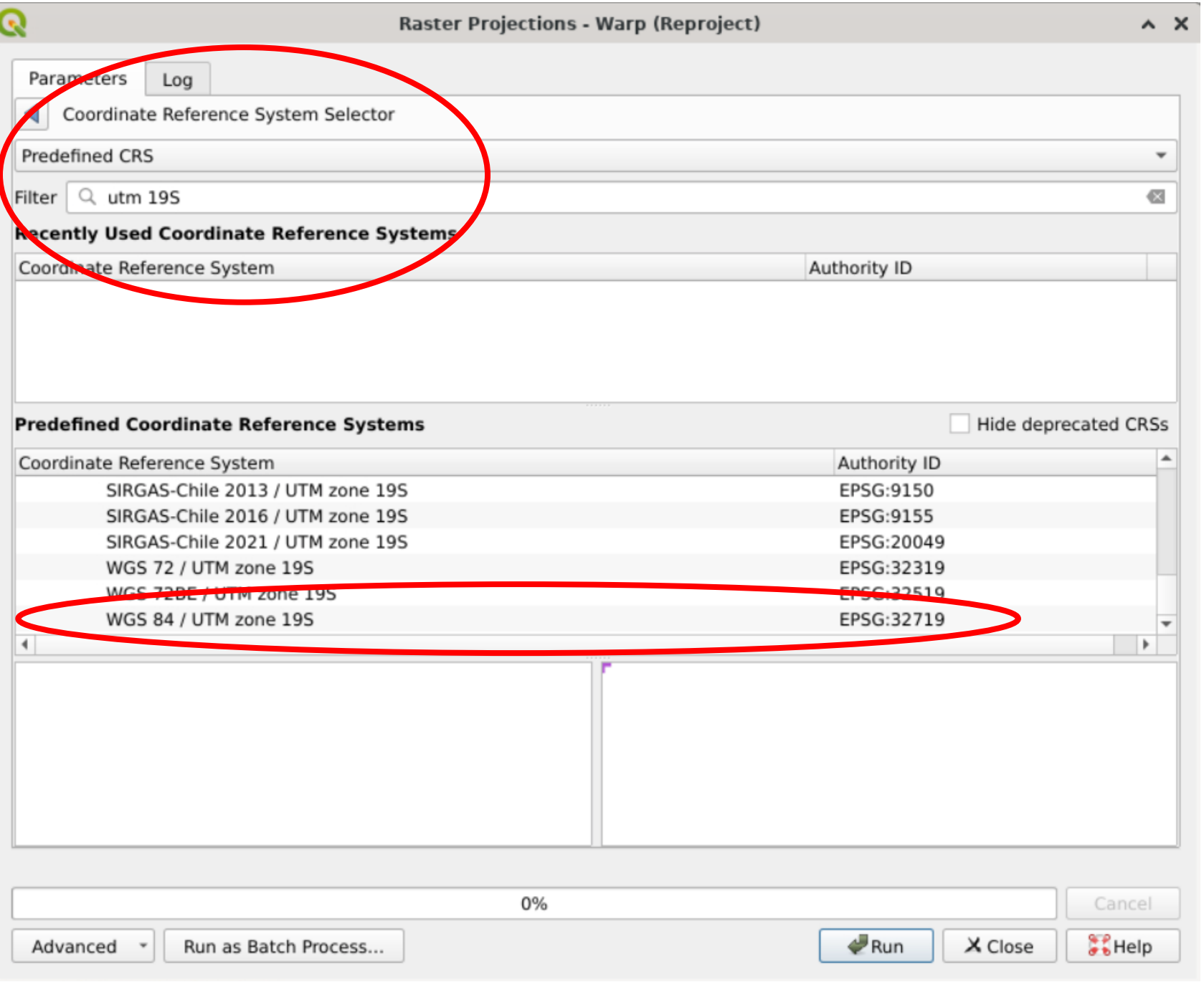
- The DEM that we used for LAHARZ was in geographic coordinate system (lat/long)
- The smaller DEM we just downloaded is also in GCS (lat/long)
- Titan2D uses UTM, so we need to use QGIS to project the DEM
- Go back to your QGIS project



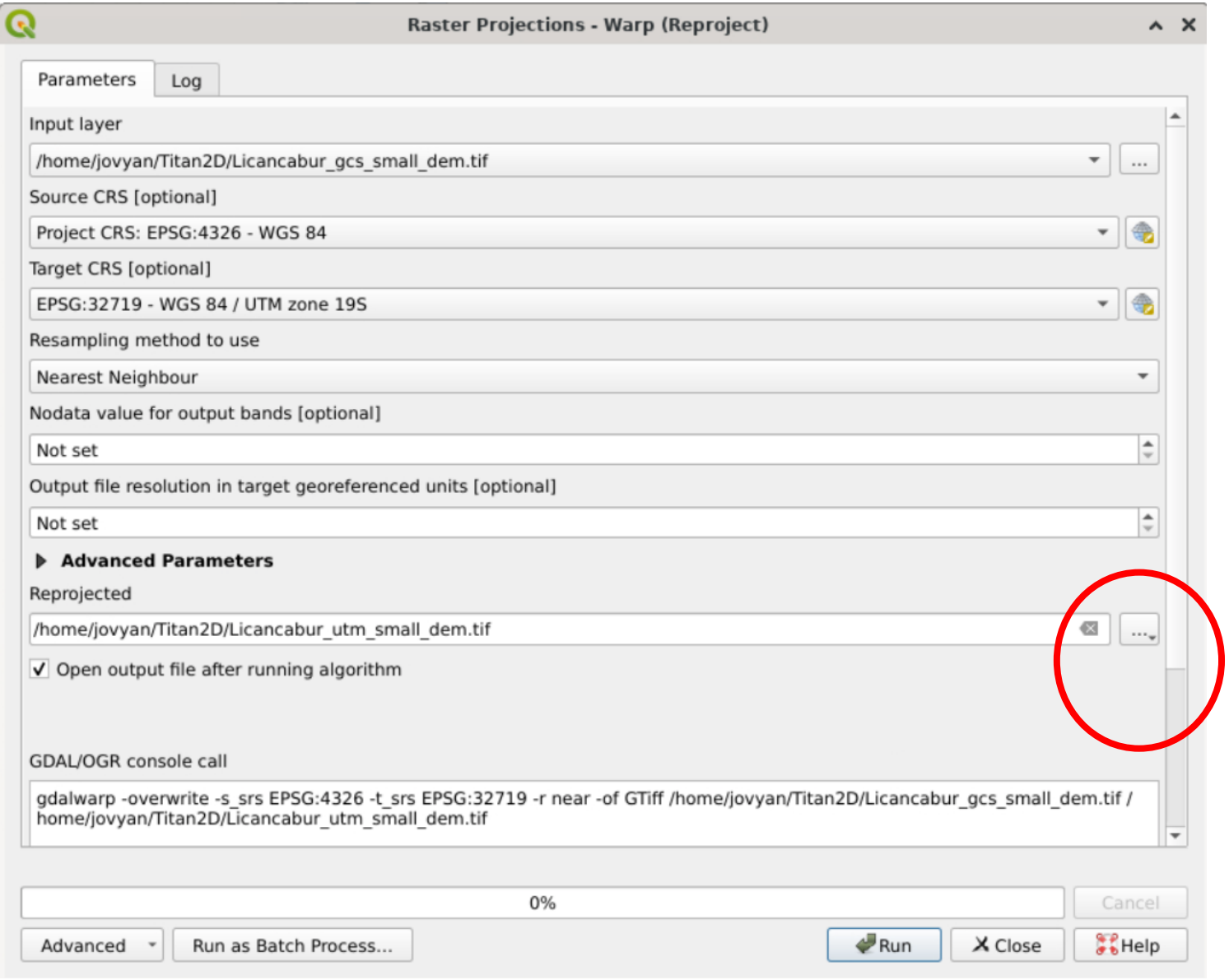
Find the tool called “Warp (reproject)” in the Processing Toolbox



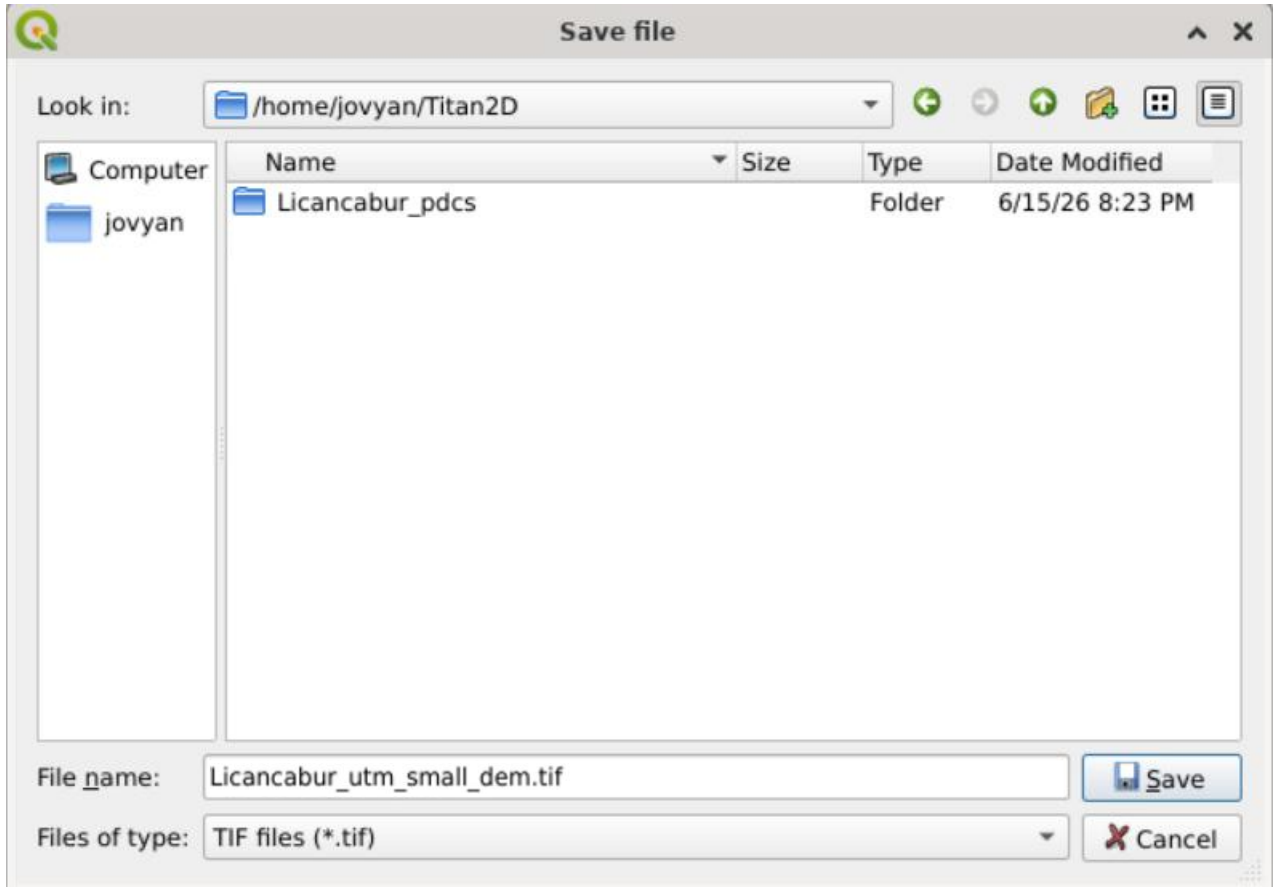
- The input is licancabur_gcs__small_dem.tif
- The source CRS is the Project CRS: EPSG: 4326 – WGS 84
- You will need to click “Select CRS” to find the Target CRS



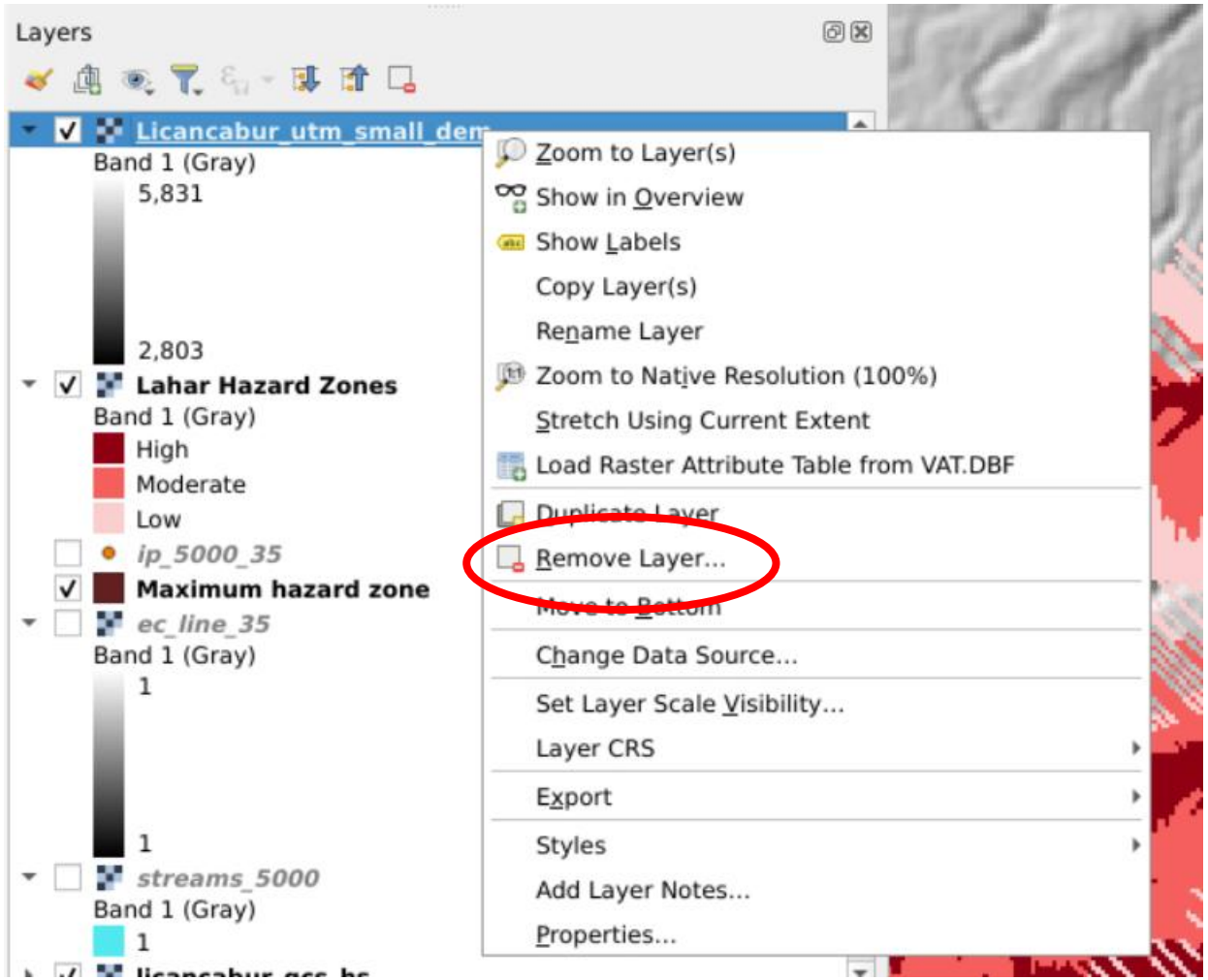
- Licancabur is in UTM zone 19S.
- **When you do your projects you will need to find the UTM zone for your volcano!!**
- Select WGS 84 / UTM zone 19S from the results (EPSG: 32719)
- Don’t click Run, but click the blue arrow in the upper left to return to the previous screen



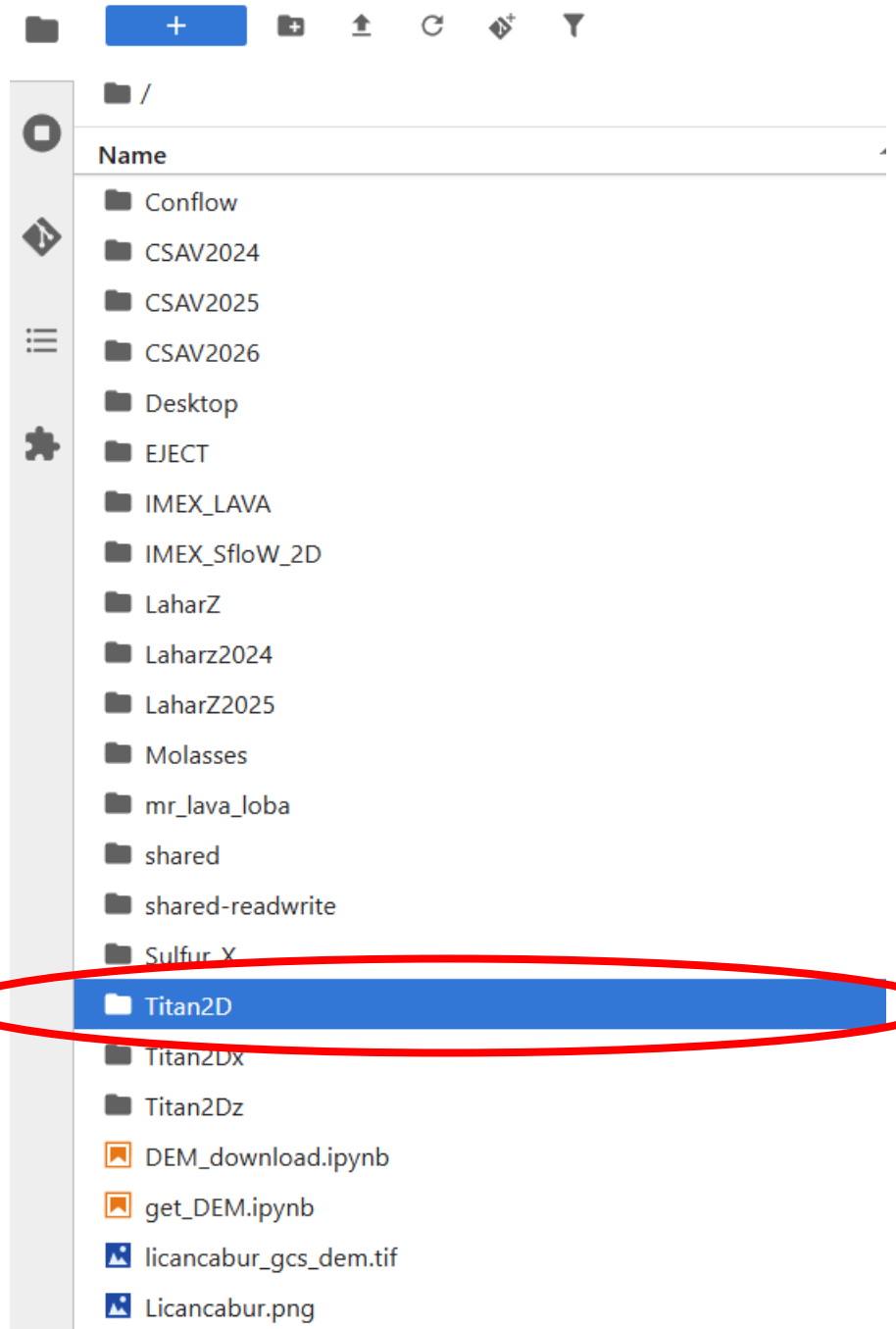
Make sure to save to a file and not a temporary layer



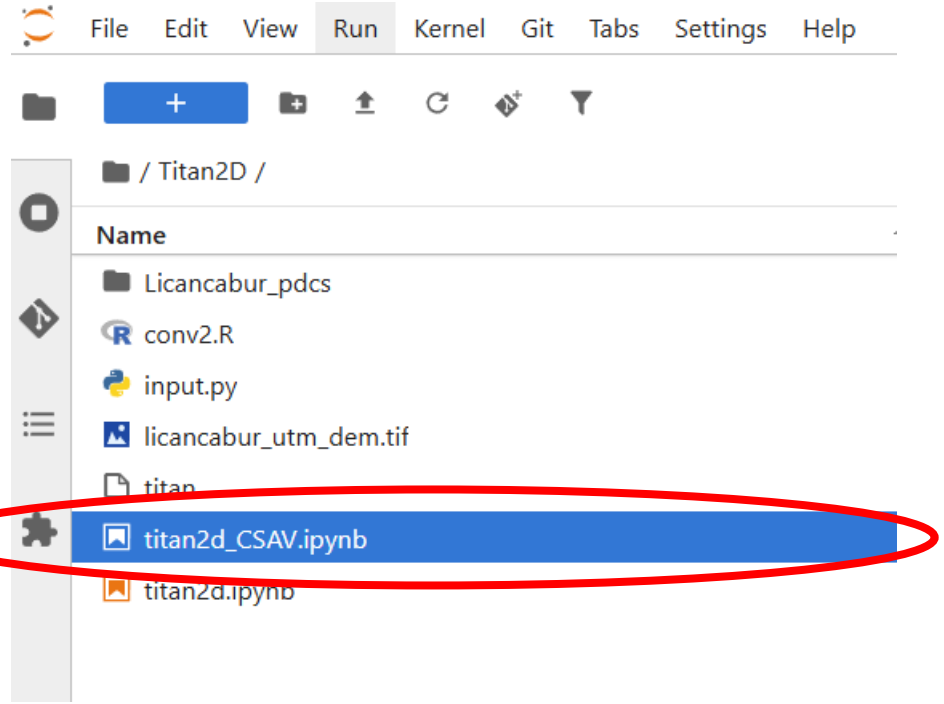
- Navigate to /home/jovyan/Titan2D
- Save the file as Licancabur_utm_small_dem.tif



- Remove the result from the QGIS project so it's not open while we use it
- Leave QGIS and return to your VICTOR files



- Navigate to /home/jovyan/Titan2D
- Double click on titan2d_CSAV.ipynb



Terminal 1 | Launcher | titan2d_CSAV.ipynb

Markdown

CSAV workflow for TITAN2D

TITAN2D is a geoflow simulation tool that, given initial parameters, will simulate granular flows over digital elev
guide is attached.

Importing Libraries [Run but don't change]

[1]:
```
#run_but_don't_change
import subprocess
import pandas as pd
import numpy as np
import sys
import matplotlib.pyplot as plt
import os
import rioxarray as rxr
import rasterio as rio
import h5py
import cartopy.crs as ccrs
import glob
```

[2]:
```
#run_but_don't_change
sys.path.insert(0, "/home/jovyan/shared/Libraries")
import victor
```

File Managment [Change these]

Notes from VICTOR team: Keep in mind that for the **GIS_GRASS** type, multiple arguments to specify file mapset/map locations are neccesary, and the DEM location fo
ascii shapefile. We highly recommend the **GDAL** format.

There are a variety of interesting but largely unecessary files that are output at the end of the simulation. Toggle the option below to remove them.

If you would like to create a gif of all the timesteps, please enable the option below. Keep in mind, this may take many minutes, depending on the number of iteration

[3]:
```
#The output files show things like momentum and thickness through time. You can set cleanup = False if you want to see these files.
#If you want a gif of the flow, you can set create_gif = True, but this will be slow. For the CSAV exercise, let's set it to False.
cleanup = False

create_gif = False
```

- Most of the rest of the instructions are in the Jupyter notebook itself!
- Read the title of the blocks to see if you will edit anything in the cells

- There is also text that starts with '#" that gives information about what parameters to choose for the exercise!
- Go to the next slide before starting!!

Run Titan here! [Run but don't change]

Titan outputs updates at **every** timestep, so we suppress the output here for clarity.

```
[*]: subprocess.run("export GDAL_DRIVER_PATH=''; ./titan -nt " + str(threads) + " input.py",shell=True, stderr=subprocess.DEVNULL)
```

```
Python home: /srv/conda/envs/notebook
Executable location: /home/jovyan/shared/Models/titan2d/bin/titan
python home: /srv/conda/envs/notebook
Titan2d 4.1.0
Executable location: /home/jovyan/shared/Models/titan2d/bin/titan
--------------------------------------------------------------------------------

Warning: This version was compiled without openmp support!
Set threads number to 1

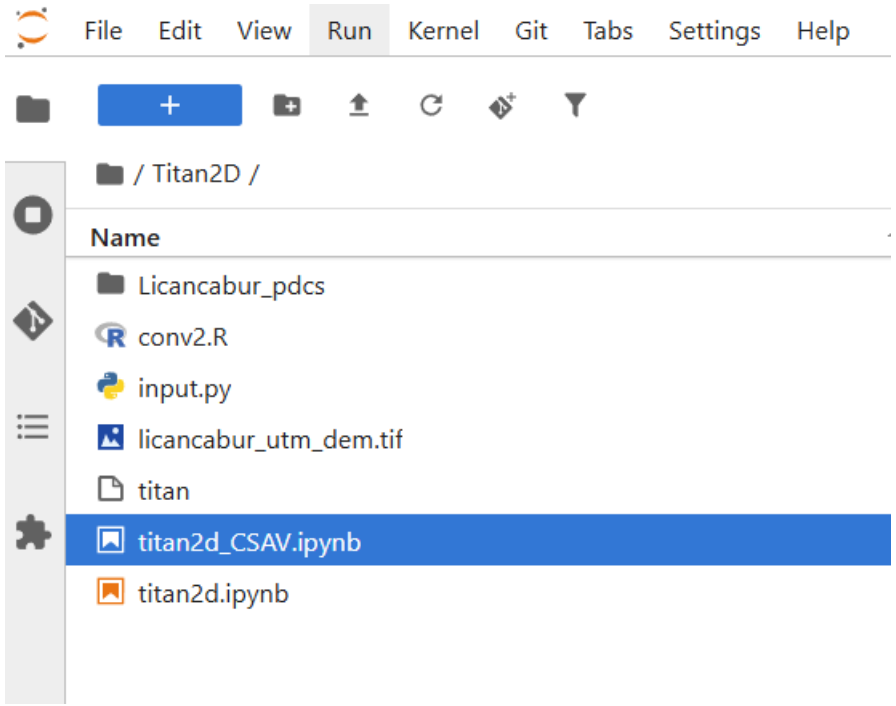

preproc...
The GIS region is:  x = 5.749191e+05 to 6.549142e+05   y = 7.434265e+06 to 7.515473e+06.
The simulation region is:  x = 5.749191e+05 to 6.549142e+05   y = 7.434265e+06 to 7.515473e+06.

GIS:
        gis_format: 2
        gis_main:
```

- Remember to wait for the cells to finish before moving on.
- When a cell is running, it will have an asterisk
- When you get to "Run Titan here!" it will take awhile!!!!
- Please let me know when you have started the Titan run!!!!
- We can take a break while these run.

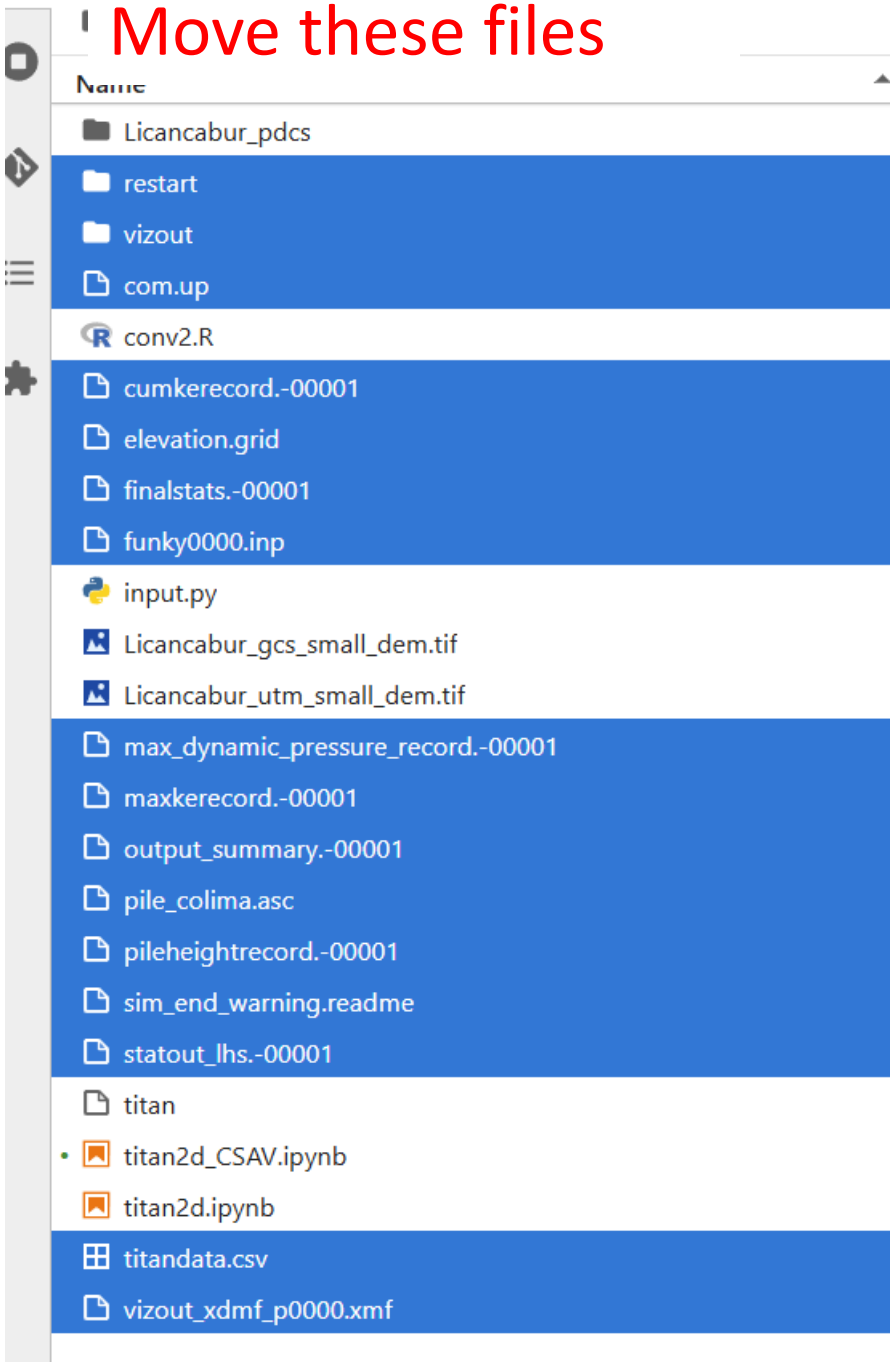
Clean-up

- To keep track of your Titan runs, move the results into your Licancabur_pdcs folder



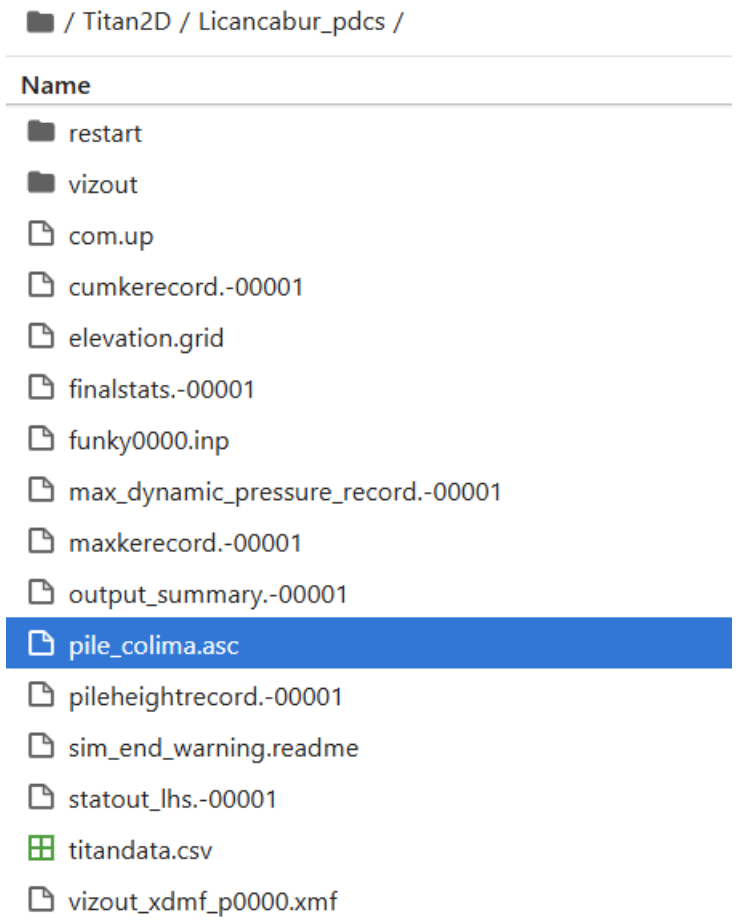
Keep these files where they are

Move these files



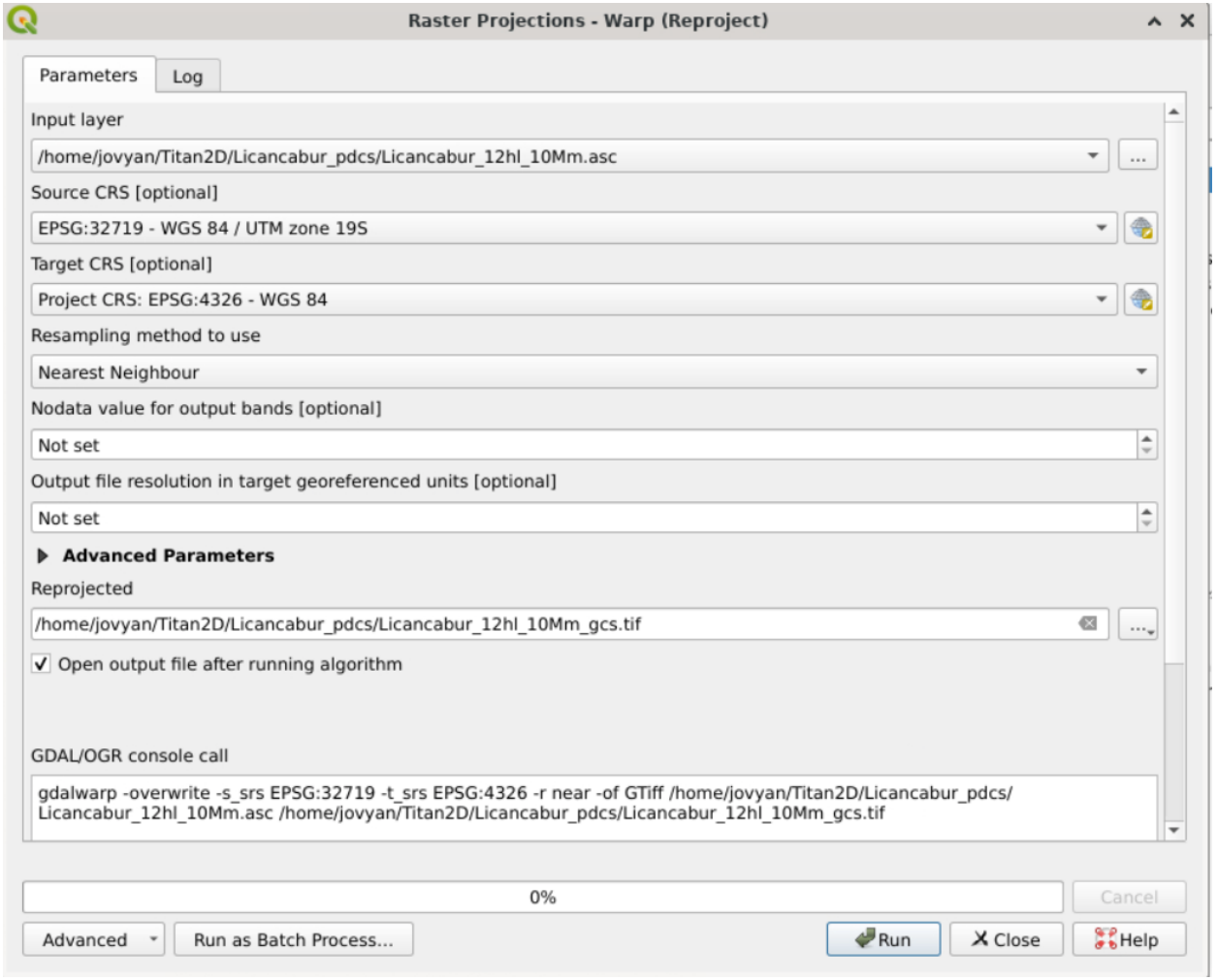
Clean-up

- Navigate into your Licancabur_pdcs folder
- Rename “pile_Colima.asc” to “Licancabur_12hl_10Mm.asc”
- This way, you have named it using the ΔH/L value and approx. volume
- You might also want to name the folders this way (with the h/l used and the volume) when you do your projects



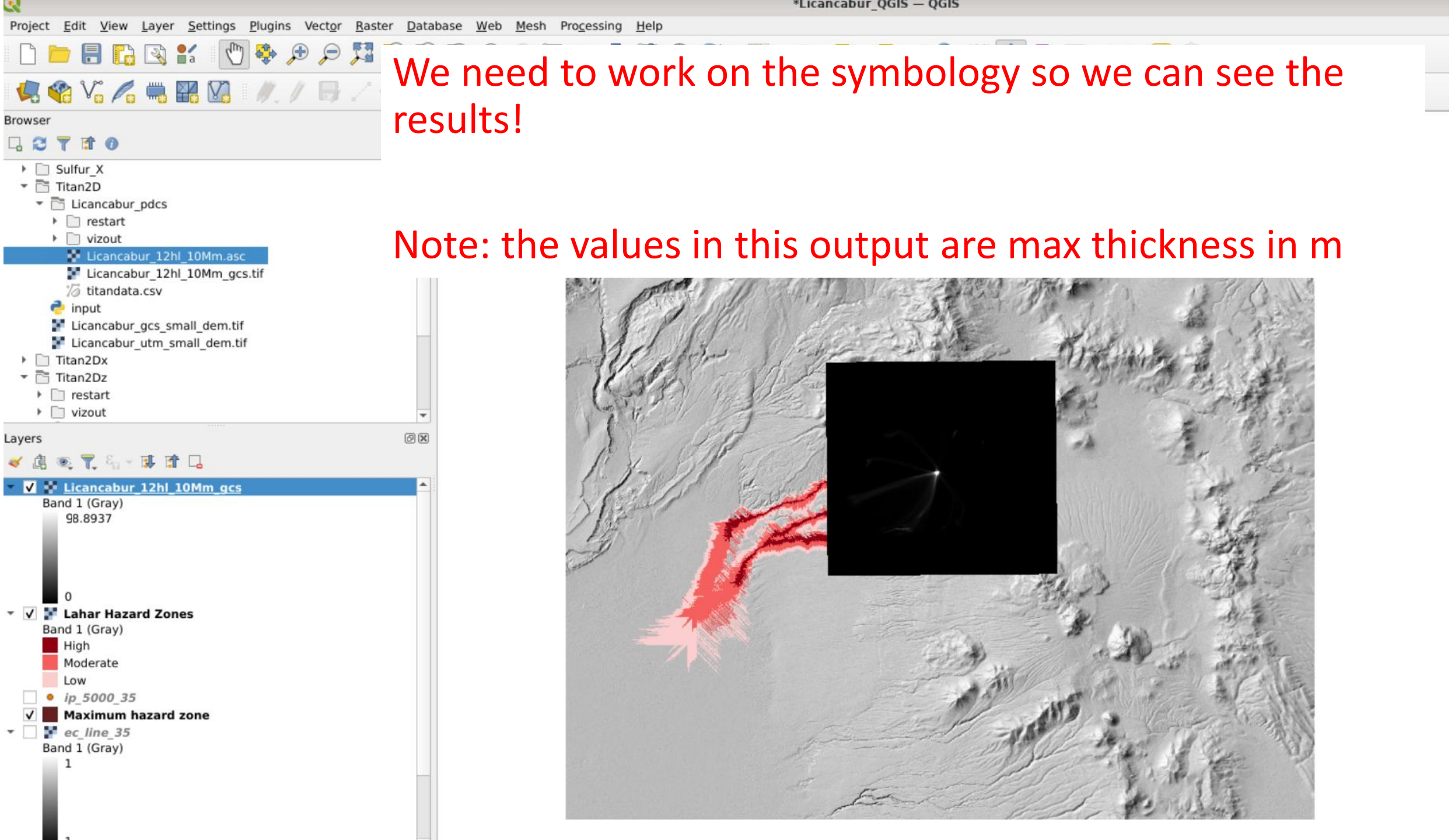
Let’s add the Titan2D runs to the hazard map

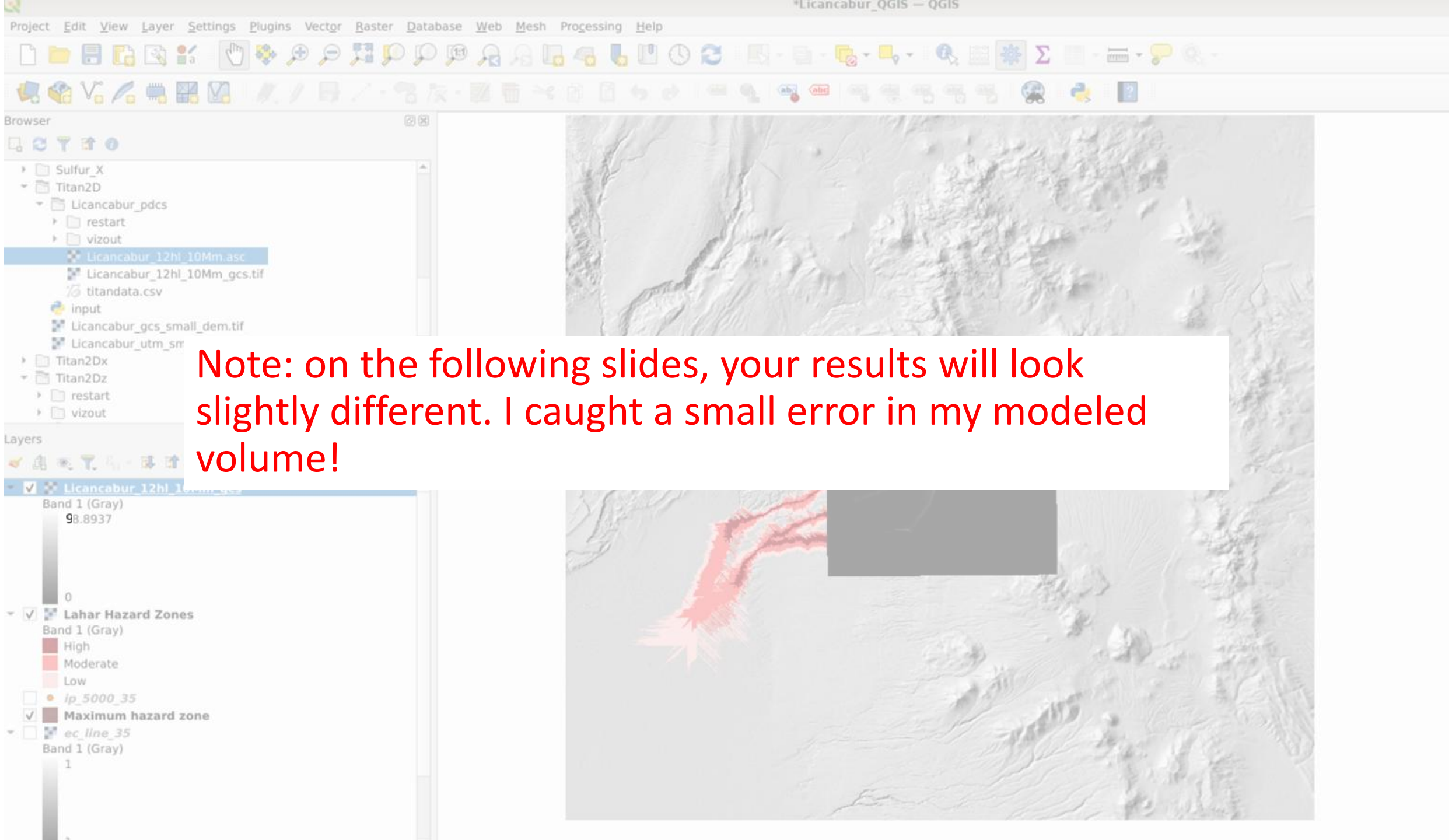
- Navigate back to your QGIS project
- We need to reproject the Titan results



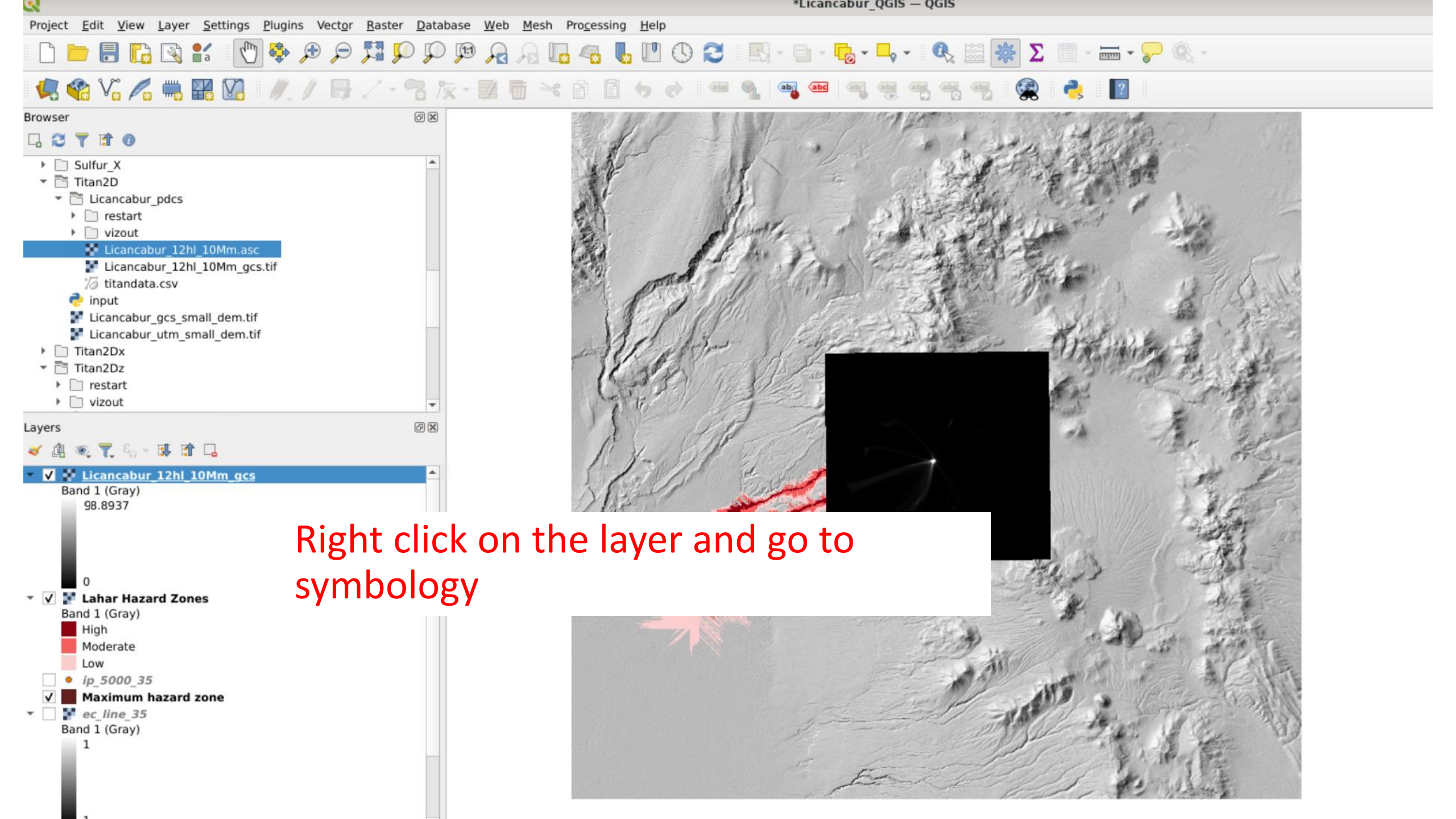
We need to work on the symbology so we can see the results!

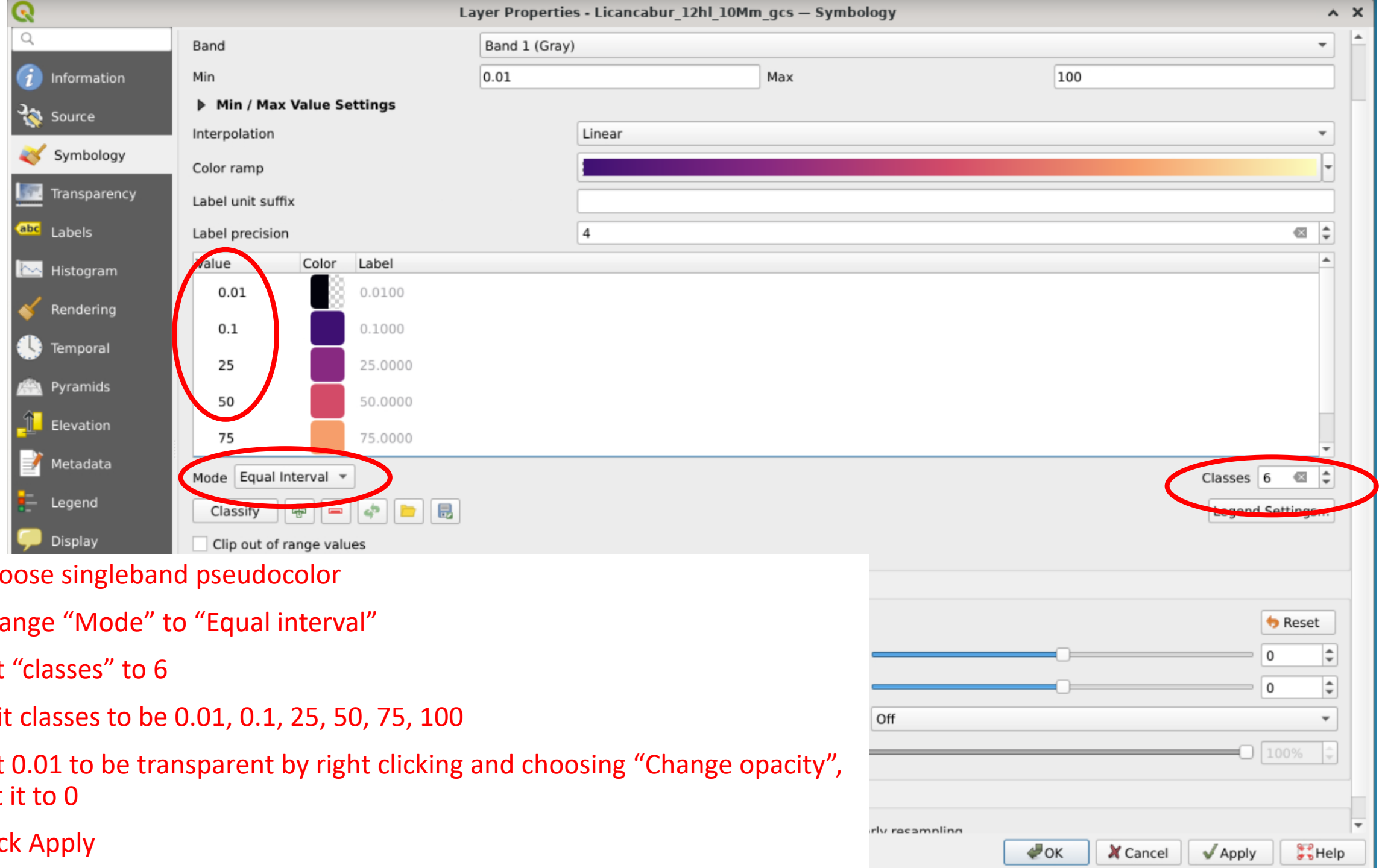
Note: the values in this output are max thickness in m



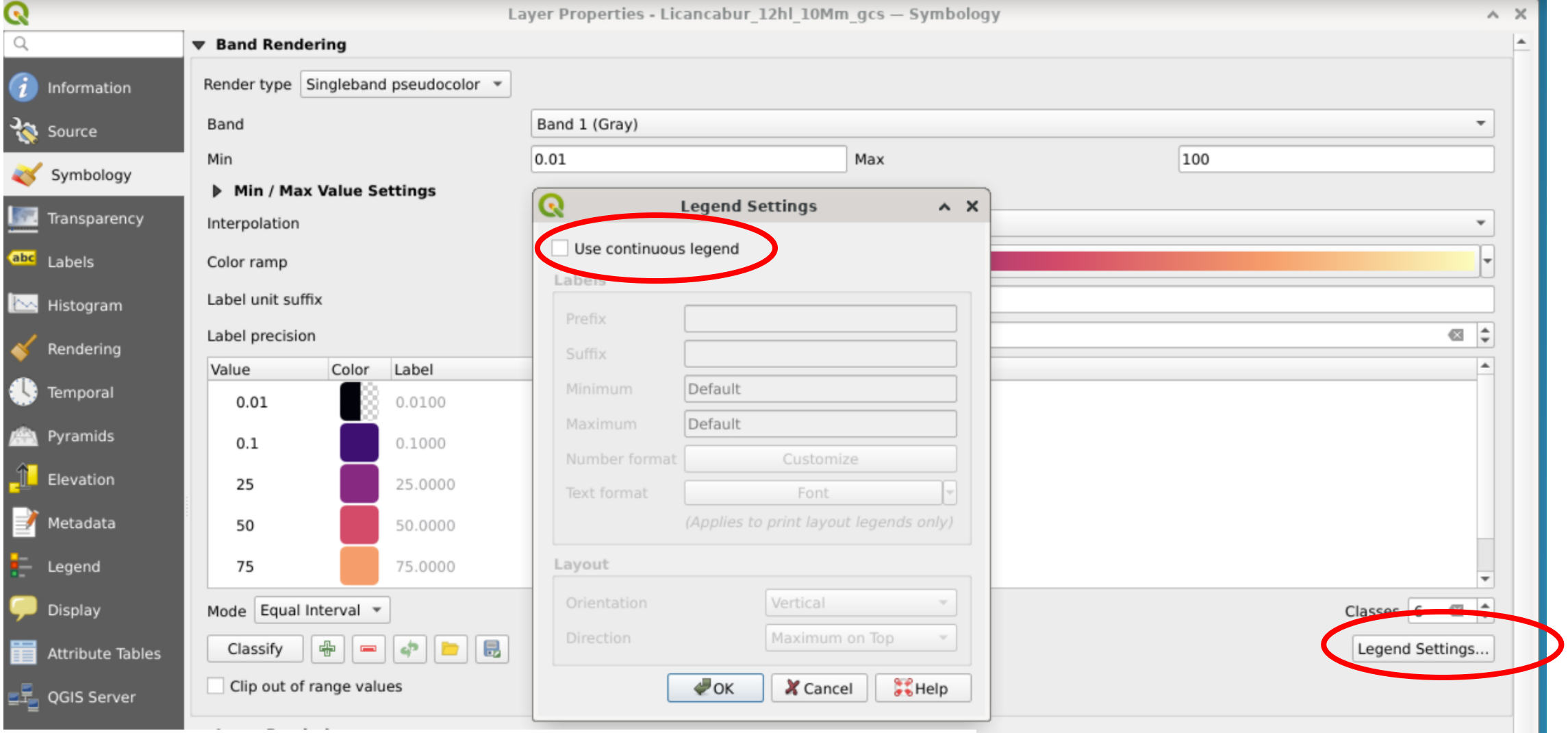


Note: on the following slides, your results will look slightly different. I caught a small error in my modeled volume!

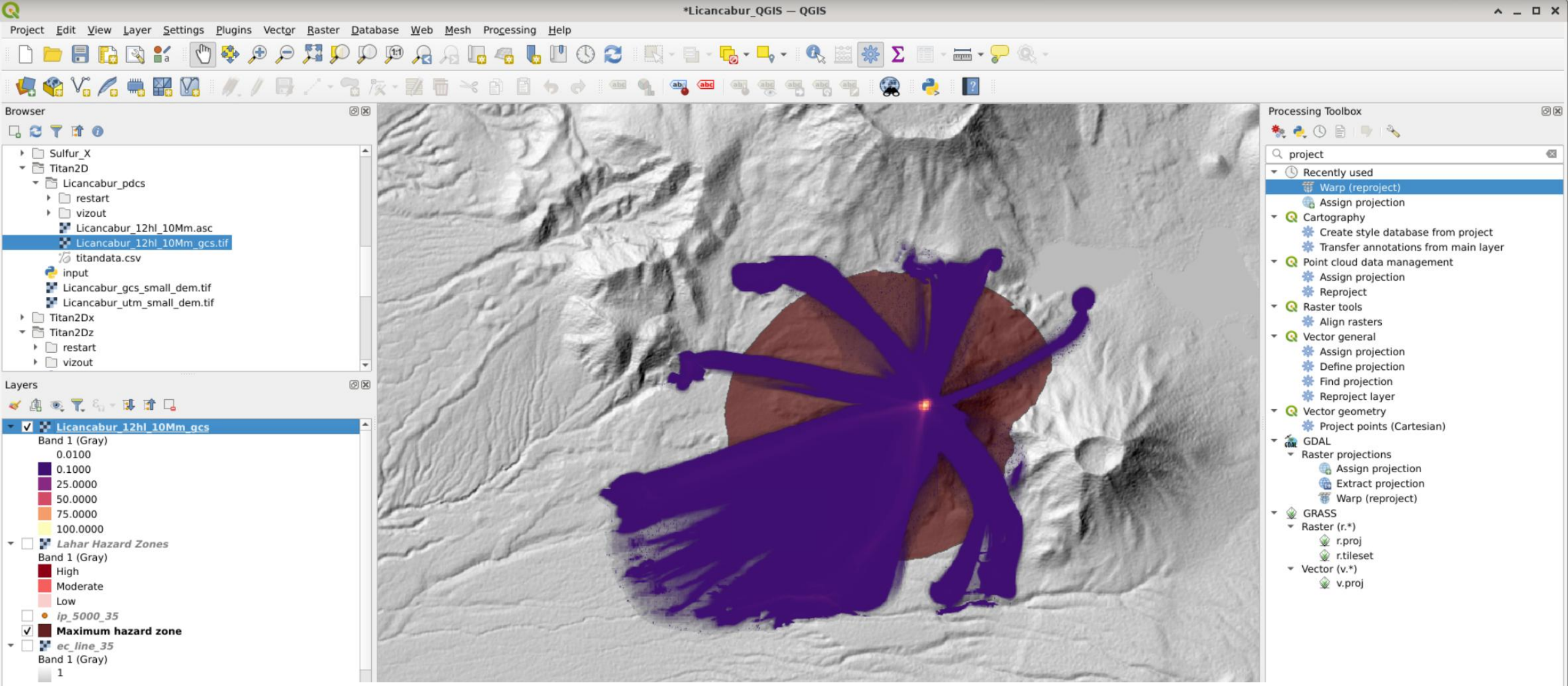

*Licancabur_QGIS — QGIS
Project Edit View Layer Settings Plugins Vector Raster Database Web Mesh Processing Help
Browser
Sulfur_X
Titan2D
Licancabur_pdcs
restart
vizout
Licancabur_12hl_10Mm.asc
Licancabur_12hl_10Mm_gcs.tif
titandata.csv
input
Licancabur_gcs_small_dem.tif
Licancabur_utm_small_dem.tif
Titan2Dx
Titan2Dz
restart
vizout
Layers
Licancabur_12hl_10Mm_gcs
Band 1 (Gray)
98.8937
0
Lahar Hazard Zones
Band 1 (Gray)
High
Moderate
Low
ip_5000_35
Maximum hazard zone
ec_line_35
Band 1 (Gray)
1
1
Right click on the layer and go to symbology



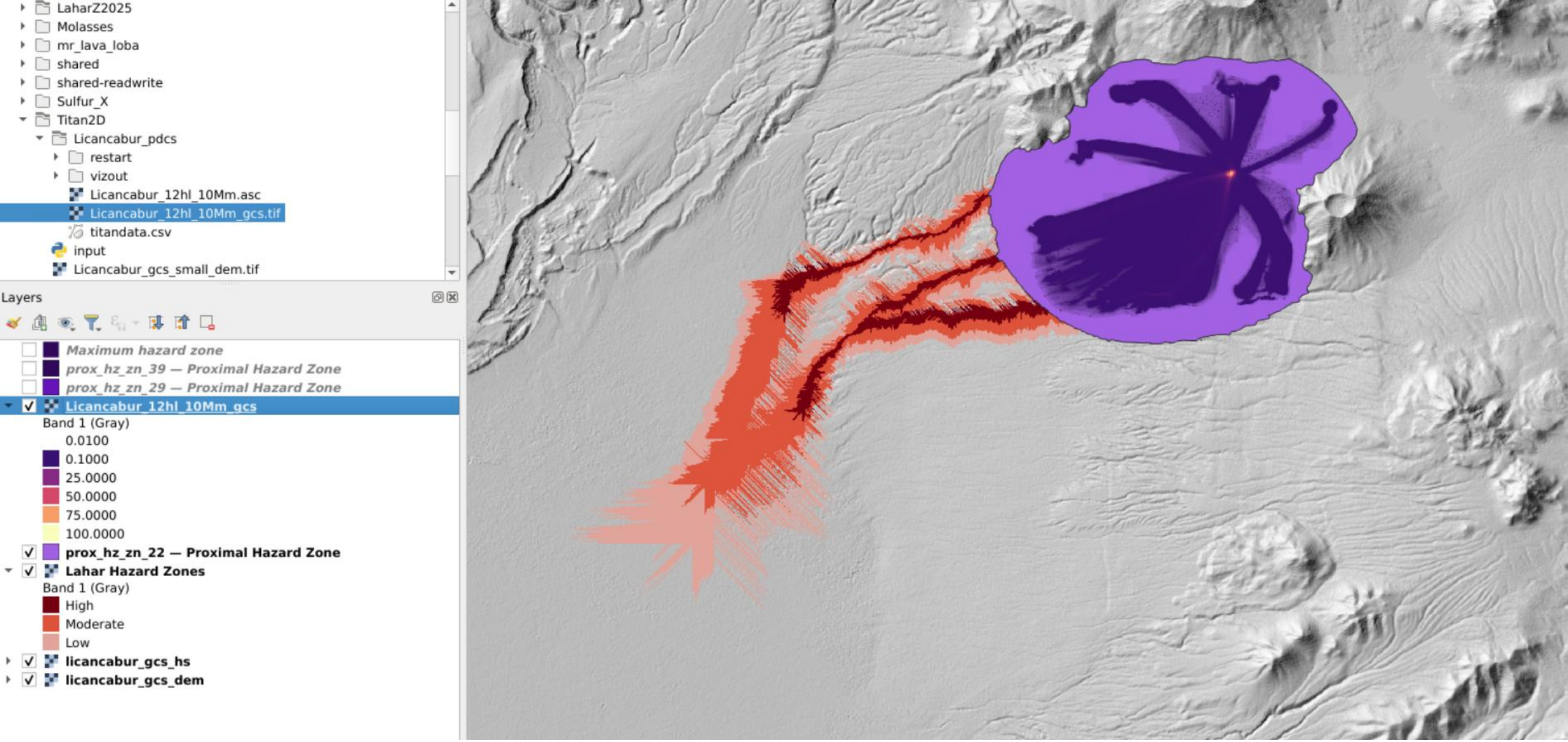
- Choose singleband pseudocolor
- Change “Mode” to “Equal interval”
- Set “classes” to 6
- Edit classes to be 0.01, 0.1, 25, 50, 75, 100
- Set 0.01 to be transparent by right clicking and choosing “Change opacity”, set it to 0
- Click Apply



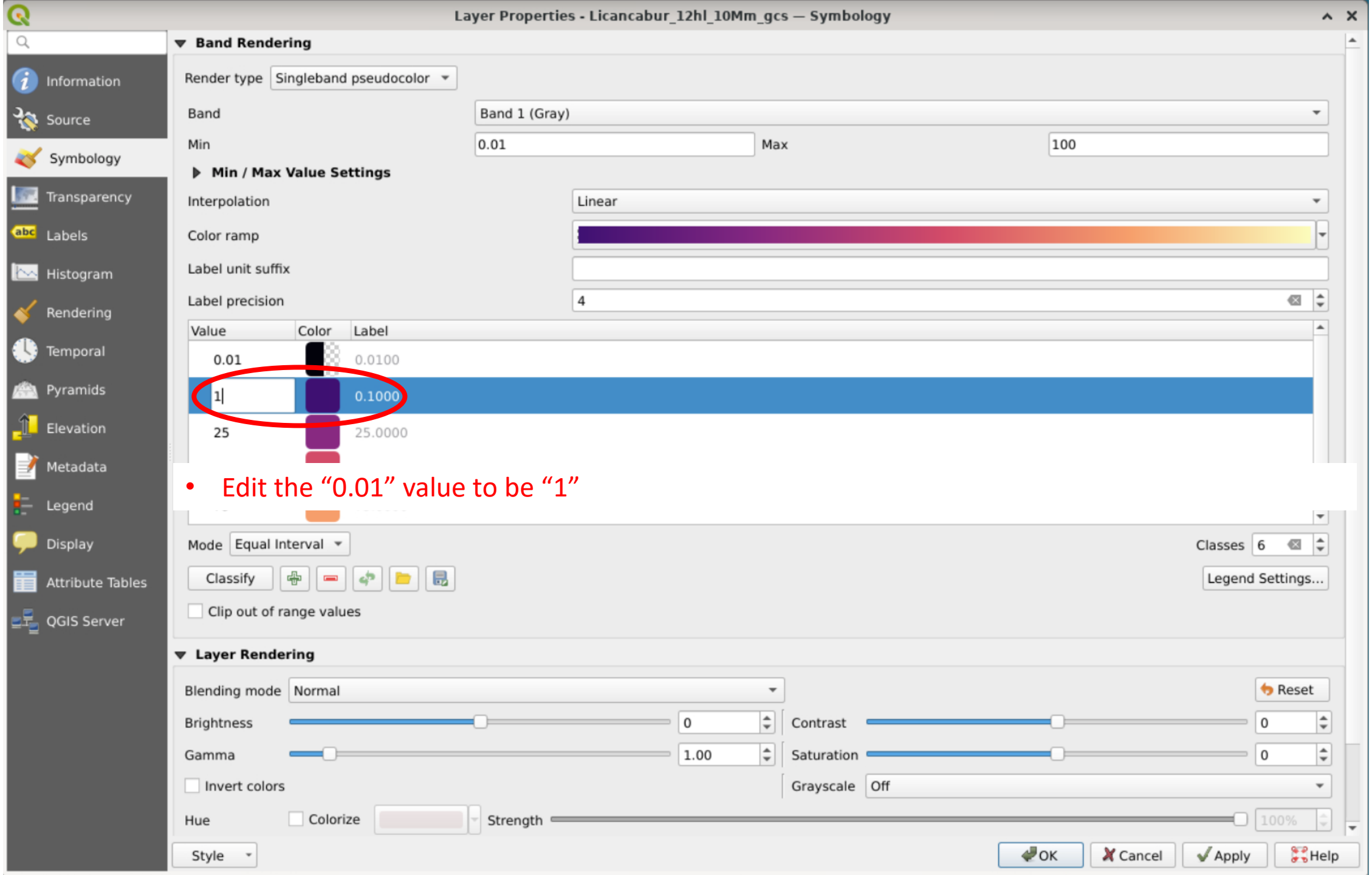
- Change the “Legend Settings”
- Unckeck the box that says “Use continuous legend”
- This lets us see the equal interval range values we just set
- Click OK
- Click Apply and OK



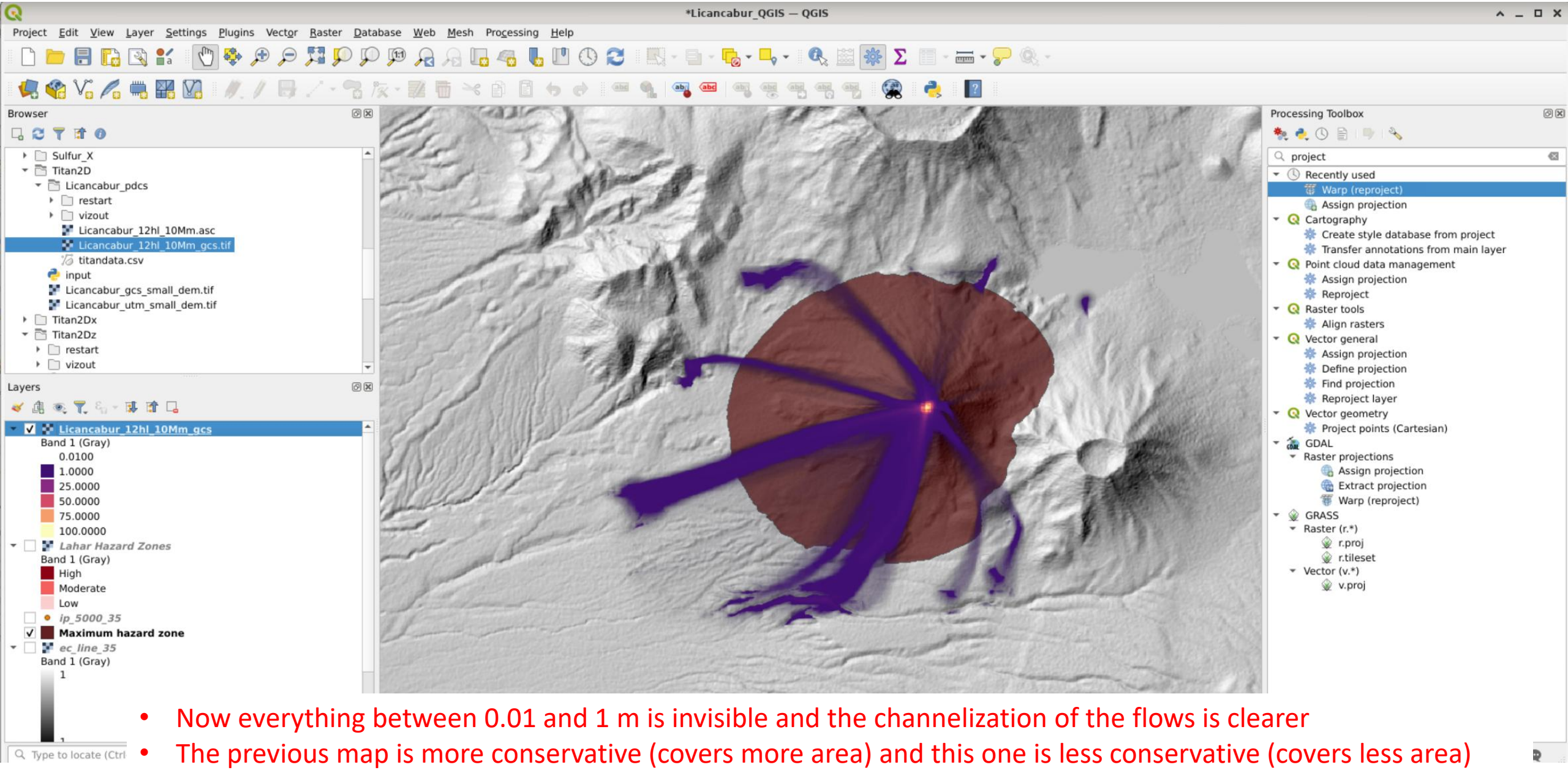
- The color ramp is for maximum thickness
- Now we can see the model runs better. Everything between 0.01 and 0.1 m is invisible
- We see that for most of the map, only a very small thickness inundated the areas
- In Titan, very thin material spreads unrealistically far, so it’s a bit of an art to decide what the “realistic” thickness is
- We can play around with these values



- If we add the energy cone we made for the same volume (10 Mm^3, 0.22, prox_hz_zn_22) we can see how they compare



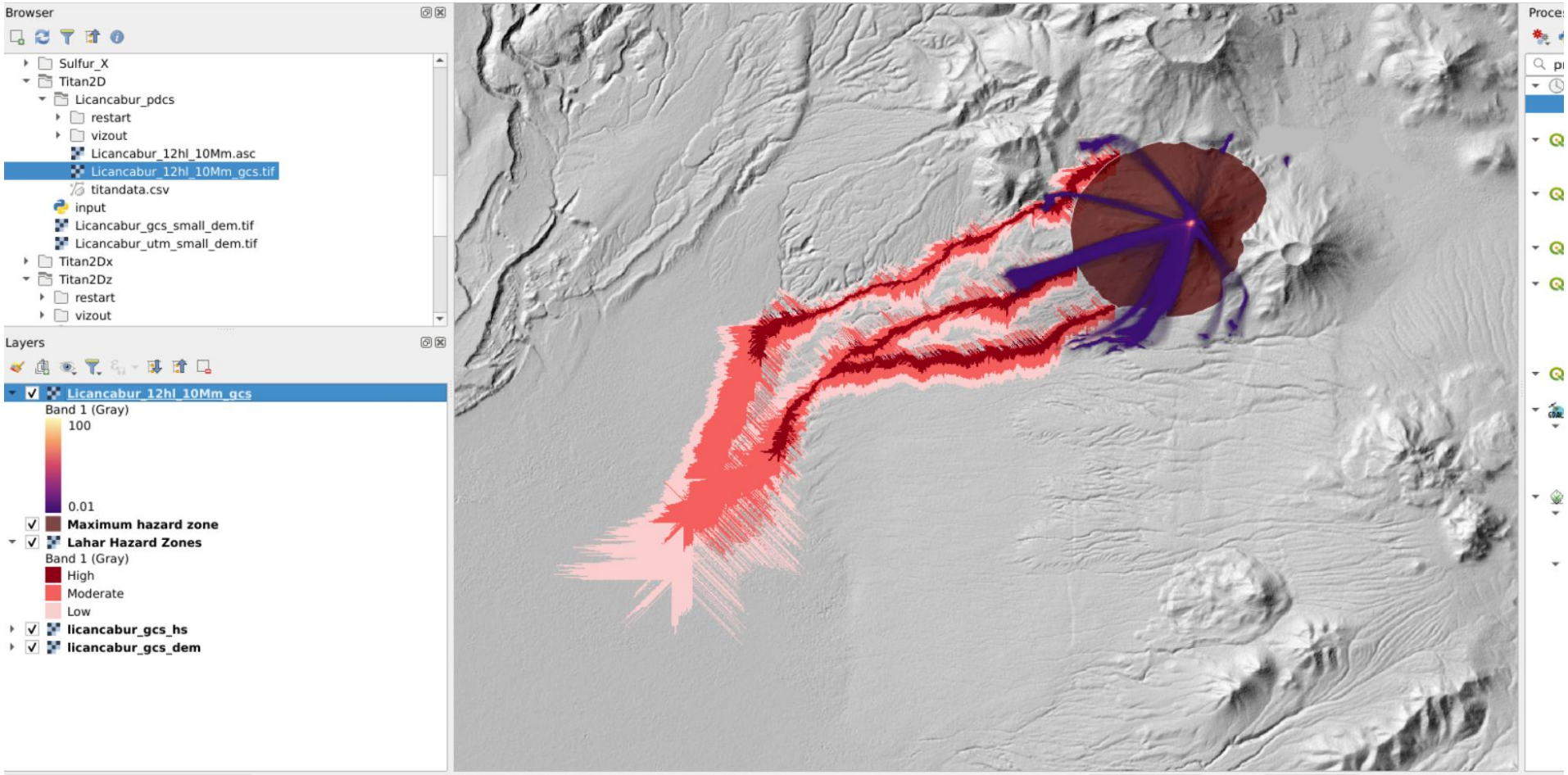
- Edit the “0.01” value to be “1”



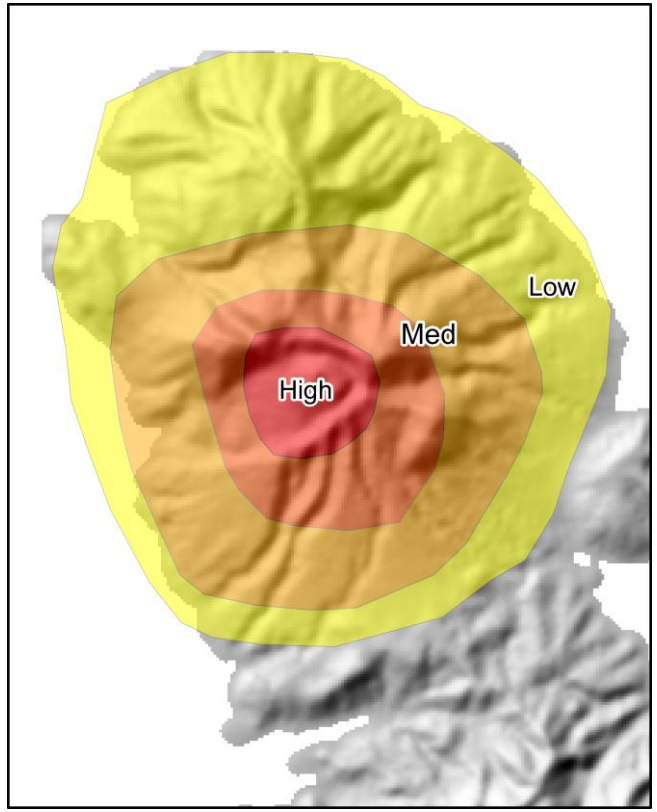
- Now everything between 0.01 and 1 m is invisible and the channelization of the flows is clearer
- The previous map is more conservative (covers more area) and this one is less conservative (covers less area)
- For a hazard map, you would need to decide which strategy to use!

Let’s add the PDC zones to the hazard map

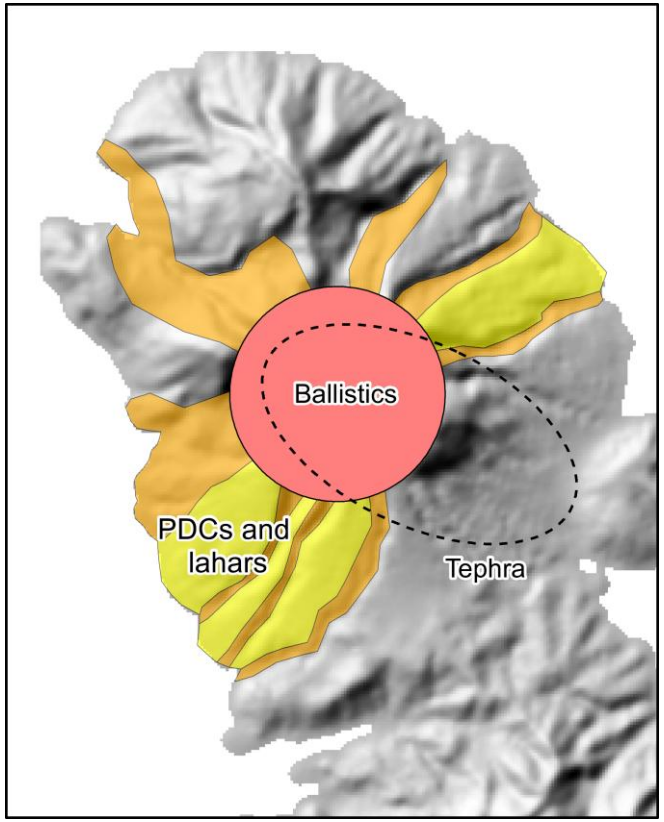
- Turn the Lahar layers back on
- Now we need to make some decisions about whether we will keep the PDC zone separate from the lahar zones or not!
- Do we want a hazard-level focused map (integrated) or a hazard-process focused map (separated)??



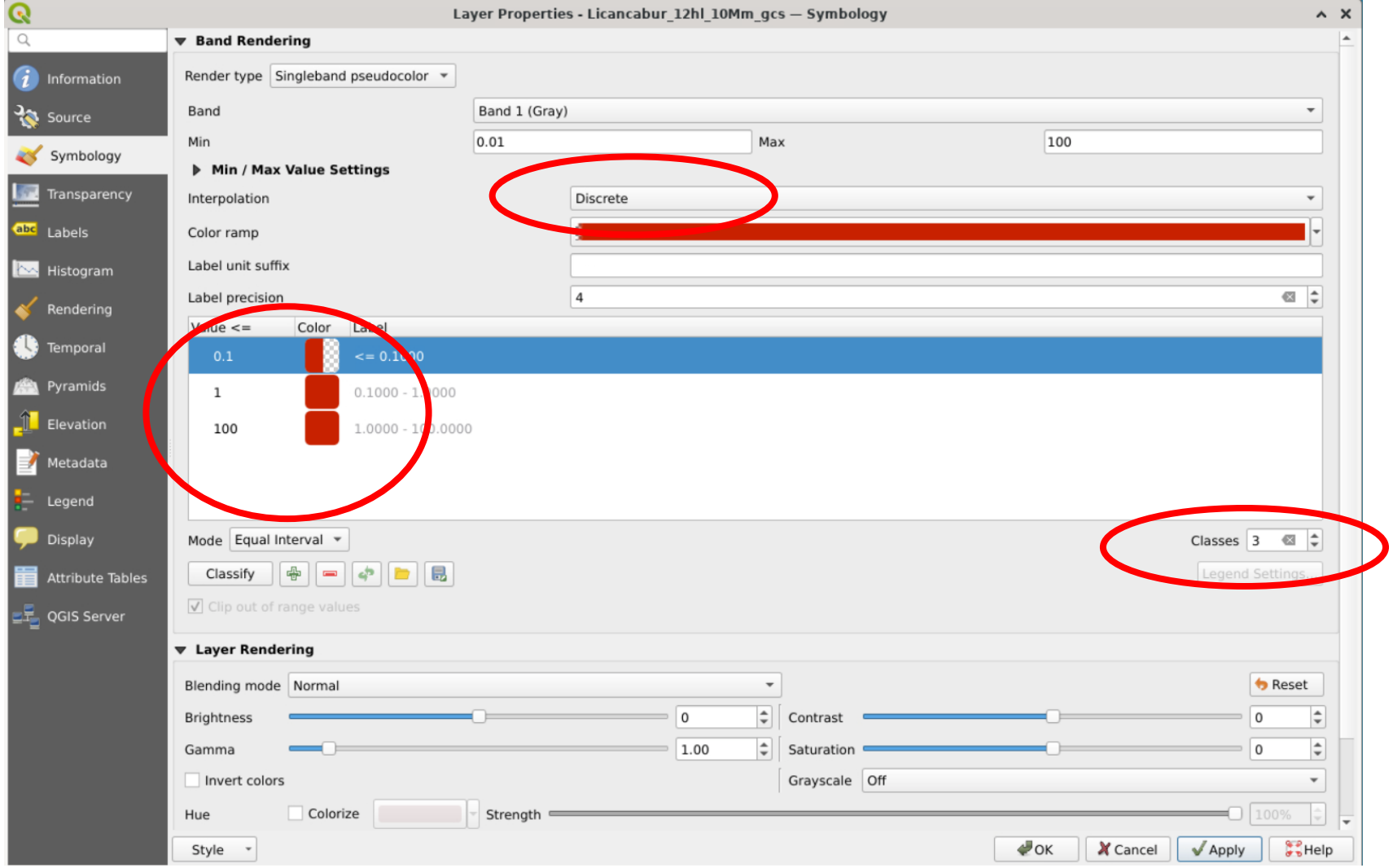
Hazard-level focused (integrated)



Hazard-process focused (separated)

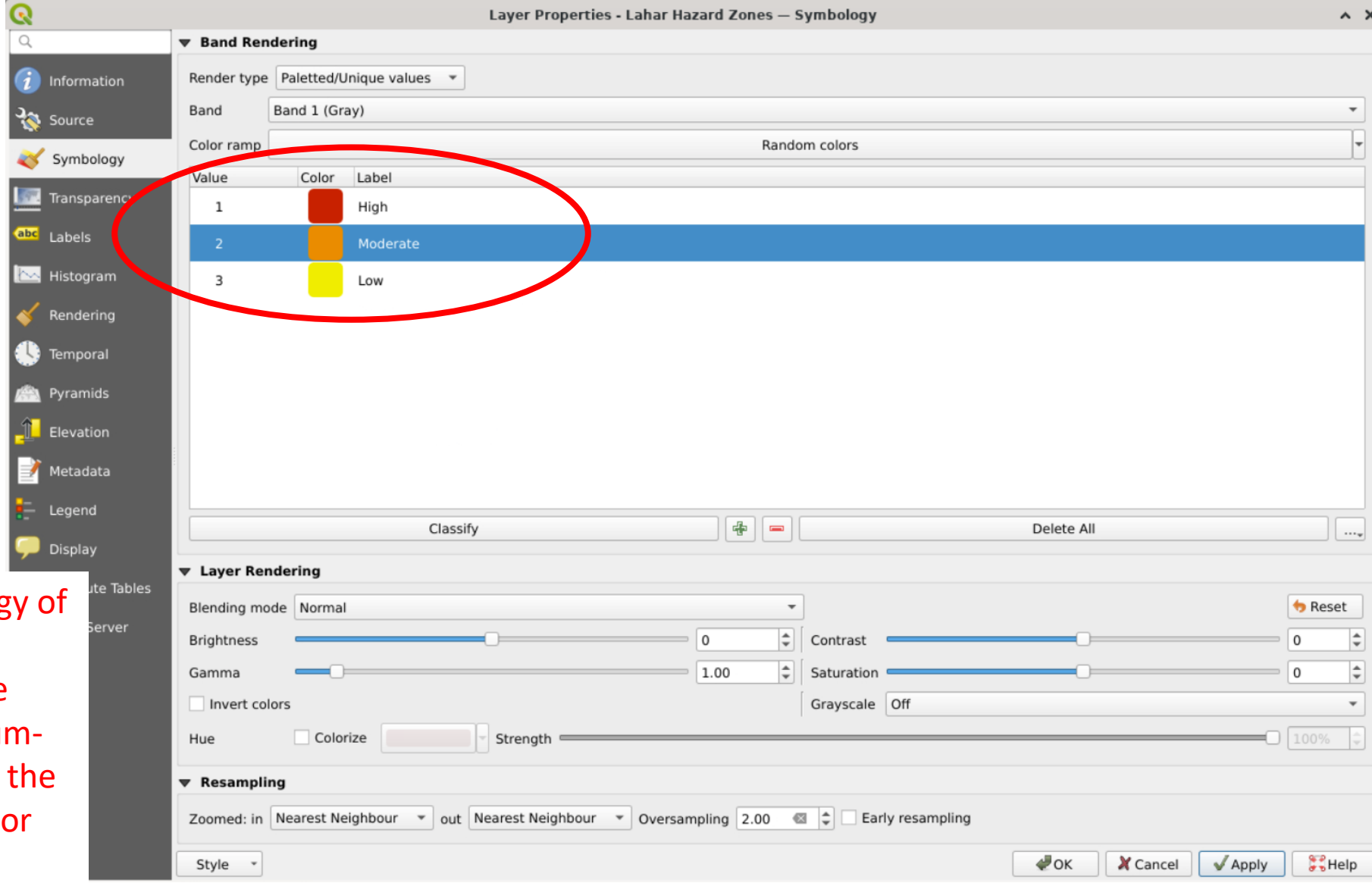


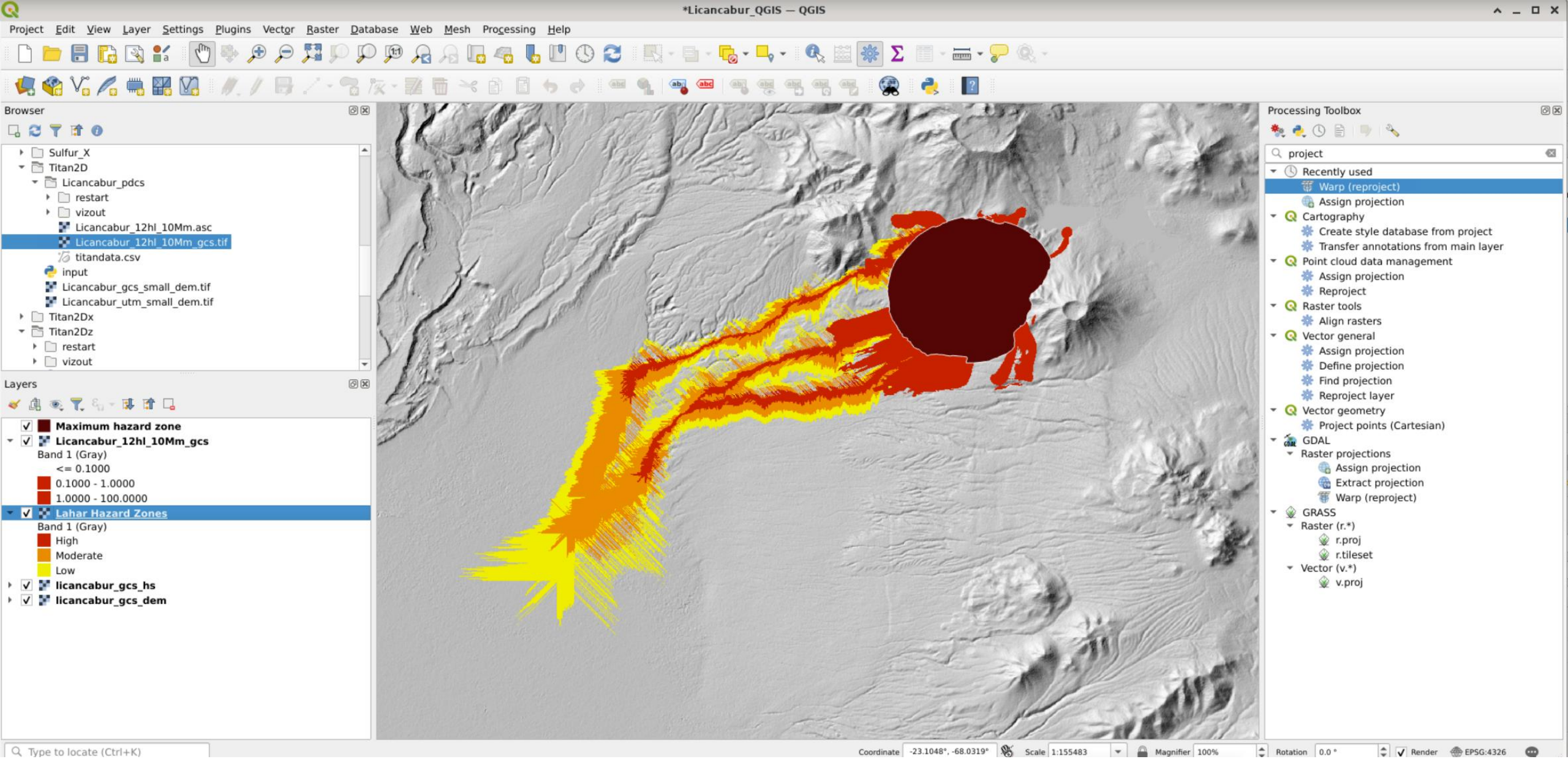
First, let’s make a hazard-level focused map (integrated) map



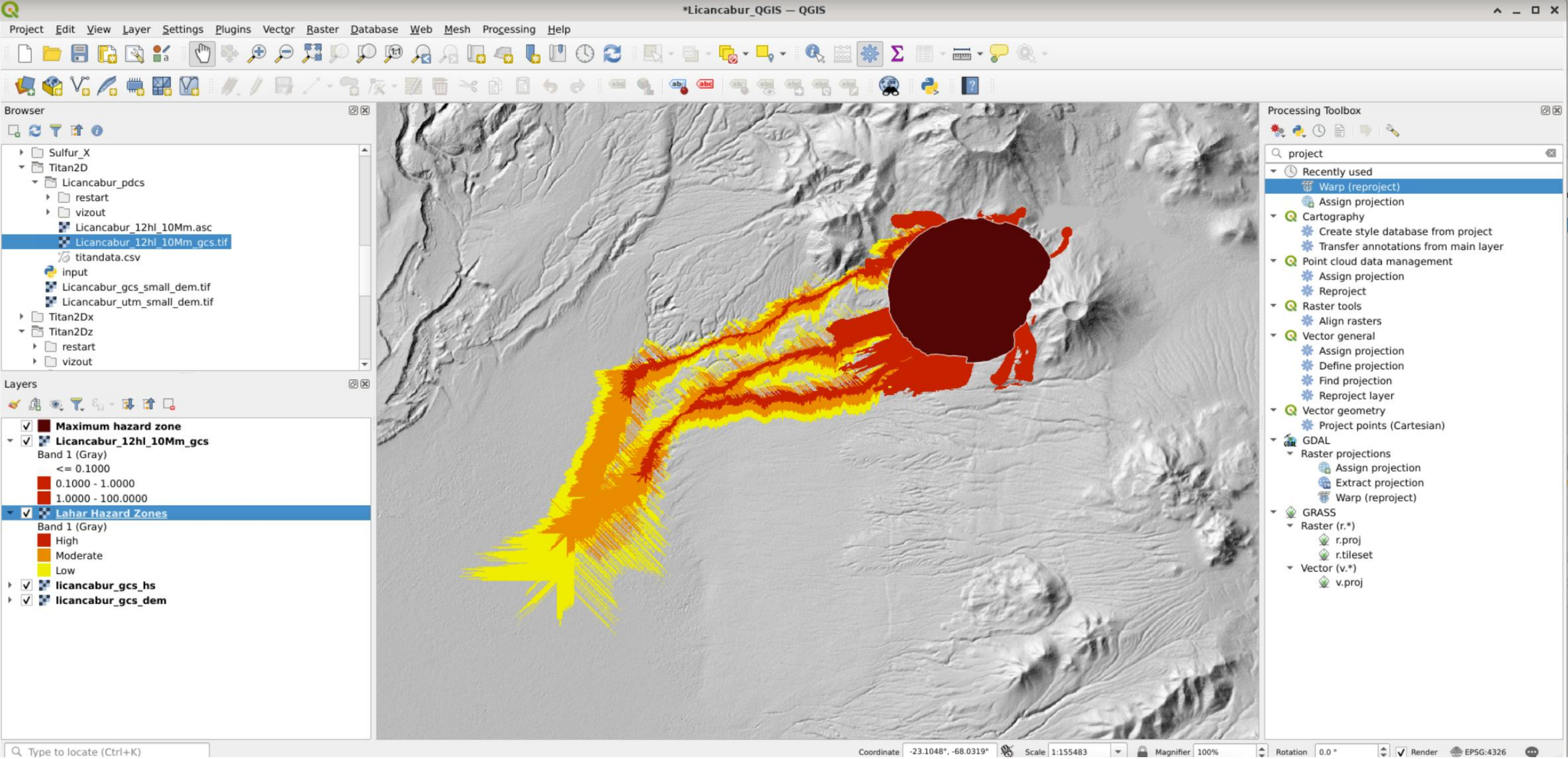
- First, let's make a hazard-level focused map (integrated) map
- We don't really care about the PDC thickness, just where the PDCs went
- Change the Interpolation to Discrete and the number of classes to 3
- Set the values to 0.1, 1, and 100
- Set them all to the same color red, but make 0.01 transparent
- Note: the easiest way to set things the same color is using the "sample

- Now we will change the symbology of the lahar layers
- Because we are integrating all the different hazards into high-medium-low hazard zones, we want to set the lahar “high” zone to the same color red as the PDC red
- The change moderate to orange and low to yellow

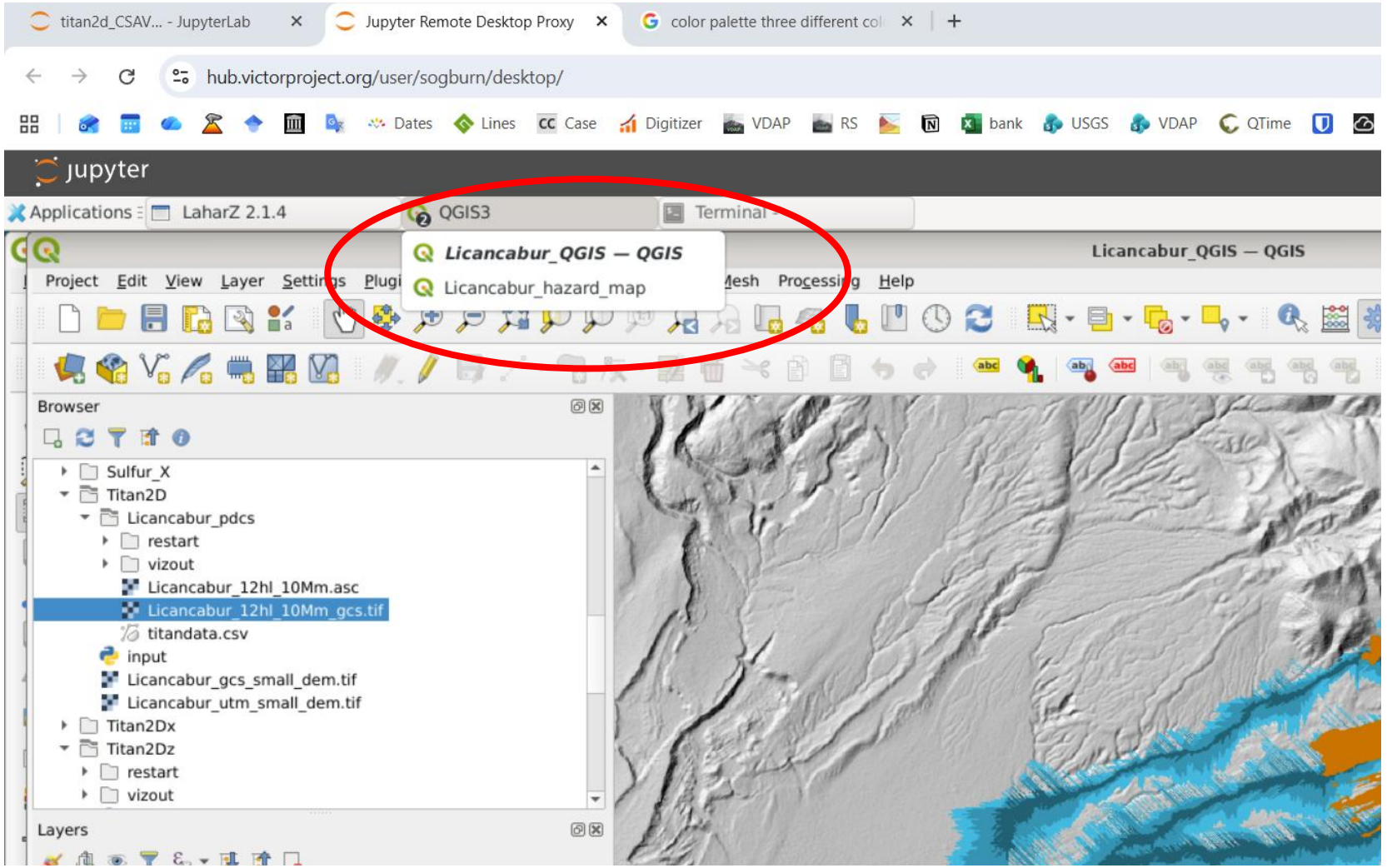




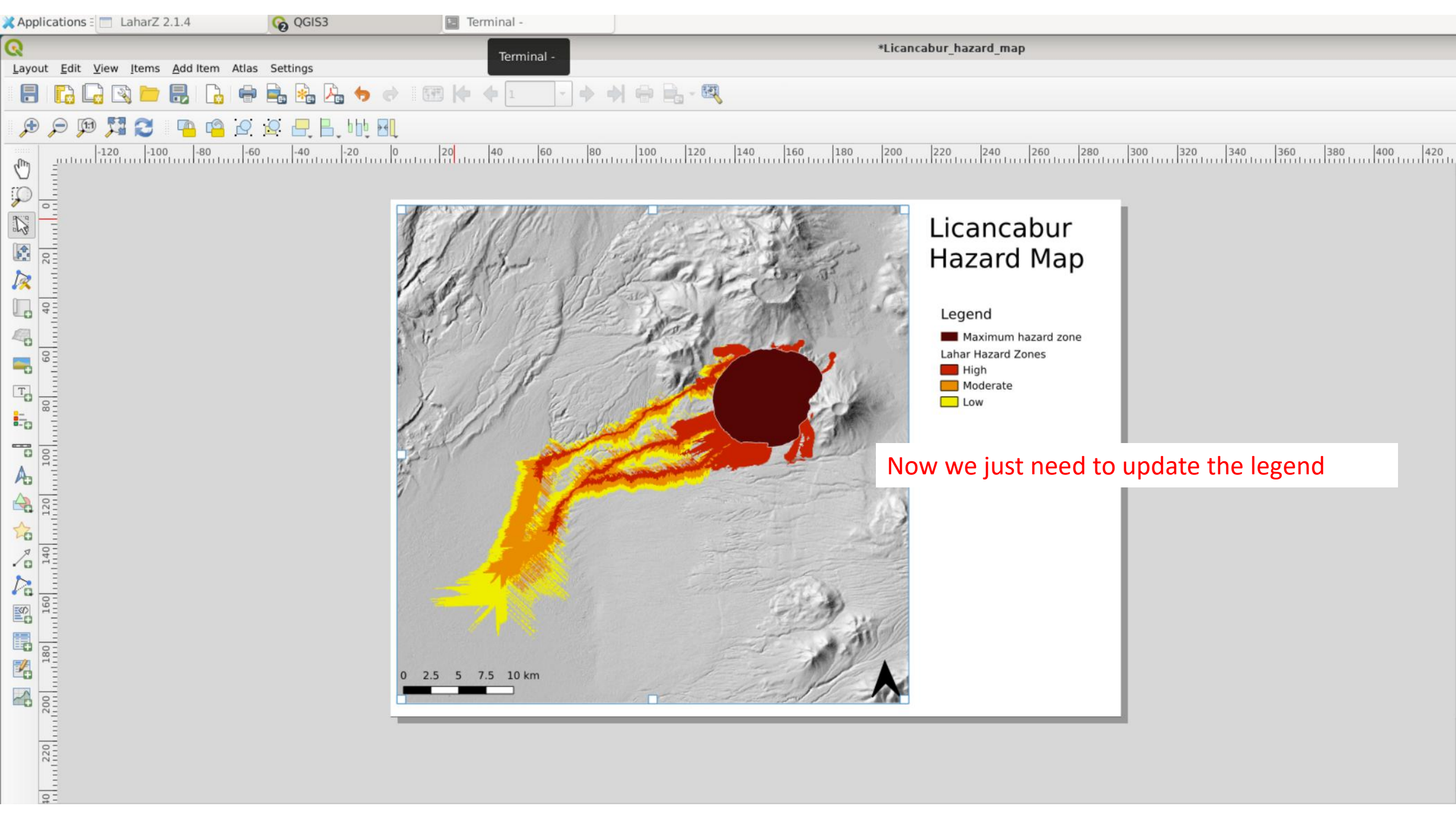
- Change the “Maximum hazard zone” to be opaque and not transparent by right clicking on it
- Go to properties and set Opacity to 100%

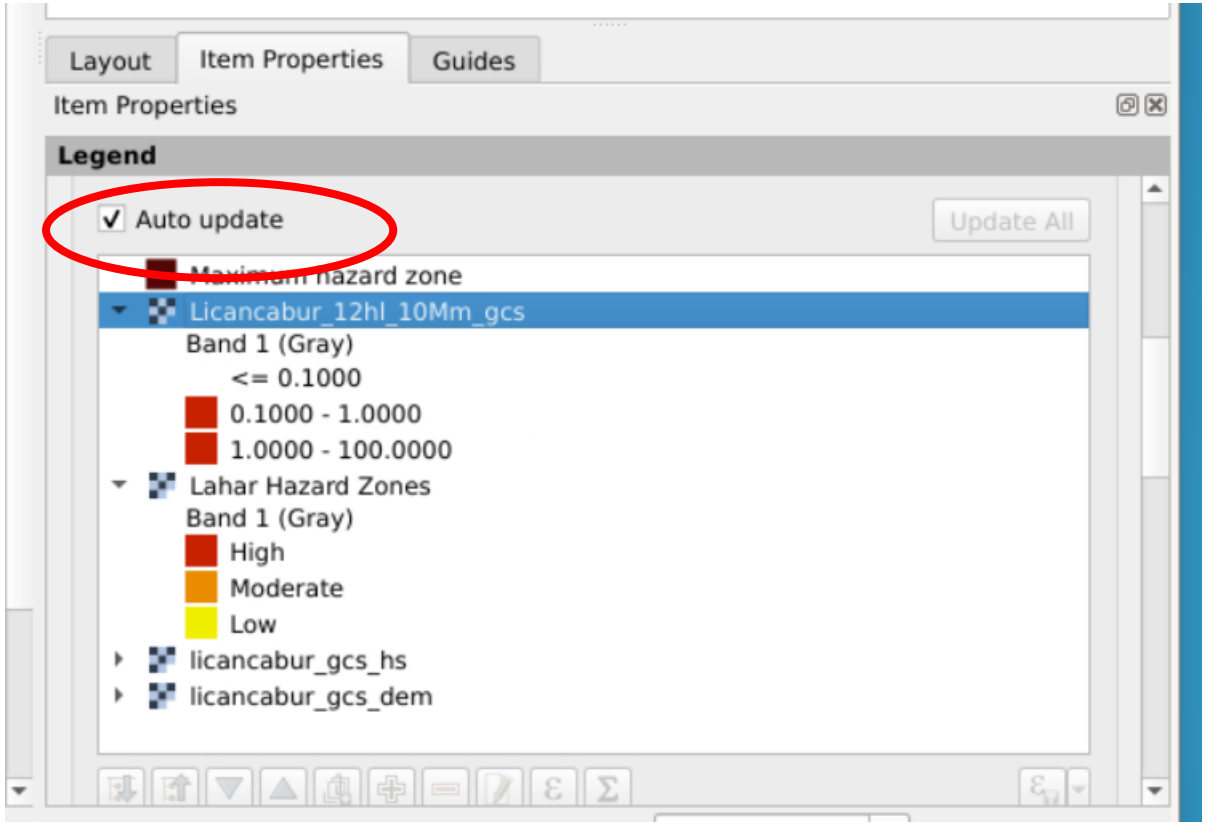


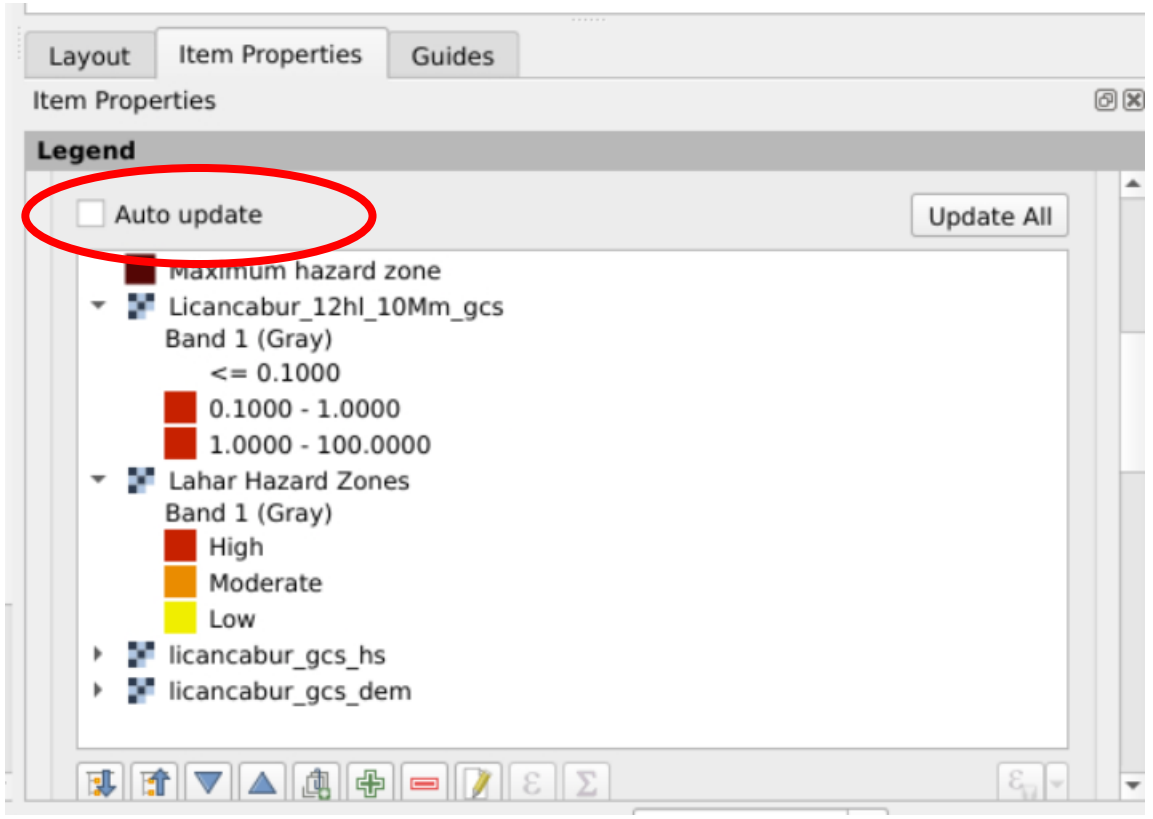
- The map should look like this
- Let’s look at our hazard map again



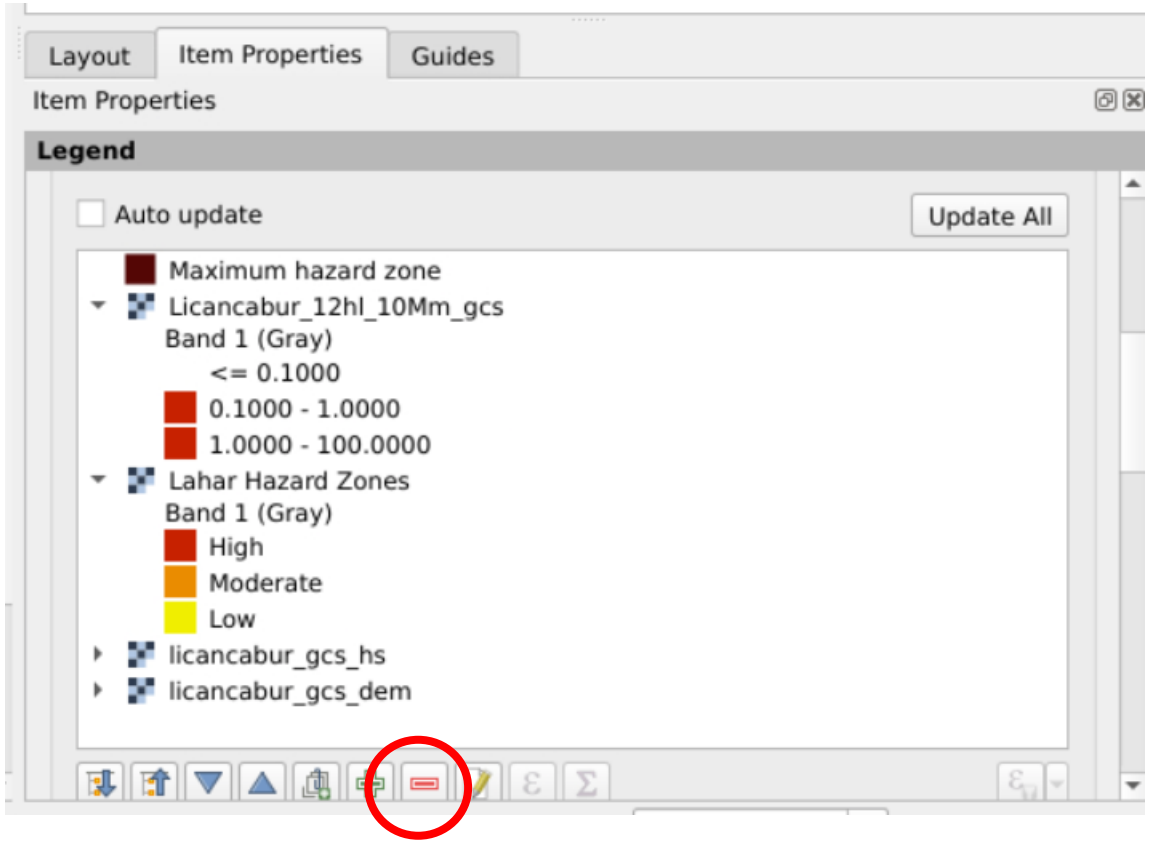
- To switch back to map view, click the QGIS icon at the top and select “Licancabur_hazard_map”


Applications
LaharZ 2.1.4
QGIS3
Terminal -
*Licancabur_hazard_map
Layout Edit View Items Add Item Atlas Settings
Licancabur
Hazard Map
Legend
Maximum hazard zone
Lahar Hazard Zones
High
Moderate
Low
0 2.5 5 7.5 10 km
Now we just need to update the legend

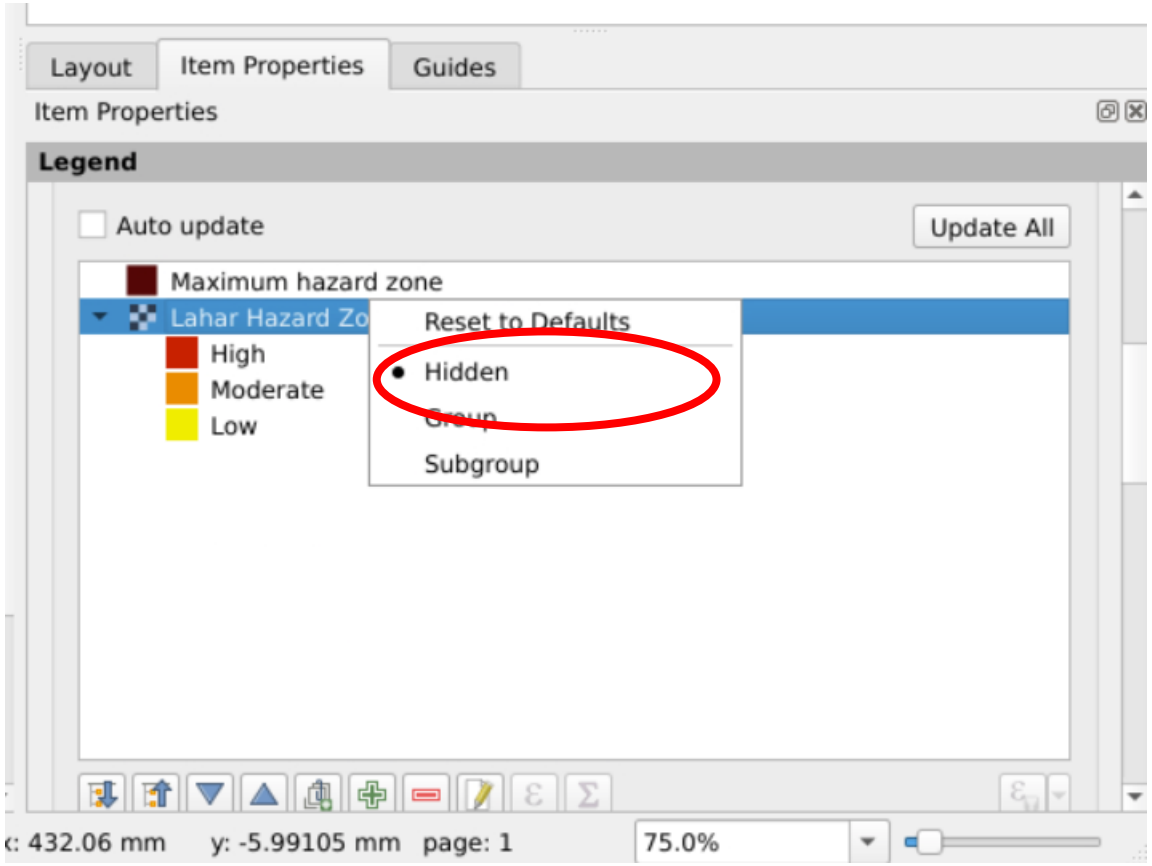




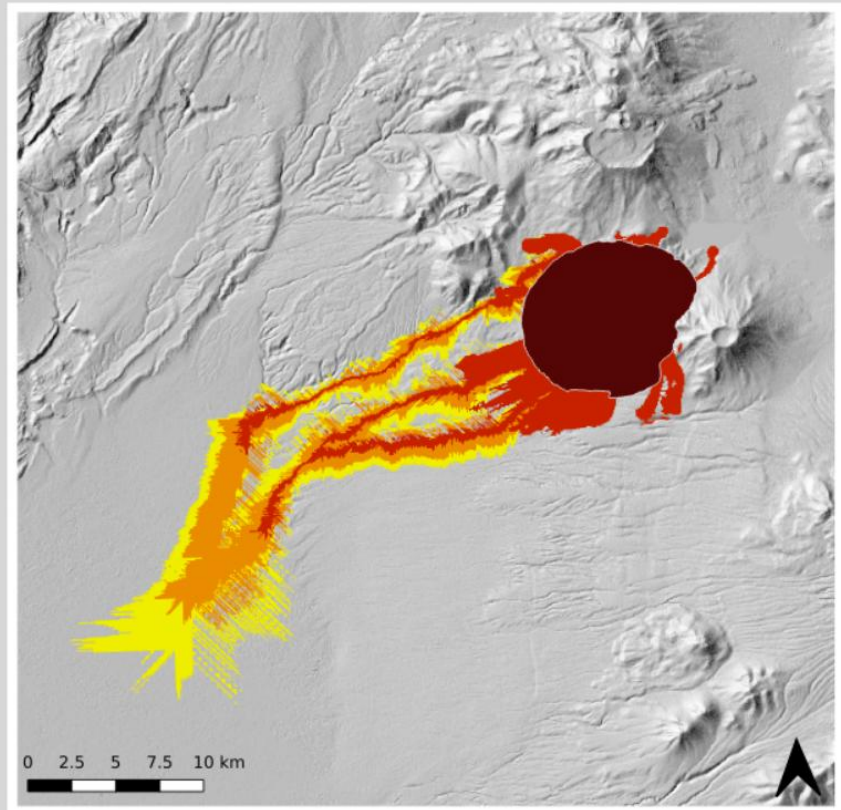
- Select the legend in the side panel.
- Click “Auto update” on and then off again to update the legend items



- Use the red minus sign to remove:
- DEMs and hillshade
- Band 1 (Gray) from lahars
- The PDC layer (we already have the red color in the legend)



- Now right click “Lahar Hazard Zones”
- Change the option to hidden



Now we have a hazard-level focused map with all the hazards integrated into Maximum to Low zones!

If we had more time, we could clean up these zones by merging them a bit better and cleaning up the LaharZ edges

Items | Undo History

Items

	Item
☑	North Arrow
☑	<Scalebar>
☑	Licancabur Hazard Map
☑	Hazard Zones
☑	Map 1

Layout | Item Properties | Guides

Item Properties

Page

Size: A4

Orientation: Landscape

Width: 297.000

Height: 210.000

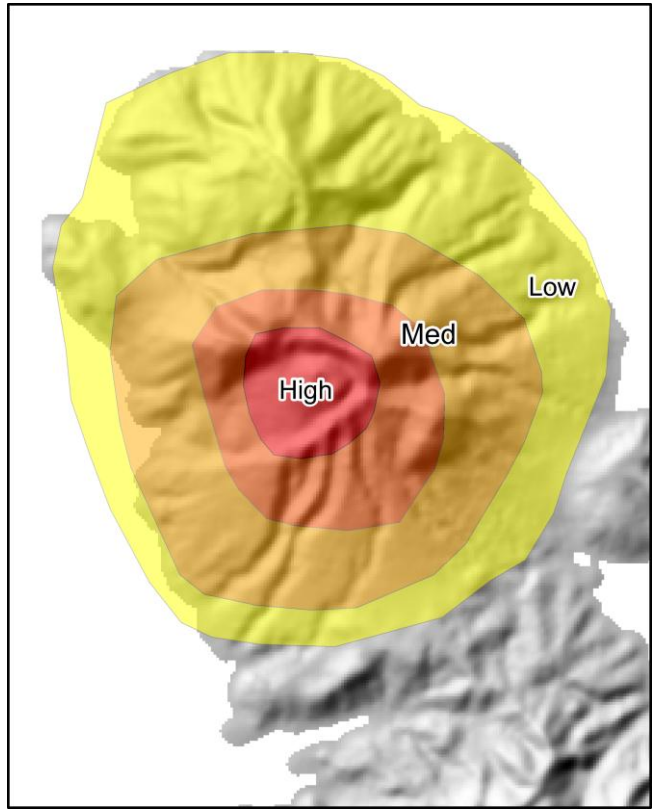
mm

Background

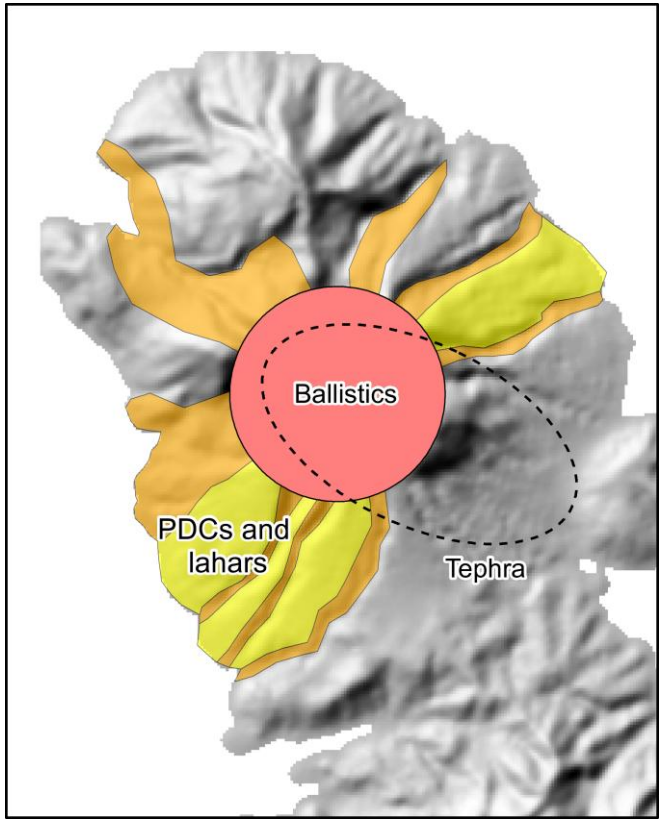
Exclude page from exports

Ok, but what if we want a hazard-level focused map where we can see all the different hazards separated?

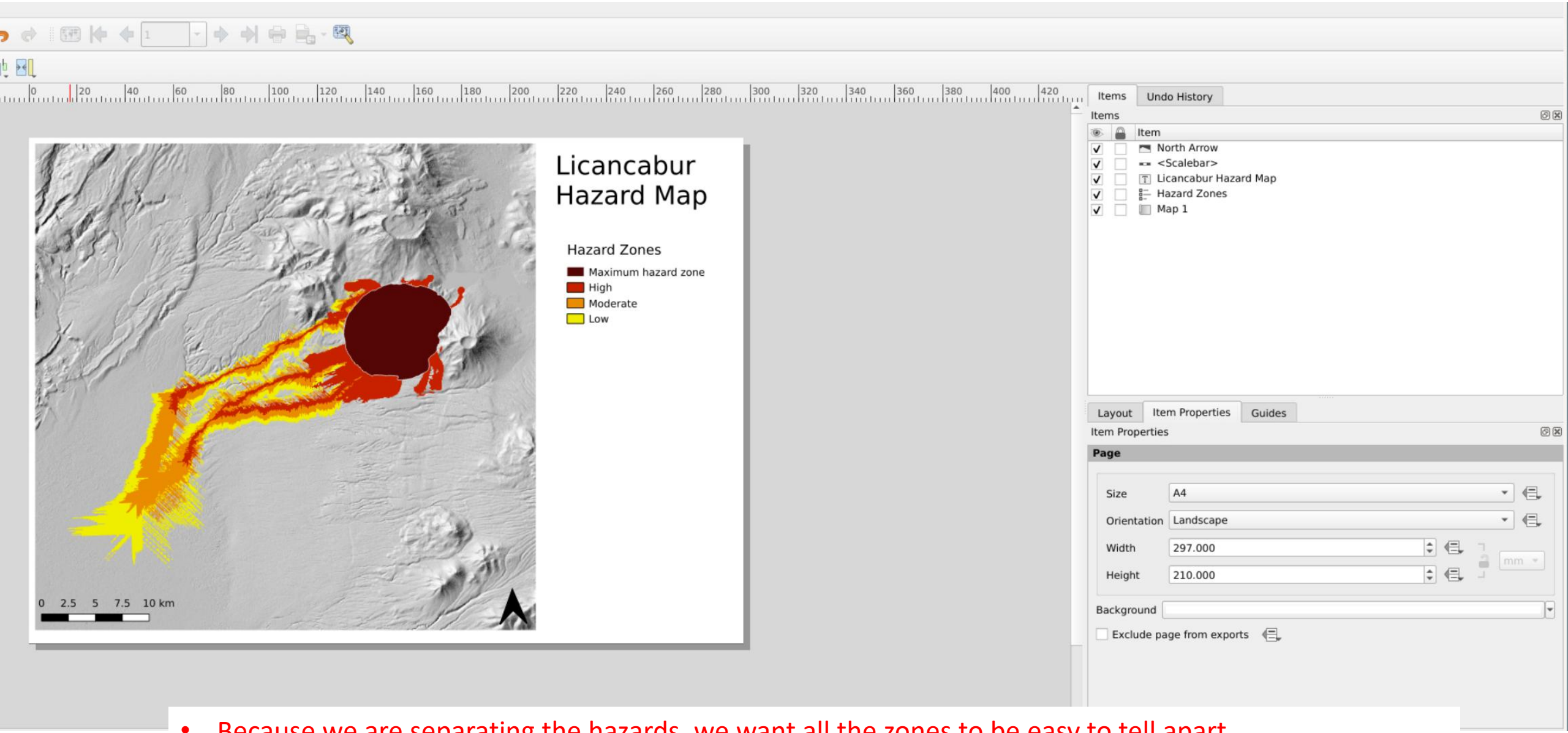
Hazard-level focused (integrated)



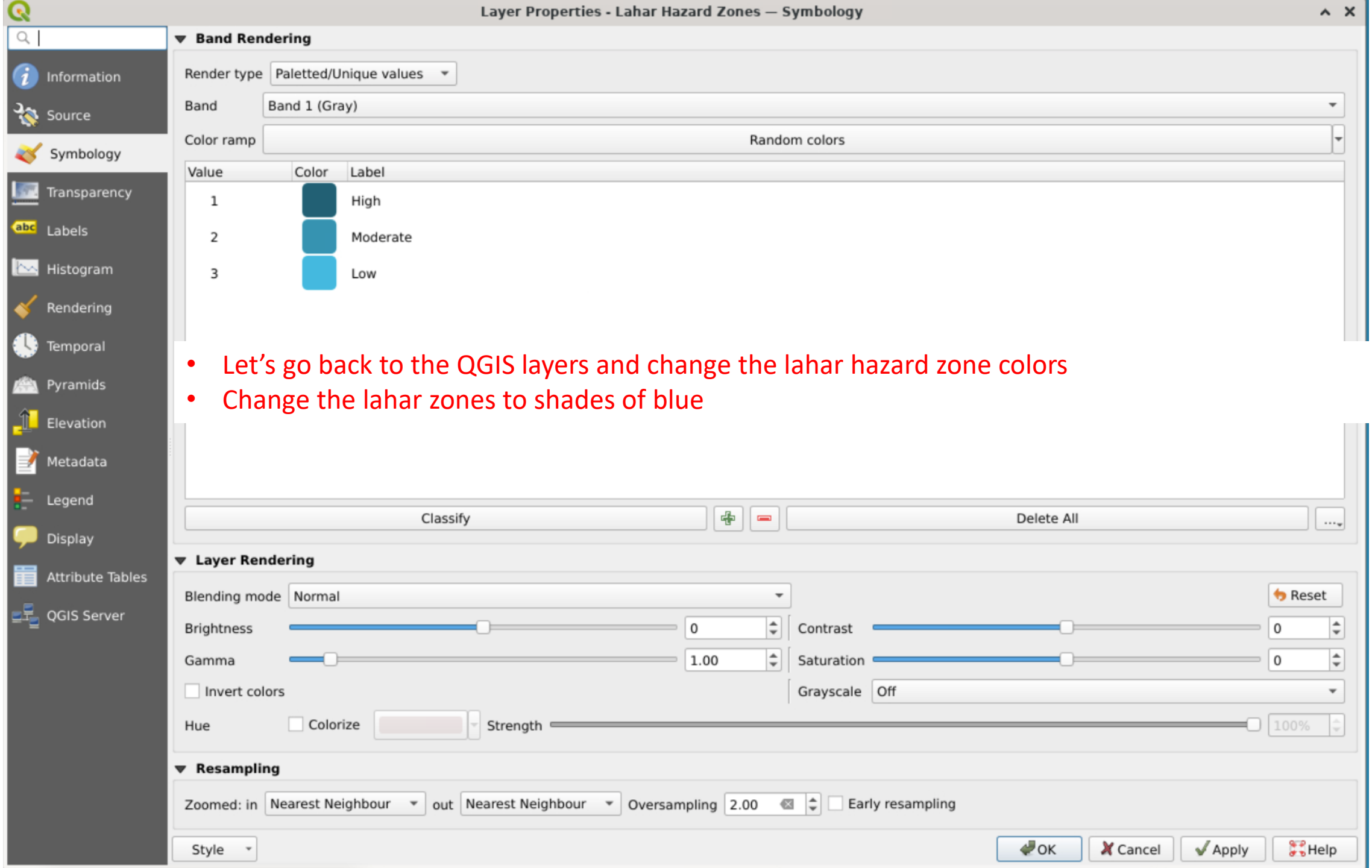
Hazard-process focused (separated)



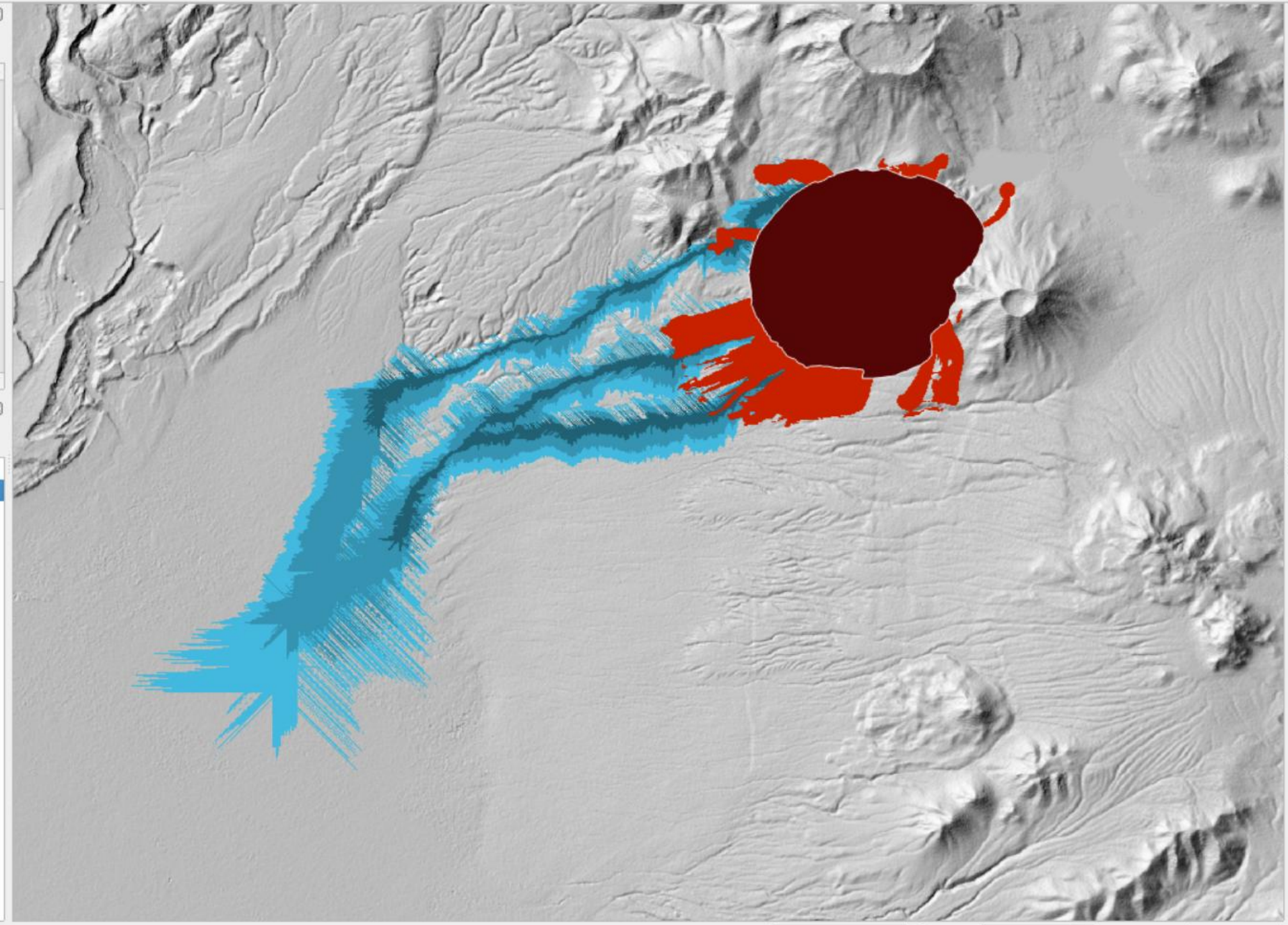
Now let's make a hazard-process focused map (separated) map



- Because we are separating the hazards, we want all the zones to be easy to tell apart
- Let’s go back to the QGIS layers and change the lahar hazard zone colors



- Let's go back to the QGIS layers and change the lahar hazard zone colors
- Change the lahar zones to shades of blue

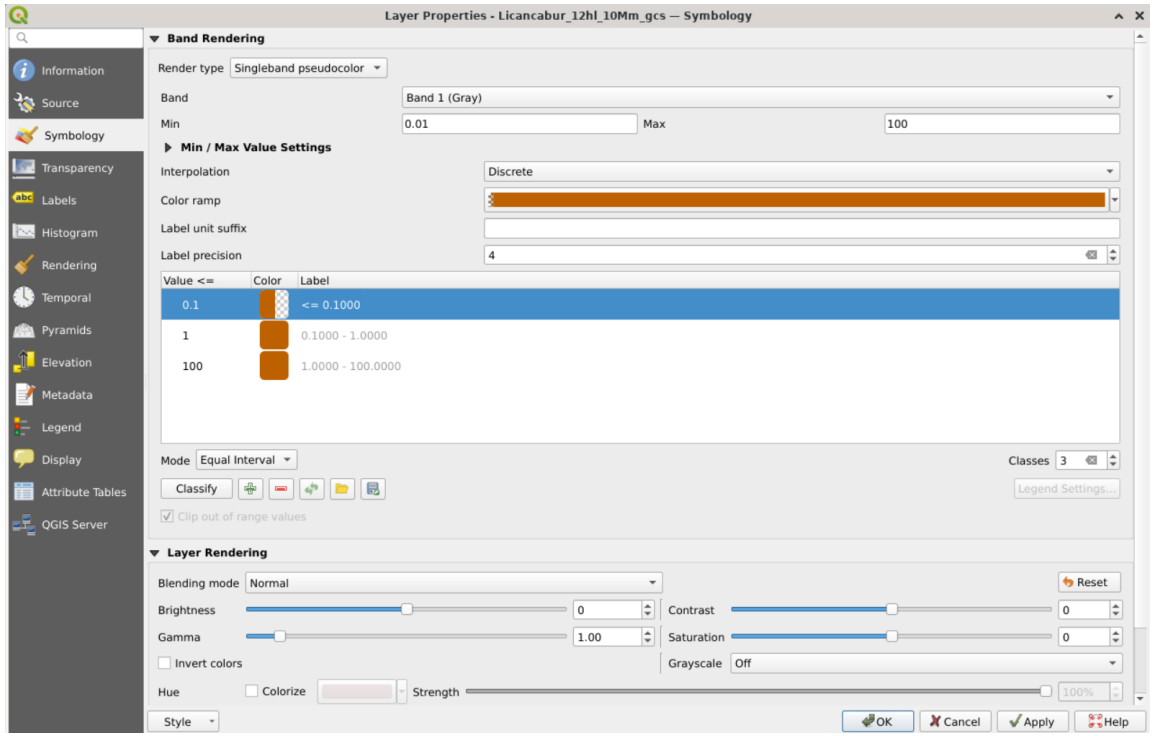


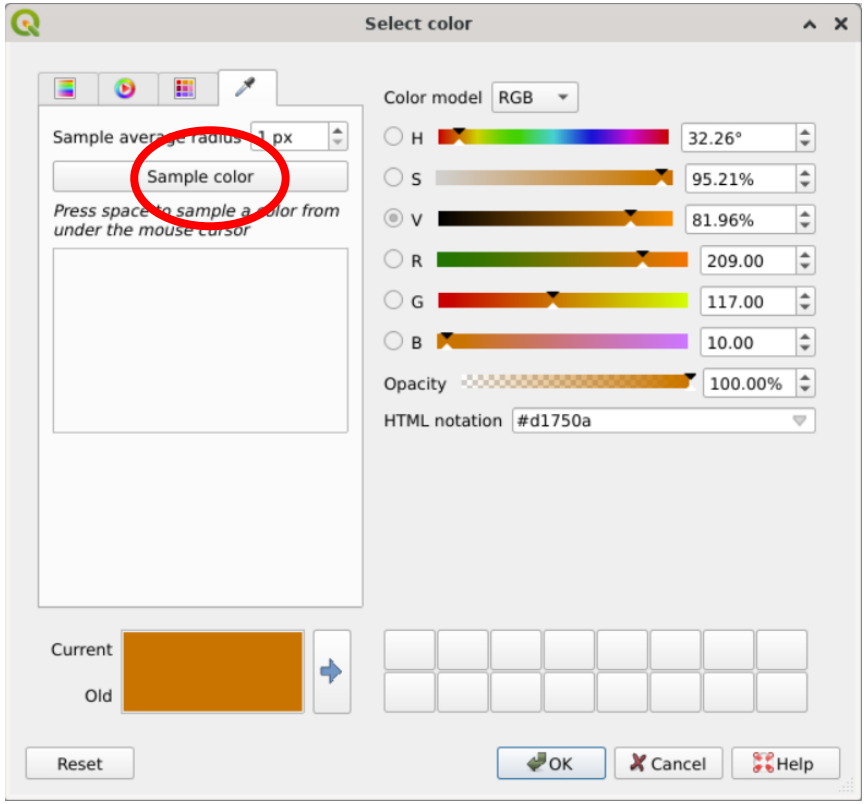
Processing Toolbox

project

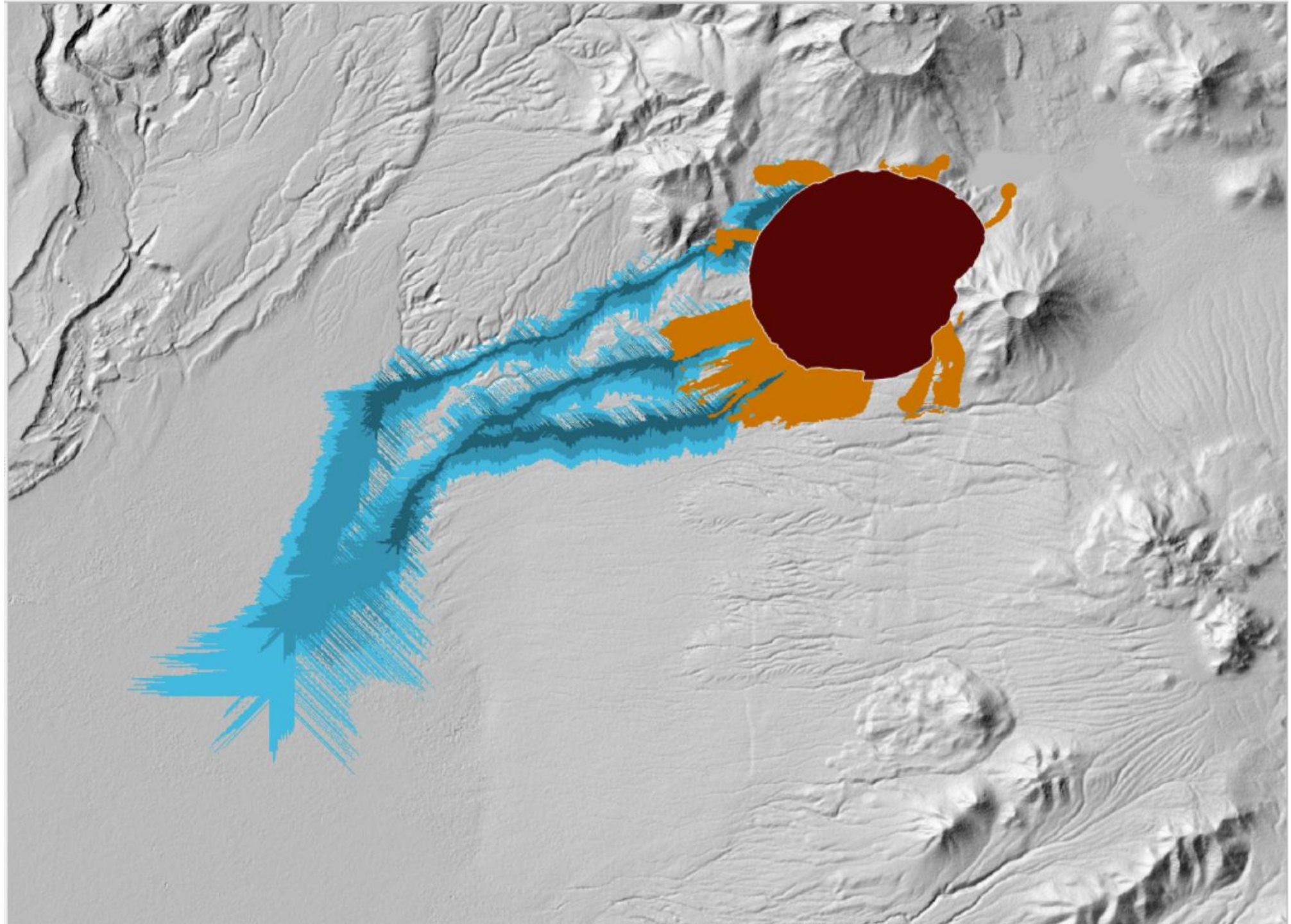
- Recently used
  - Warp (reproject)
  - Assign projection
- Cartography
  - Create style database from project
  - Transfer annotations from main layer
- Point cloud data management
  - Assign projection
  - Reproject
- Raster tools
  - Align rasters
- Vector general
  - Assign projection
  - Define projection
  - Find projection
  - Reproject layer
- Vector geometry
  - Project points (Cartesian)
- GDAL
  - Raster projections
    - Assign projection
    - Extract projection
    - Warp (reproject)
- GRASS
  - Raster (r.*)
    - r.proj
    - r.tileset
  - Vector (v.*)
    - v.proj

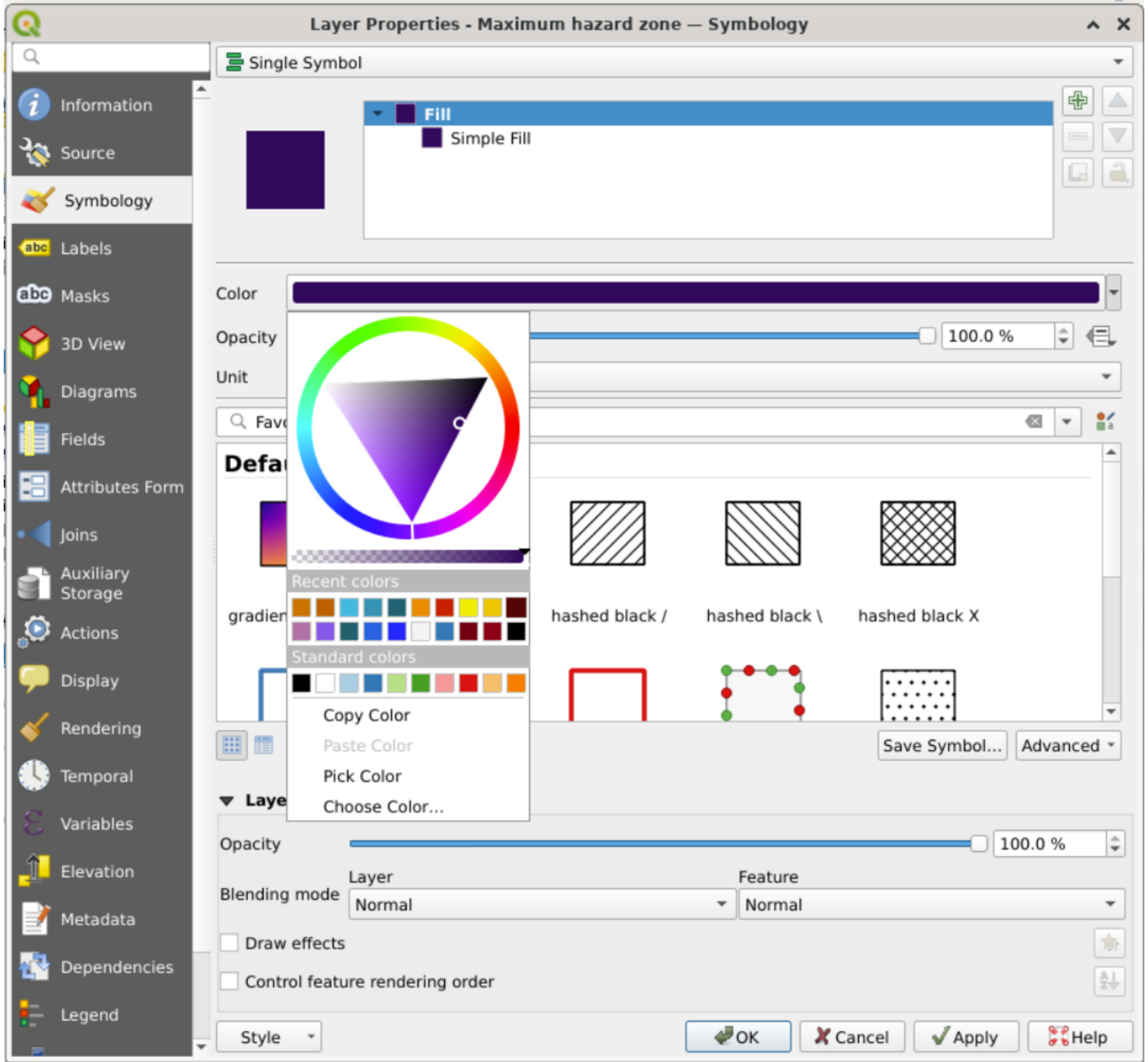
Coordinate -22.9946°, -68.2373° Scale 1:155483 Magnifier 100% Rotation 0.0 ° Render EPSG:4326



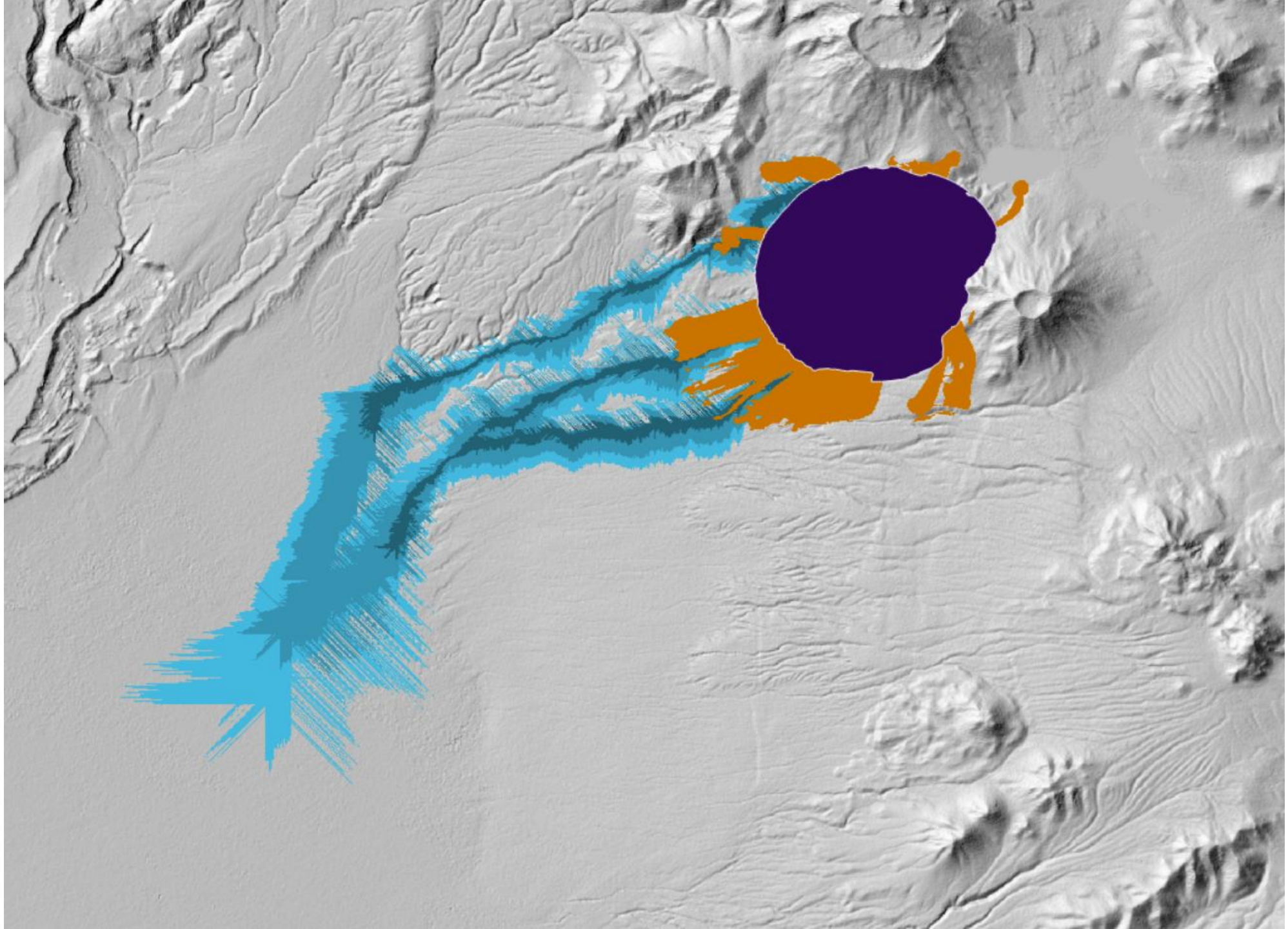


- Let’s set the PDC colors to oranges
- Set them all to the same color, but make 0.01 transparent
- Note: the easiest way to set things the same color is using the “sample color” eye dropper tool



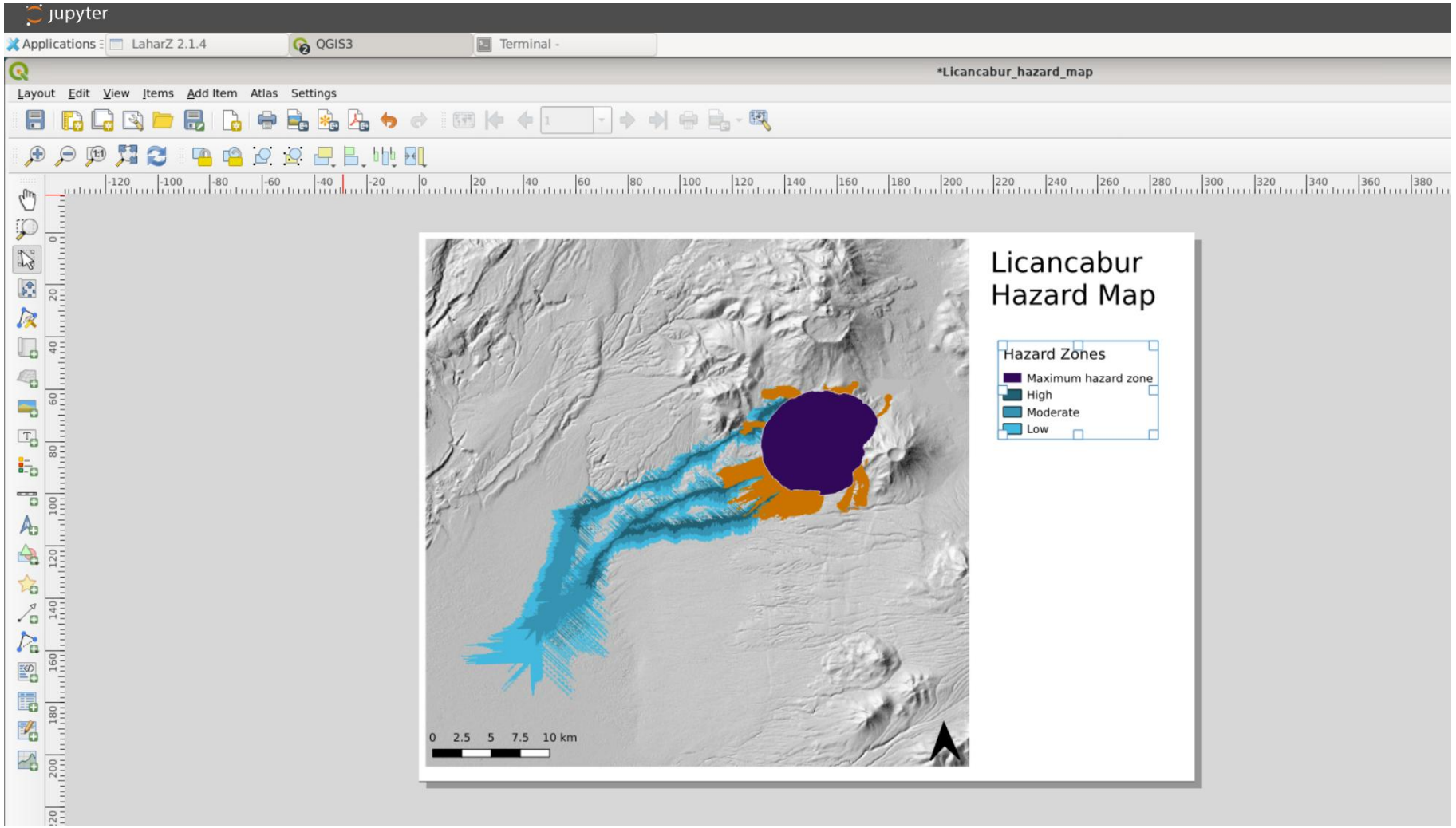


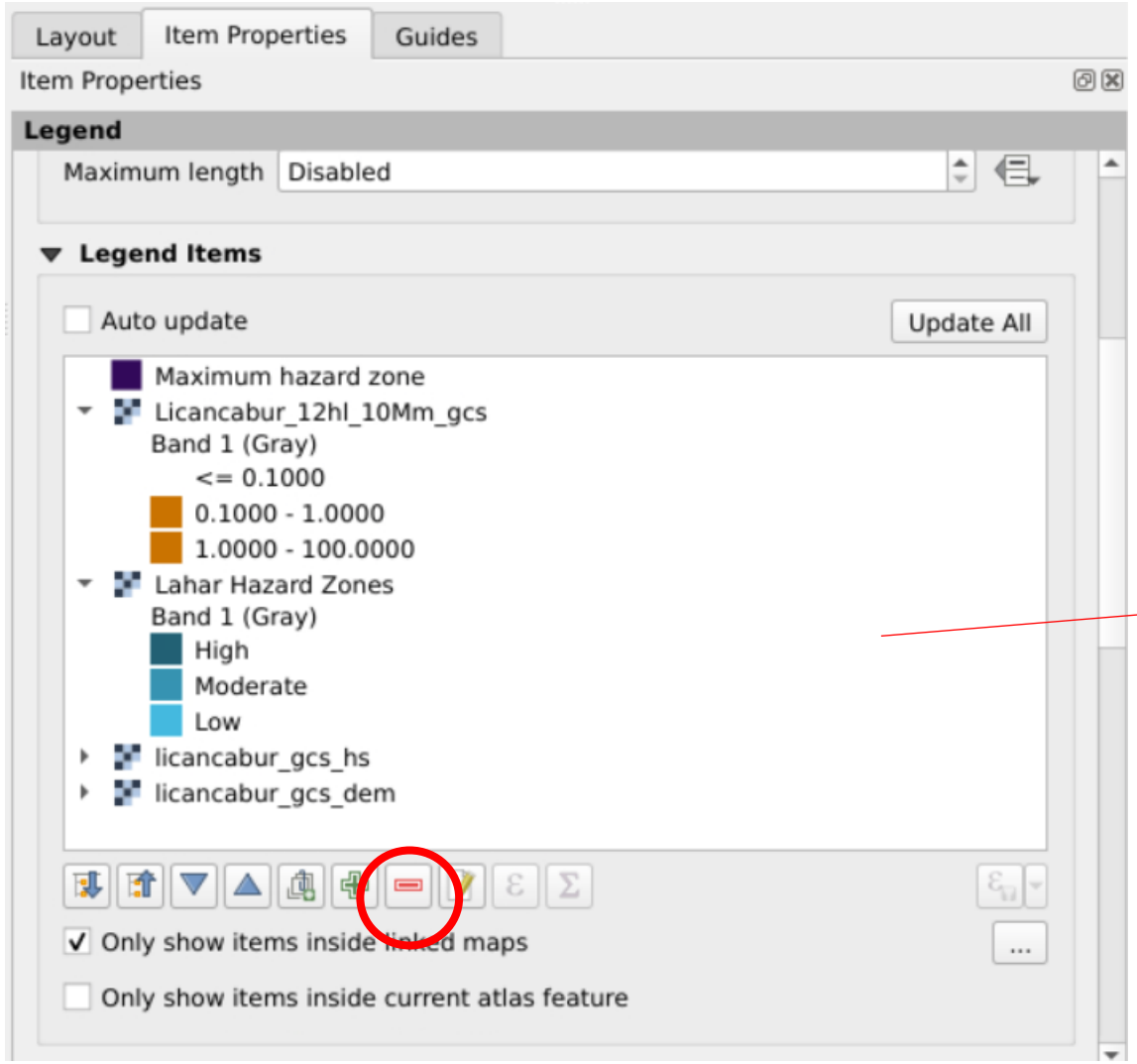
- Change the maximum hazard zone to purple

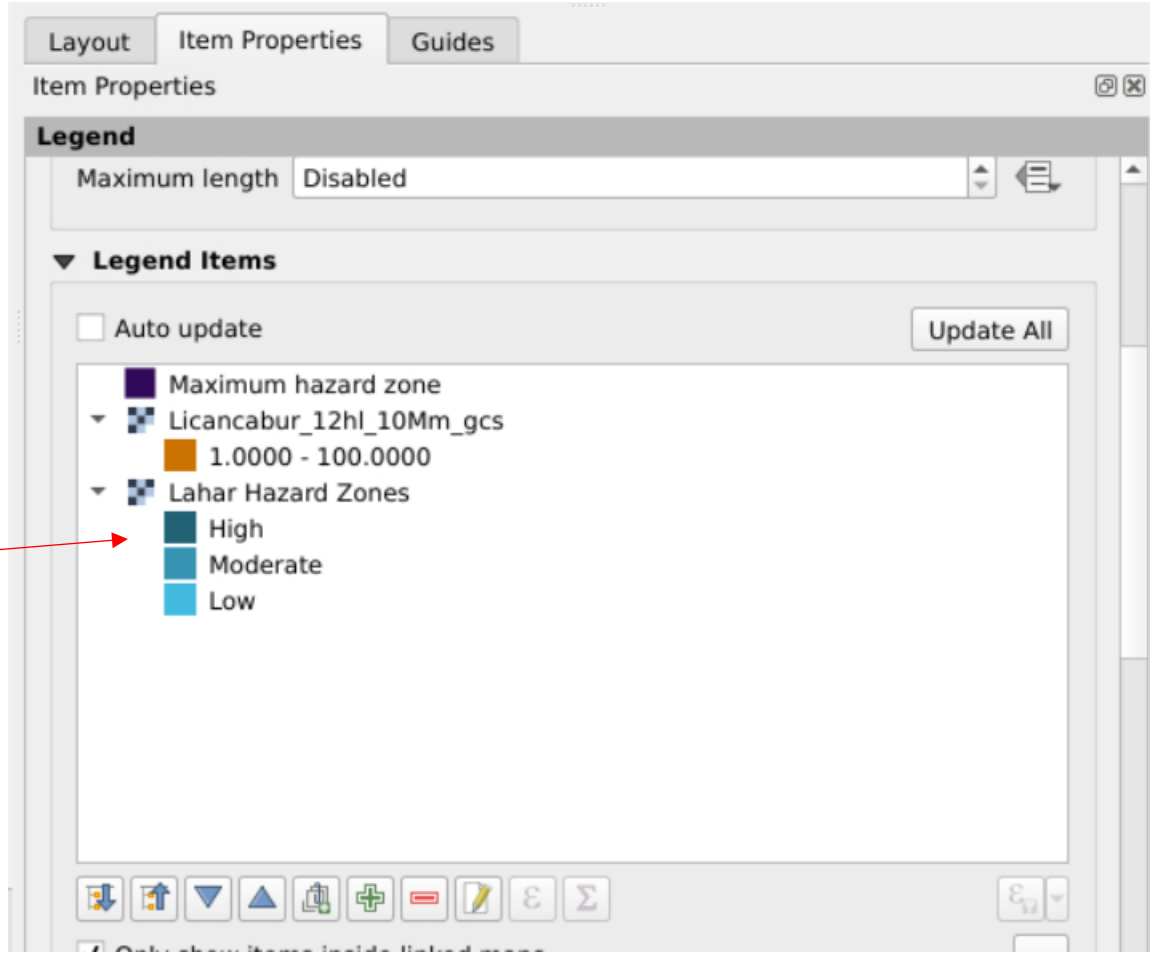


Let’s check the map

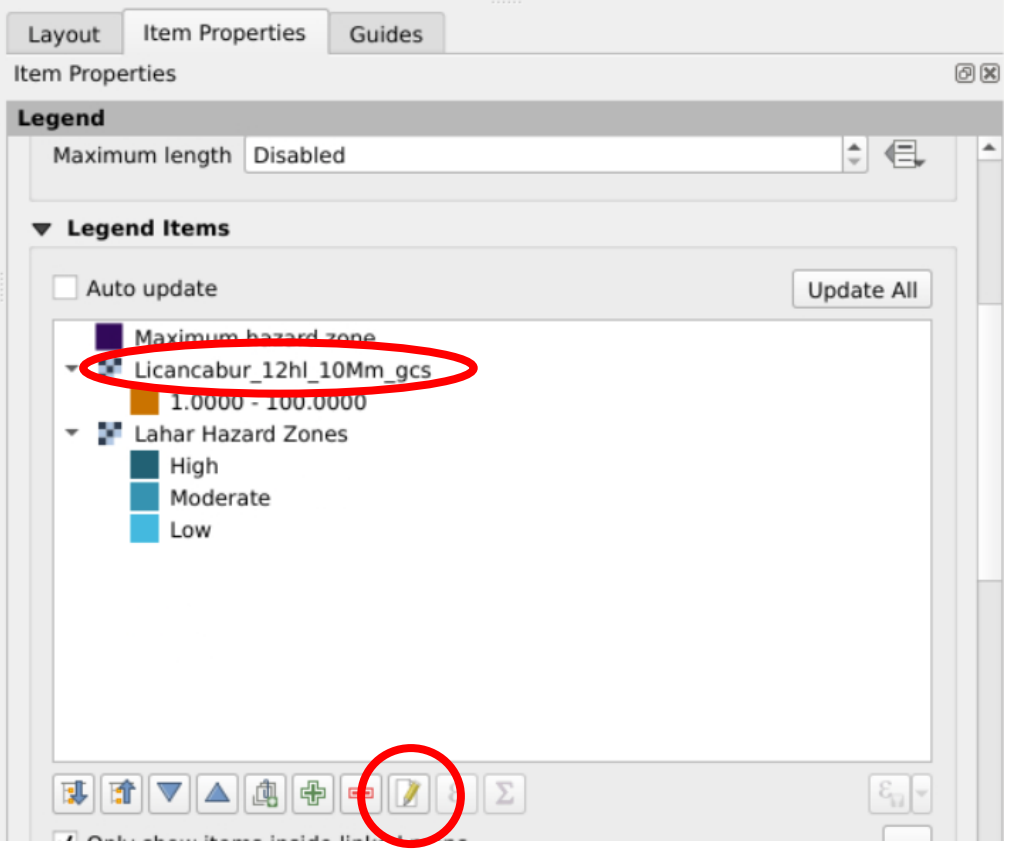
- We need to fix the legend

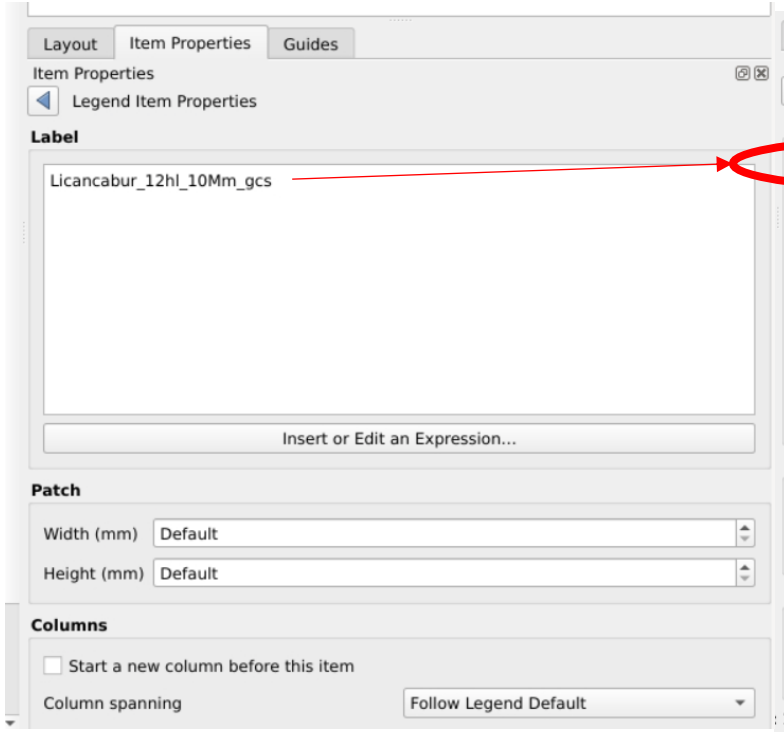


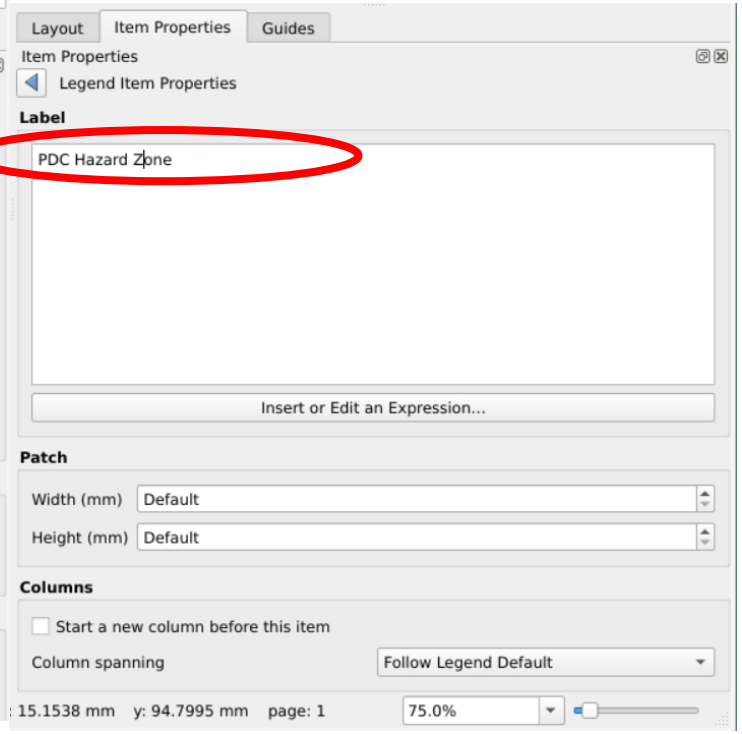




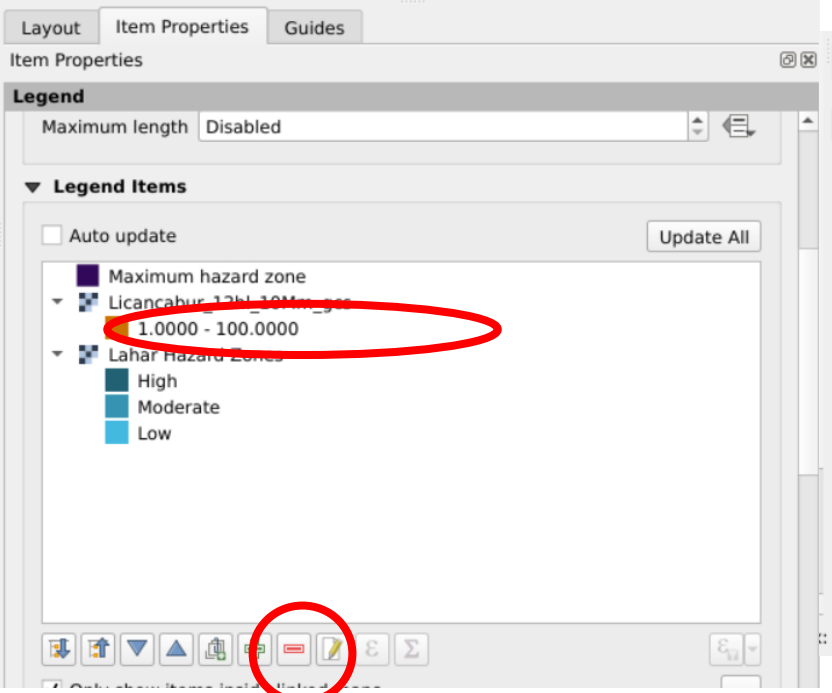
- Click auto update on and off
- Remove the DEM and hillshade
- Remove the “Band 1 (Gray)” from the lahar and PDC layers
- Remove all but one of the orange PDC zones

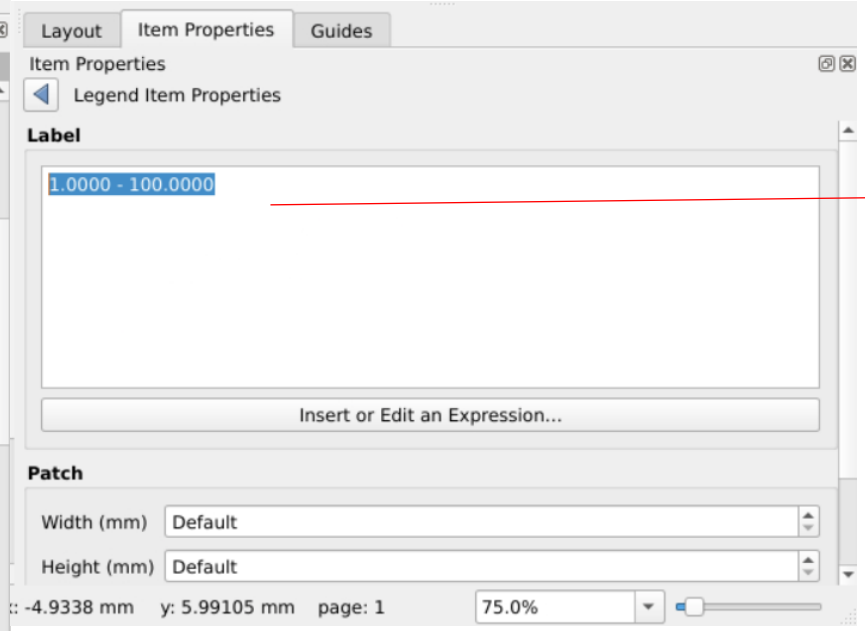


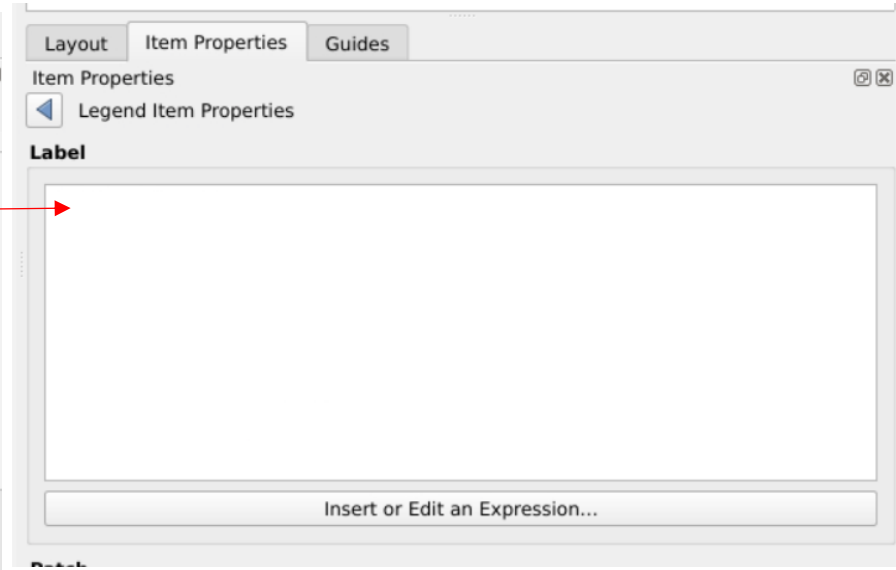


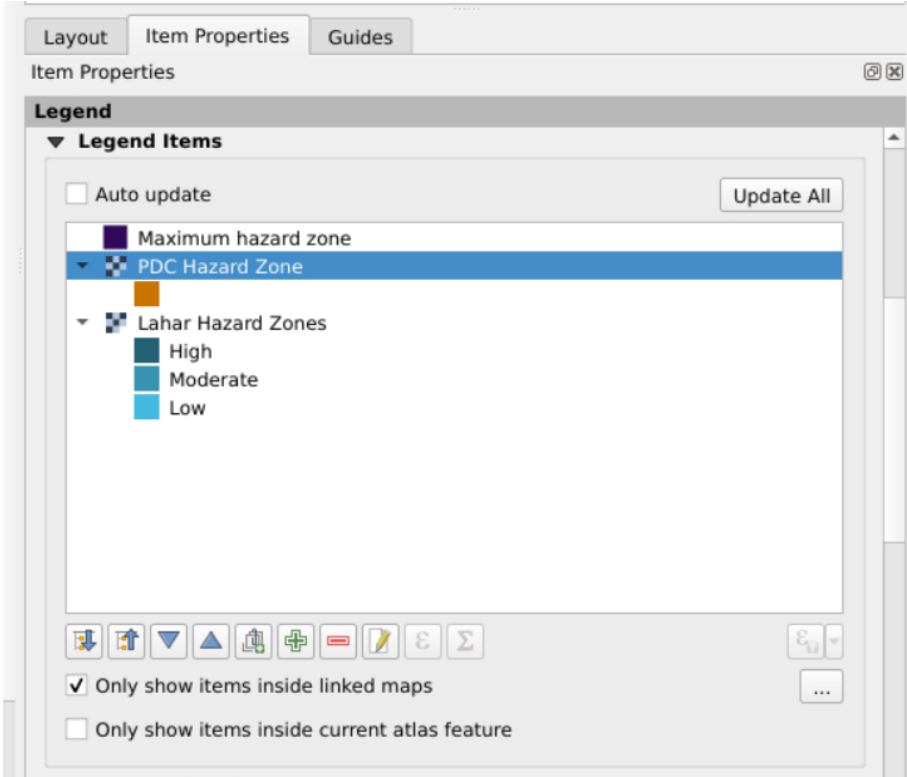


- Now let's rename some layers
- Click on the Licancabur_12hl_10Mm_gcs layer and click the pencil button
- In the edit box, change the name to "PDC Hazard Zone"
- Then click the blue back arrow

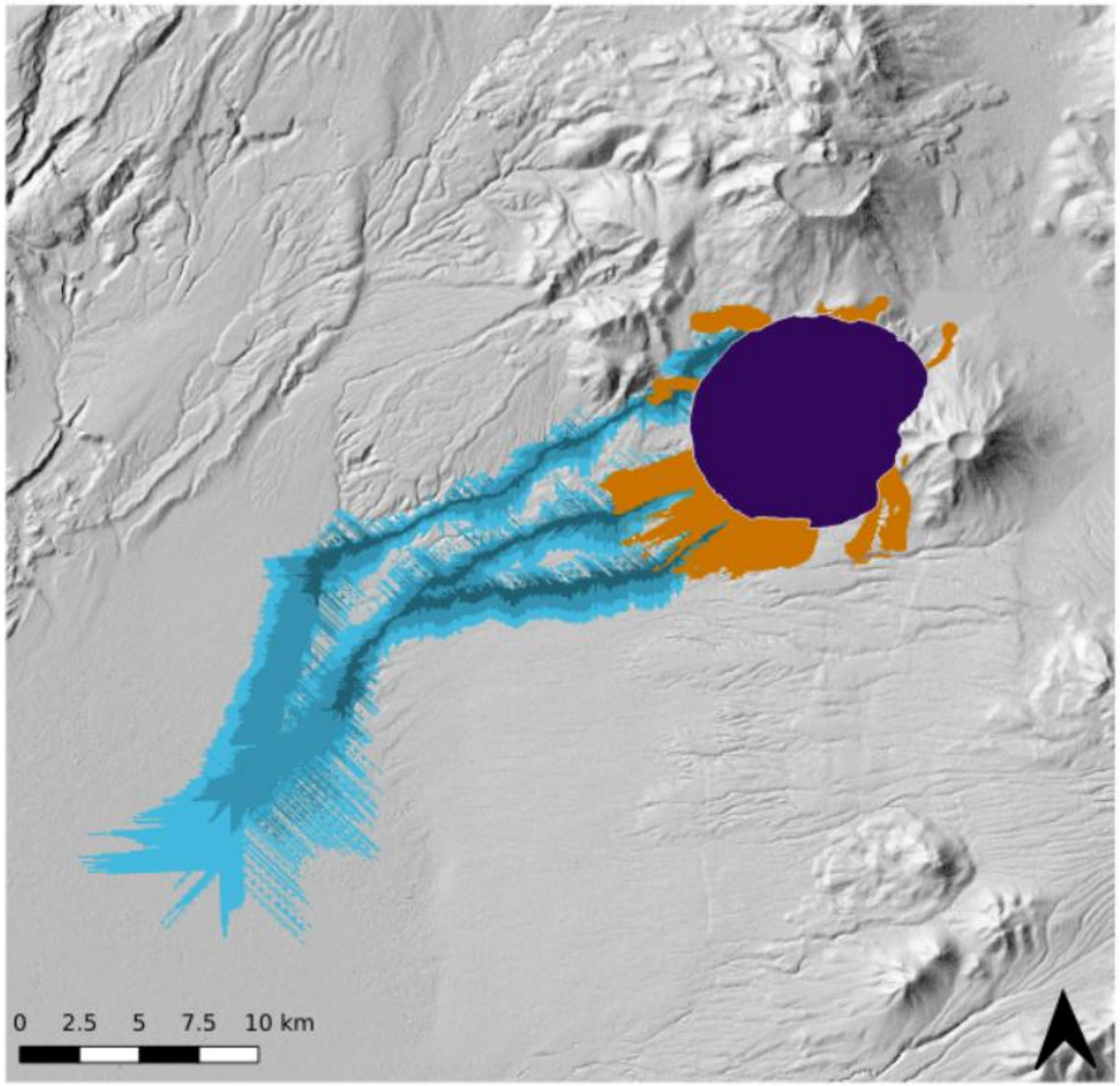




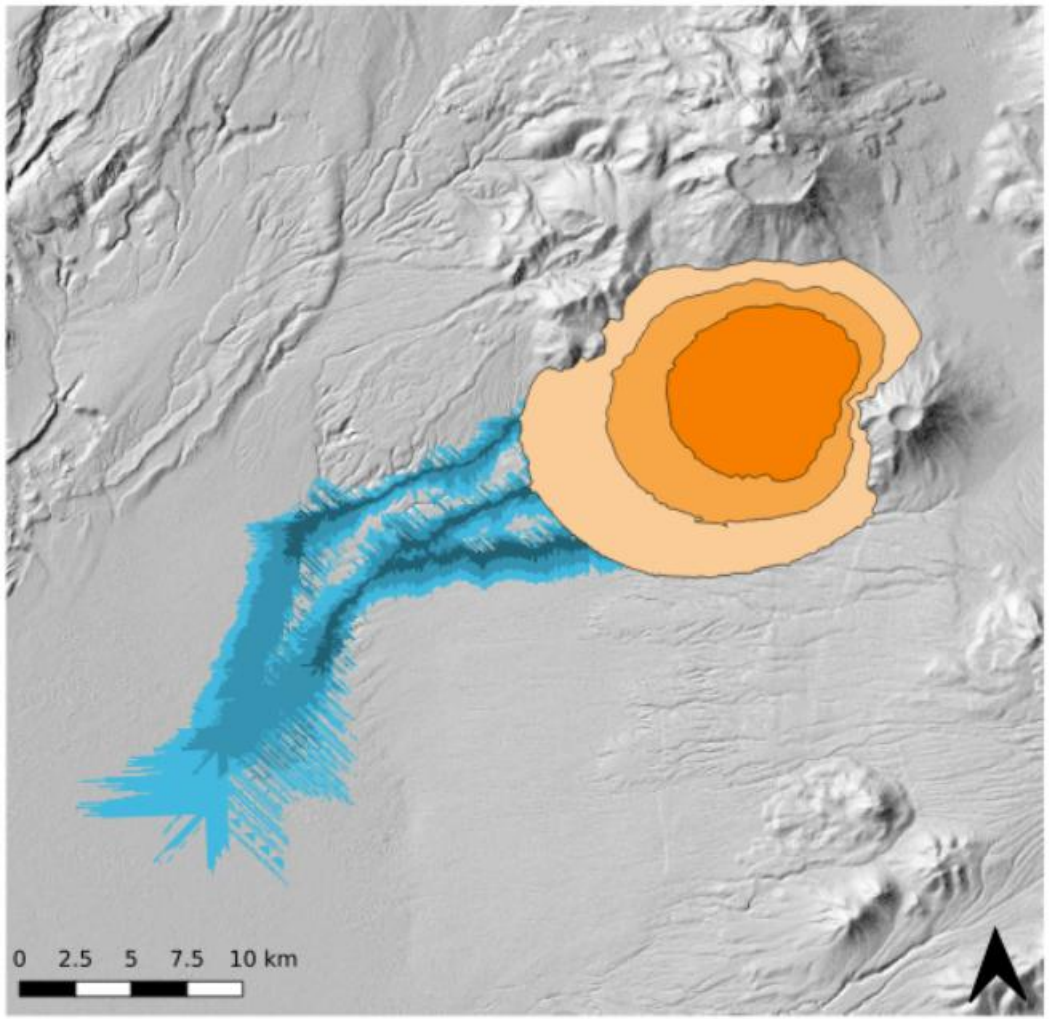




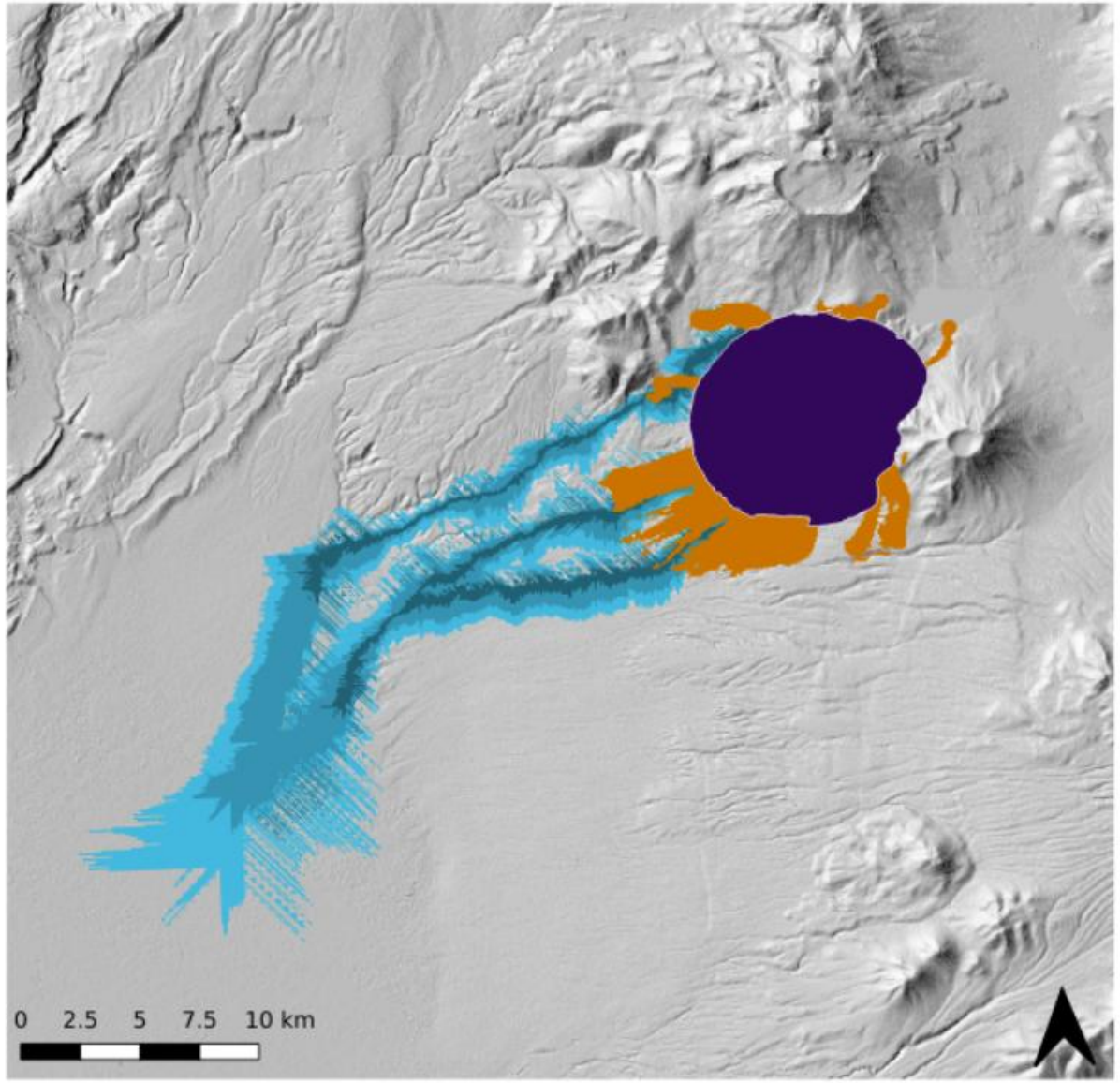
- Do the same for the PDC layer name
- In the edit box, change the name to a blank space
- Then click the blue back arrow
- Now our legend is cleaned up



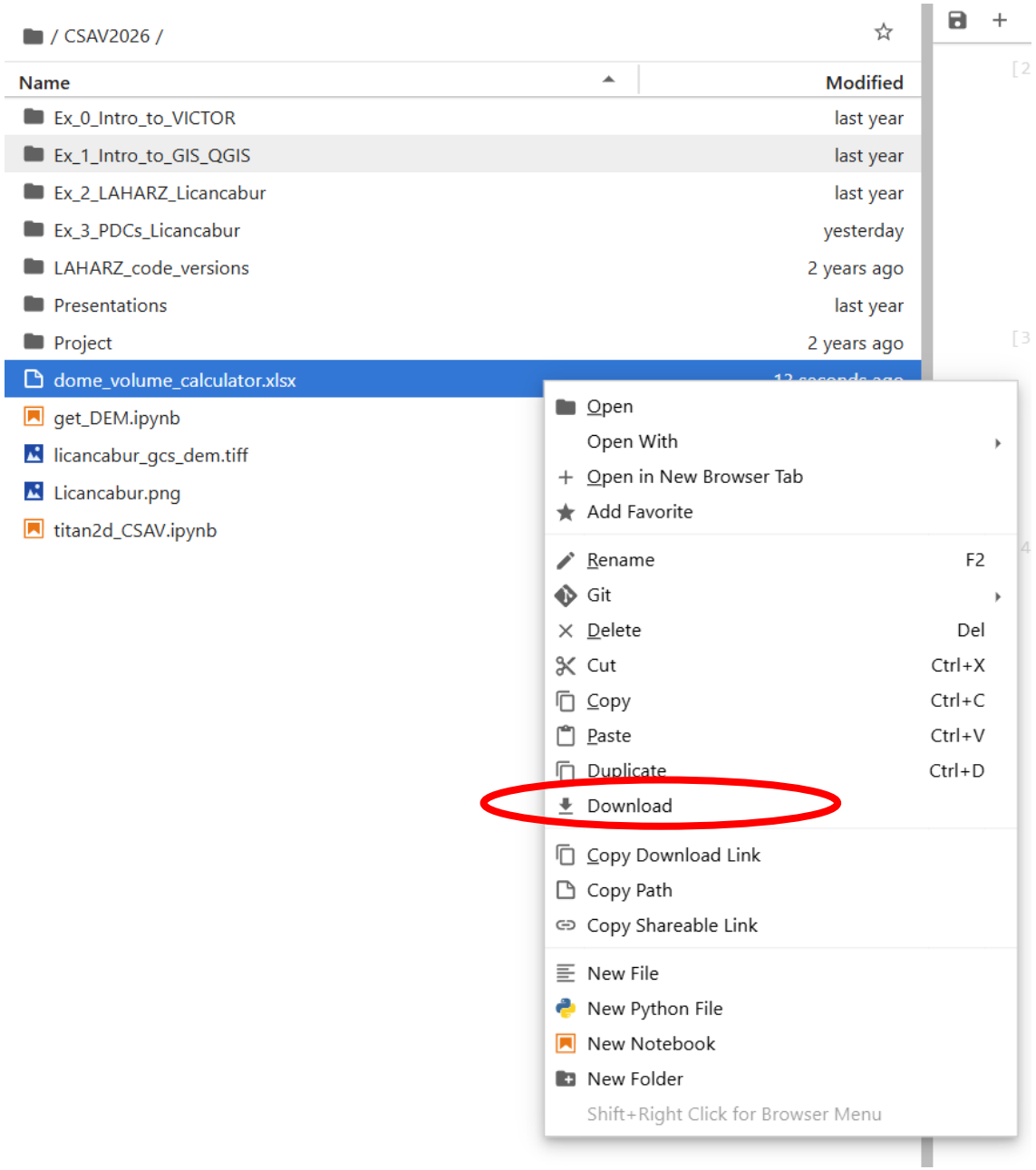
- Now we have a hazard map with separate zones for PDCs and Lahars



- We could do the same thing with our energy cone models!



- We made these maps very quickly!
- With more time, I would think about:
  - Adjusting transparency of the layers to see the topography
  - Adding a symbol for the volcano summit
  - Cleaning up the LaharZ runs to smooth the jagged edges
  - Adding additional information to the map
    - Explanations of the zones or information about these hazard types
    - Where to get more information
    - Roads, towns, infrastructure, streams
    - And many more!
  - We will talk about more hazard map design aspects on Wednesday



- For your projects, there is an Excel spreadsheet in the CSAV folder called “dome_volume_calculator.xlsx
- Download it to your computer if you want to use it for your projects
- Volume=1/2*pi*major_radius*minor_radius*height